

Evaluation of 1–14-Day Machine-Learning Forecasts of Marine Heatwaves in the Northern Bay of Bengal Against Persistence Benchmarks

B.S. Project (2025)

Abu Basir Mojumder

Exam Roll No: 238109

Registration No: 2021012705 Session: 2021-22

Department of Meteorology

Faculty of Earth and Environmental Sciences

University of Dhaka

DECLARATION

This is to certify that the work presented in this project entitled “Evaluation of 1–14-Day Machine-Learning Forecasts of Marine Heatwaves in the Northern Bay of Bengal Against Persistence Benchmarks” in partial fulfilment of the requirements for the award of the degree of Bachelor of Science (B.S.) in Meteorology from the University of Dhaka, Bangladesh, is an authentic work carried out by me and has not been submitted anywhere else for the award of any other degree.

Date: 05-11-2026

……

Abu Basir Mojumder
Registration number: 2021-012-705
Session: 2021-22
Department of Meteorology
University of Dhaka

CERTIFICATION

This is to certify that the work presented in this project entitled “Evaluation of 1–14-Day Machine-Learning Forecasts of Marine Heatwaves in the Northern Bay of Bengal Against Persistence Benchmarks” in partial fulfilment of the requirements for the degree of Bachelor of Science (B.S.) in Meteorology from the University of Dhaka, Bangladesh, is an authentic work carried out by ABU BASIR MOJUMDER, Registration number: 2021-012-705, Session: 2021-22, under my supervision and guidance, and has not been submitted anywhere else for the award of any other degree.

To the best of my knowledge, the content of this project does not form a basis for the award of any previous degree from anywhere else.

……

Sabista Yildiz
Supervisor
Lecturer
Department of Meteorology
Faculty of Earth and Environmental Sciences
University of Dhaka

ACKNOWLEDGEMENT

First of all, I am grateful to Almighty Allah for keeping me in good health and giving me the patience to complete this work.

I would like to express my sincere respect and gratitude to my supervisor, Sabista Yildiz, Lecturer, Department of Meteorology, University of Dhaka, for guidance and valuable advice throughout this project.

I am thankful to all the teachers of the Department of Meteorology for what they have taught me and for their helpful comments. I also acknowledge the NOAA Physical Sciences Laboratory for the OISST data, the Copernicus Climate Change Service and ECMWF for the ERA5 reanalysis, the ESA Climate Change Initiative and the Copernicus Marine Service for the CCI/C3S SST analysis, the NOAA Climate Prediction Center for the Niño 3.4 index, the Oceanic Niño Index and the GODAS ocean reanalysis, and the Australian Bureau of Meteorology for the RMM index.

Finally, I thank my family and friends for their encouragement and support throughout my studies.

ABSTRACT

Marine heatwaves (MHWs) harm marine ecosystems and the people who depend on them, and machine-learning (ML) models are now often used to forecast them a few days to a few weeks in advance. These forecasts are usually compared with persistence, but persistence is defined in different ways, and under the widely used Hobday et al. (2016) definition a day can only be labelled as part of an MHW once the following days are known. This study examines how much MHW forecast skill is really persistence, at lead times of 1 to 14 days in the northern Bay of Bengal (15–23°N, 85–93°E). Daily NOAA OISST v2.1 sea surface temperature (SST), ERA5 surface fluxes, winds and air temperature, and four climate indices were used for 1982–2025. MHWs were detected on detrended SST (the main target) and on raw SST with a 1982–2021 baseline, which gave 109 and 133 events respectively. Two persistence forecasts were compared: label-based persistence, which copies the official MHW label of the forecast day, and honest persistence, which uses only information available on that day. Exceedance persistence, which predicts an MHW whenever SST is above the threshold on the forecast day, was added afterwards as a post-hoc robustness test. Logistic regression and random forest models were tested with leave-one-year-out cross-validation, using purging, a 30-day embargo, tuning and calibration inside each training fold, and year-block bootstrap confidence intervals, under decision rules set before the results were known. Label-based persistence raised the Matthews correlation coefficient (MCC) by 0.28 at a 1-day lead and by 0.11 at 7 days, roughly three extra days of hindsight. This inflation was significant at every lead in the Bay of Bengal, the Levantine Sea and the South China Sea, and it was larger still for individual grid cells (median 0.31 at 1 day across 805 cells, significant in every cell at 1–7 days). For the regional-mean SST series, the models gained 0.13–0.18 MCC over honest persistence at 1–5 days, but at 1–3 days they added at most about 0.025 MCC to exceedance persistence, for both detrended and raw labels. Skill beyond exceedance persistence was small (0.02–0.06 MCC) and clearest at about 7 days; at 10–14 days it was weak and depended on the SST product. Repeated with the satellite-only ESA CCI/C3S SST (1982–2024), the analysis reproduced the inflation and the short-lead results but not the gain over persistence at 10–14 days, and the two products agreed on only about half of the MHW days. ERA5 predictors did not improve the forecasts of new events at any lead, and random forest

performed no better than logistic regression. A post-hoc lasso selection retained the Indian Ocean basin-wide warming index at every lead from 3 to 14 days, but no selected predictor significantly improved skill over SST alone, and upper-ocean temperature and isothermal-layer depth from the GODAS reanalysis also added no skill at 7–14 days with either SST product. At the yearly scale, MHW days were more than twice as frequent after El Niño winters, whereas the Indian Ocean Dipole had no effect of its own. For this regional-scale problem, the choice of benchmark therefore decides whether ML looks useless, impressive or trivial. MHW forecasting studies should report honest and exceedance persistence, explain how the five-day minimum is treated, evaluate their models with blocked designs that limit leakage, and test their results with more than one SST product.

TABLE OF CONTENTS

Declaration ..... **I**
Certification ..... **II**
Acknowledgement ..... **III**
Abstract ..... **IV**
List of Figures ..... **IX**
List of Tables ..... **X**
List of Abbreviations ..... **XII**
Chapter 1 Introduction **1**
1.1 Background of the study 1
1.2 Rationale and contribution of the study 1
1.3 Research objectives 2
1.4 Study area 3
1.5 Conceptual framework 4
1.6 Organization of the study 5
1.7 Work plan 5
Chapter 2 Literature Review **6**
2.1 Theoretical background 6
2.1.1 Defining marine heatwaves 6
2.1.2 Drivers of marine heatwaves 6
2.1.3 Forecast verification 7
2.1.4 Machine learning and evaluation leakage 7
2.2 Review of previous studies 8
2.2.1 Marine heatwaves in the Bay of Bengal 8
2.2.2 Machine-learning forecasts of marine heatwaves 9
2.2.3 How previous studies benchmark against persistence 9
2.3 Research gap 10
Chapter 3 Data and Methodology **11**
3.1 Data collection 11
3.1.1 Sea surface temperature 11
3.1.2 Atmospheric data 12
3.1.3 Climate indices 12
3.1.4 Ocean reanalysis 13
3.2 Data processing 13
3.3 Marine heatwave detection 13
3.4 Persistence benchmarks 14

3.5 Predictors....15
3.6 Models....16
3.7 Evaluation design....16
3.8 Verification metrics....17
3.9 Statistical inference and decision rules....18
3.10 Interpretation methods....18
3.11 Comparison regions....20
3.12 Grid-cell analysis....20
3.13 Validation with a second SST dataset....20
3.14 Upper-ocean heat memory test....22
3.15 Previous-year ENSO and IOD....22
3.16 Software and computing....23
Chapter 4 Results....24
4.1 Marine heatwave record of the northern Bay of Bengal....24
4.2 Atmospheric and oceanic precursors of MHW onset....26
4.3 Benchmark leakage (RQ1)....27
4.3.1 Regional mean and comparison regions....27
4.3.2 Grid-cell scale....29
4.4 Forecast skill by lead (RQ2)....31
4.5 A stricter baseline: exceedance persistence (post-hoc)....33
4.6 ERA5 and new-event skill (RQ3)....34
4.7 What the SST model uses....35
4.8 Case studies....37
4.9 Validation with a second SST dataset....40
4.9.1 Agreement between OISST and CCI/C3S....40
4.9.2 Benchmark leakage and forecast skill with both products....41
4.10 Individual predictor selection (post-hoc)....44
4.11 Upper-ocean heat memory (post-hoc)....46
4.12 Previous-year ENSO and IOD (post-hoc)....47
Chapter 5 Discussion....50
5.1 Why label-based persistence flatters persistence....50
5.2 What the regional-mean models learn....51
5.3 Raw and detrended labels....52
5.4 Dependence on the SST product....53
5.5 Comparison with previous studies....54
5.6 Limitations of the study....55
Chapter 6 Conclusion....57

6.1 Major findings.....57
6.2 Recommendations for future study.....58
References.....60
Appendices.....65
Appendix A Supplementary tables.....65
Appendix B Data and code availability.....67
Appendix C Main processing code.....67
C.1 Regional SST and MHW labels.....68
C.2 ERA5 daily fields and regional means.....70
C.3 Climate indices and the daily predictor table.....71
C.4 Persistence benchmarks and benchmark leakage.....73
C.5 Predictors.....74
C.6 Evaluation framework.....76
C.7 Models, skill by lead and exceedance persistence.....78
C.8 Coefficients and drop-one-group test.....79
C.9 Grid-cell analysis.....80
C.10 Second SST dataset.....82
C.11 Individual predictor selection with the L1 penalty.....85
C.12 Upper-ocean heat memory test (GODAS).....86
C.13 Previous-year ENSO and IOD.....90

LIST OF FIGURES

Figure	Title	Page
Figure 1.1	Study area (dashed box, 15–23°N, 85–93°E), coloured by the correlation of each grid cell's detrended daily SST anomaly with the regional mean, 1982–2025	3
Figure 1.2	Conceptual framework of the study	4
Figure 3.1	Schematic of the three persistence benchmarks for an idealised event	15
Figure 4.1	Regional-mean SST of the study box, 1982–2025	24
Figure 4.2	Annual MHW statistics for raw and detrended labels	25
Figure 4.3	Number of MHW events by onset month for raw and detrended labels, 1982–2025	25
Figure 4.4	Composites of standardised anomalies from 30 days before to 10 days after the onset of detrended MHWs	27
Figure 4.5	MCC of label-based and honest persistence by lead in the Bay of Bengal	28
Figure 4.6	Inflation of label-based over honest persistence (MCC difference) by lead in three regions	28
Figure 4.7	Grid-cell leakage for detrended labels in the study box (dashed)	30
Figure 4.8	Inflation by lead in the 805 grid cells (median and 10th–90th percentile) compared with the regional mean	31
Figure 4.9	Forecast skill by lead for the detrended labels	32
Figure 4.10	Standardised coefficients of the logistic SST model by lead	36
Figure 4.11	Forecasts for the three strongest detrended MHWs at 7-day (left) and 14-day (right) leads	38
Figure 4.12	Comparison of OISST and CCI/C3S, 1982–2024	43
Figure 4.13	Share of 100 half-samples of the years in which each predictor was selected by the L1 logistic model, by lead (detrended labels, OISST)	45
Figure 4.14	Change in MCC from adding the GODAS predictors to the logistic SST model, for (a) all forecast days and (b) new-event days	47
Figure 4.15	Previous-winter ENSO and MHW activity in the study box, 1983–2025	49

LIST OF TABLES

Table	Title	Page
Table 1.1	Work schedule of the study, May–October 2026	5
Table 2.1	Persistence benchmarks in recent ML forecasting studies of marine heatwaves	9
Table 3.1	Datasets used in the study	11
Table 3.2	Predictor groups	16
Table 3.3	Models and hyperparameter grids	16
Table 4.1	MHW event statistics, 1982–2025, for the three regions and both label types	26
Table 4.2	Inflation of label-based over honest persistence (detrended labels), with 95% CIs, and the lead it effectively gains	28
Table 4.3	Inflation of label-based over honest persistence in the 805 grid cells of the study box, compared with the regional mean	30
Table 4.4	MCC by lead for honest persistence and the four models (detrended labels)	32
Table 4.5	Exceedance persistence and the gain of the logistic SST model over it, for detrended and raw labels (MCC, with 95% CIs for the gain)	33
Table 4.6	Gain from adding ERA5 to SST predictors (95% CIs) in MCC over all days and in new-event MCC	34
Table 4.7	Seven-day comparison of all models and predictor sets (detrended labels)	35
Table 4.8	Drop-one-group test for the logistic SST model	37
Table 4.9	Case-study scores for the three strongest events	38
Table 4.10	Detrended SST and MHW statistics of the two products over the common period 1982–2024	40
Table 4.11	Agreement between the two products by period	41
Table 4.12	Inflation of label-based over honest persistence with the two products, 1982–2024 (MCC, with 95% CIs)	41
Table 4.13	MCC of honest persistence, exceedance persistence and the logistic SST model with the two products, 1982–2024, and the gain of the model over each benchmark (95% CIs)	42
Table 4.14	L1 logistic model with all 83 predictors compared with the logistic SST model (detrended labels, OISST)	44
Table 4.15	MCC of the logistic SST model with and without the four GODAS predictors (14-day delay), 1982–2024, with the	47

	gain and its 95% CI	
Table 4.16	Spearman correlation (p-value) between previous-year climate indices and detrended MHW activity, 1983–2025	48
Table A.1	Inflation of label-based over honest persistence for raw labels (MCC difference, 95% CI), and lead lost in days	65
Table A.2	Skill of honest persistence and the logistic SST model by lead (detrended labels)	65
Table A.3	Raw-label comparison of honest persistence, exceedance persistence and the logistic SST model (MCC)	66
Table A.4	MCC of the logistic SST model with and without the four GODAS predictors with a 28-day delay, 1982–2024, with the gain and its 95% CI	66
Table A.5	Spearman correlation (p-value) between previous-year climate indices and raw-label MHW activity, 1983–2025	66
Table B.1	Result files behind the tables and figures of Chapter 4	67

LIST OF ABBREVIATIONS

AVHRR	Advanced Very High Resolution Radiometer
BoB	Bay of Bengal
BSS	Brier skill score
C3S	Copernicus Climate Change Service
CCI	Climate Change Initiative (of ESA)
CI	Confidence interval
DMI	Dipole Mode Index
ECMWF	European Centre for Medium-Range Weather Forecasts
ENSO	El Niño–Southern Oscillation
ERA5	Fifth-generation ECMWF atmospheric reanalysis
ESA	European Space Agency
GODAS	Global Ocean Data Assimilation System (NCEP)
IOBW	Indian Ocean Basin-wide warming
IOD	Indian Ocean Dipole
LOYO	Leave-one-year-out
MCC	Matthews correlation coefficient
MHW	Marine heatwave
MJO	Madden–Julian Oscillation
ML	Machine learning
NCEP	National Centers for Environmental Prediction
NOAA	National Oceanic and Atmospheric Administration
OISST	Optimum Interpolation Sea Surface Temperature
ONI	Oceanic Niño Index
RF	Random forest
RMM	Real-time Multivariate MJO index
RMS	Root mean square
RQ	Research question
SST	Sea surface temperature

CHAPTER 1
INTRODUCTION

1.1 Background of the study

A marine heatwave (MHW) is a long spell of unusually warm sea water. The definition used by most researchers today comes from Hobday et al. (2016): an MHW occurs when sea surface temperature (SST) stays above the 90th percentile of a fixed seasonal climatology for at least five days in a row. Such events have caused coral bleaching, mass deaths of marine animals, shifts in the distribution of fish and heavy losses for fisheries and aquaculture (Smith et al., 2021). Over the last century MHWs have become longer and more frequent worldwide (Oliver et al., 2018), and they are expected to become more common as the ocean continues to warm (Frölicher et al., 2018).

The Bay of Bengal (BoB) is a semi-enclosed basin in the northern Indian Ocean, surrounded by India, Bangladesh and Myanmar. Many communities along its coasts depend on fishing, and its coral reefs and mangroves are sensitive to heat stress. Several recent studies have reported that MHWs in the bay have become more frequent since the early 1980s and that their year-to-year variation is linked to the El Niño–Southern Oscillation (ENSO) and the Indian Ocean Dipole (IOD) (Gao et al., 2022; Lin et al., 2023; Kumar et al., 2023; Liang et al., 2024). Within the study area of this thesis, the regional-mean SST rose by 0.21 °C per decade between 1982 and 2025, and the number of raw MHW days increased sharply after 2015.

MHWs build up over days to weeks, so a forecast issued a few days or weeks ahead could give fishers and coastal managers time to prepare. Dynamical seasonal forecasts already show useful skill for MHWs in some regions (Jacox et al., 2022). In parallel, a growing number of studies have used machine learning (ML) to forecast MHWs from observed predictors (Giamalaki et al., 2022; Sun W. et al., 2023; Bonino et al., 2024; Sun D. et al., 2024; Parasyris et al., 2025; Mi et al., 2025; Howard et al., 2026; Welandawe et al., 2025).

1.2 Rationale and contribution of the study

A forecast is only worth using if it beats a simple reference. For MHWs the obvious reference is persistence, which assumes that whatever is happening today will still be

happening on the target day. Because MHWs last many days, persistence is hard to beat at short lead times.

The Hobday definition, however, makes persistence harder to define than it first appears. Whether a given day counts as part of an MHW depends on what happens after it: a warm spell only becomes an MHW once it has lasted five days, and short gaps between events are filled in afterwards. The official label of a day is therefore not known on that day. If a persistence forecast simply copies the official label of the forecast day (called label-based persistence in this thesis), it quietly uses information from after the forecast was made, and any comparison of a model against it becomes distorted. This problem has received very little attention.

This study contributes in three ways. First, it measures how much label-based persistence inflates forecast skill at lead times of 1 to 14 days, and checks whether the same happens in two other seas and in each of the 805 grid cells of the study area. Second, it measures the skill of logistic regression and random forest models against honest persistence, which uses only what is known on the forecast day, and, as a post-hoc robustness test, against a stricter baseline called exceedance persistence. Third, it tests whether atmospheric predictors from the ERA5 reanalysis help to forecast new events better than SST alone. The main results are then checked with a second, largely independent SST product. Finally, three further analyses, added after the main results were known, test whether individual predictors selected with an L1 penalty, upper-ocean information from the GODAS ocean reanalysis, or the state of ENSO and the IOD in the previous year carry information that SST alone does not.

1.3 Research objectives

The study answers three research questions (RQs):

- **RQ1.** How much does label-based persistence inflate forecast skill, at which lead times, and does this also happen outside the Bay of Bengal?
- **RQ2.** Can statistical and ML models beat honest persistence, and at which lead times?
- **RQ3.** Do ERA5 surface predictors improve the forecasts of new events compared with SST alone?

The lead times (1, 3, 5, 7, 10 and 14 days), the models and the rule used to answer each question were decided before the corresponding results were seen, and results are reported for every lead, whether or not the rule was met. The robustness checks and the three additional analyses (Sections 3.10, 3.14 and 3.15) were added after the main results were known. Each of them has its own decision rule, which was fixed before that analysis was run, and each is reported as post-hoc.

1.4 Study area

The study area is the northern Bay of Bengal between 15–23°N and 85–93°E (Fig. 1.1). This box contains 805 ocean grid cells of the 0.25° OISST grid and touches the coasts of Bangladesh, eastern India and western Myanmar. The forecasting work uses a single area-weighted mean SST for the whole box, while the benchmark analysis of RQ1 is also repeated for every grid cell. Using one regional series is reasonable because SST in the box varies largely together: the detrended daily SST anomaly of individual cells correlates with the regional mean at a median of r = 0.73, 62% of cells have r ≥ 0.7, and even the weakest cell reaches 0.47.

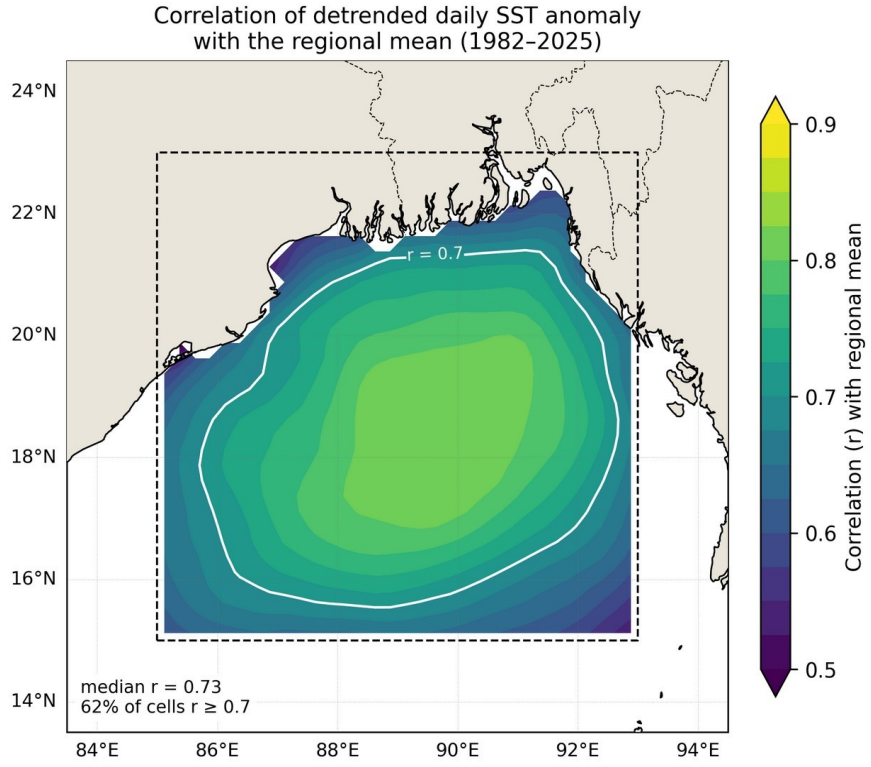


Figure 1.1 Study area (dashed box, 15–23°N, 85–93°E), coloured by the correlation of each grid cell's detrended daily SST anomaly with the regional mean, 1982–2025. The white contour marks r = 0.7.

Two other regions are used only to check whether the inflation found for RQ1 holds elsewhere: the Levantine Sea in the eastern Mediterranean (32–35.5°N, 26–32°E) and the central South China Sea (10–18°N, 111–119°E). ML forecasts of MHWs have already been published for both seas, and their events last longer than those in the BoB. The Levantine Sea is also warming about twice as fast.

1.5 Conceptual framework

The structure of the study is shown in Fig. 1.2. Three data sources feed into a common processing and evaluation design, which is then used to answer the three research questions. All three questions rely on the same MHW labels and the same evaluation design, which is built to limit leakage. The models built for RQ2 are examined further through their coefficients, a drop-one-group test, three case studies and a post-hoc selection of individual predictors, and the main results are checked with a second SST product, an ocean reanalysis and the previous year's climate modes.

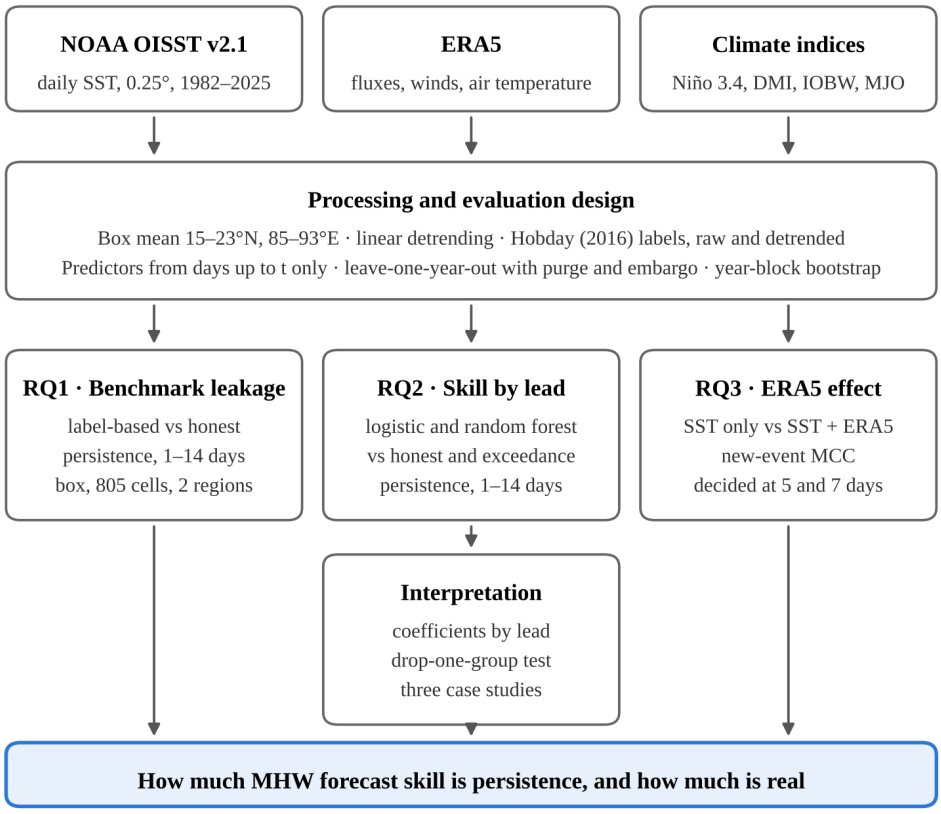


Figure 1.2 Conceptual framework of the study.

1.6 Organization of the study

The rest of the thesis is arranged as follows. Chapter 2 reviews how MHWs are defined, what drives them, how forecasts are verified and evaluated, and what earlier MHW forecasting studies have done, including the benchmarks they used. Chapter 3 describes the data, the detection of MHWs, the persistence benchmarks, the predictors, the models, the evaluation design and the additional analyses. Chapter 4 presents the results in the order of the research questions, checks them with a second SST dataset and then reports three post-hoc analyses: the selection of individual predictors, the test of upper-ocean heat memory and the link with the previous year's ENSO and IOD. Chapter 5 discusses what the results mean, compares them with earlier work and sets out the limitations, and Chapter 6 summarises the main findings and suggests directions for future work.

1.7 Work plan

The study was carried out over six months, from May to October 2026, following the schedule in Table 1.1. The literature review continued alongside the early stages of the work, while the data were collected and processed during the middle of the period. Model development, evaluation and the writing of this report took place in the final two months.

Table 1.1 Work schedule of the study, May–October 2026.

Activities	May	Jun	Jul	Aug	Sep	Oct
Literature search and review	■	■	■	■		
Topic selection and proposal preparation	■	■				
Proposal approval			■			
Data collection (OISST, ERA5, climate indices)			■	■		
Data processing and MHW detection				■	■	
Model development and evaluation					■	■
Report writing and submission						■

CHAPTER 2
LITERATURE REVIEW

2.1 Theoretical background

2.1.1 Defining marine heatwaves

Most MHW studies now follow the method proposed by Hobday et al. (2016). For every calendar day, a climatological mean and a 90th-percentile threshold are calculated from a fixed baseline period, using all values within five days either side of that date, and both are then smoothed with a 31-day moving average. A day exceeds the threshold when its SST is higher than the threshold for that date, and an MHW is a run of at least five such days. If two events are separated by a gap of two days or less, they are treated as a single event. Each event can then be described by its duration, its mean and maximum intensity (the SST anomaly above the climatological mean) and its cumulative intensity, which is the sum of the daily anomalies over the event and is expressed in °C·days. Hobday et al. (2018) later added four categories (moderate, strong, severe and extreme) based on multiples of the difference between the threshold and the climatological mean.

Two parts of this definition are important for forecasting. The first is the choice of baseline. With a fixed baseline, long-term warming by itself produces more and longer events in recent years. If the trend is removed before detection, events instead measure departures from the warming trend, which is closer to what a short-range forecast can realistically predict (Oliver et al., 2021). The second is that the label of a day depends on later days. The first four days of an event only become MHW days once the fifth day has occurred, and a gap is only filled once the event has resumed. This is not a problem when analysing past events, but it does matter for forecasting, because the official label of the forecast day is not yet known when the forecast is issued.

2.1.2 Drivers of marine heatwaves

An MHW develops when heat builds up in the ocean mixed layer faster than it is lost. Holbrook et al. (2019) separated the local causes, which are anomalies in the air–sea heat flux and in ocean heat transport, from the remote causes, which are large-scale climate modes that set up those local anomalies. Weak winds reduce evaporation and

latent heat loss and also weaken vertical mixing, clear skies let more sunlight reach the sea surface, and a shallow mixed layer warms faster for the same amount of heat. Once a warm anomaly has formed it tends to fade slowly, because the upper ocean stores heat for weeks, and this is what gives MHWs their persistence (Holbrook et al., 2020).

In the Bay of Bengal the main remote influences are ENSO, the IOD and the Indian Ocean Basin-wide warming (IOBW) mode (Gao et al., 2022; Lin et al., 2023; Liang et al., 2024). The IOBW mode is the warming of the whole tropical Indian Ocean that tends to follow an El Niño event (Yang et al., 2007). The Madden–Julian Oscillation (MJO), usually tracked with the RMM index of Wheeler and Hendon (2004), changes cloud cover and winds over periods of a few weeks and is a possible source of predictability on subseasonal time scales.

2.1.3 Forecast verification

When an MHW forecast is made for a fixed lead time, it gives a yes-or-no answer for each target day, and it can be checked with a contingency table of hits (TP), false alarms (FP), misses (FN) and correct negatives (TN) (Jolliffe and Stephenson, 2012). Because MHW days are rare, a forecast that never predicts an event can still have high accuracy simply by getting the many quiet days right. The Matthews correlation coefficient (MCC; Matthews, 1975) avoids this problem, as it uses all four cells of the table, equals zero for a forecast with no skill and is not misled by class imbalance (Chicco and Jurman, 2020). It is therefore used as the main score in this study.

Forecasts that give a probability are checked with the Brier score, which is the mean squared difference between the forecast probability and what actually happened (Brier, 1950). The Brier skill score compares this with a reference forecast, normally climatology (Wilks, 2011). It is worth stressing that skill is always relative: a forecast only has skill if it does better than a reference that would have been available at the time. Persistence and climatology are the two references most often used (Jolliffe and Stephenson, 2012).

2.1.4 Machine learning and evaluation leakage

Logistic regression describes the log-odds of an event as a linear combination of the predictors, and a penalty term keeps the coefficients small to reduce overfitting

(Hastie et al., 2009). An L1 (lasso) penalty goes further and sets the coefficients of uninformative predictors exactly to zero, so that it also selects predictors (Tibshirani, 1996); because the selection can change from one sample to another, its stability is usually checked by refitting the model on subsamples of the data (Meinshausen and Bühlmann, 2010). A random forest averages many decision trees, each grown on a bootstrap sample of the data with a random subset of predictors, so it can capture non-linear effects and interactions (Breiman, 2001). Gradient-boosted trees such as XGBoost build trees one after another, each correcting the errors of the previous ones (Chen and Guestrin, 2016). The raw scores from these classifiers are not always reliable probabilities, but isotonic regression can be used to calibrate them (Zadrozny and Elkan, 2002).

Whatever the model, its skill only means something if it is measured on data that the model has not seen. Leakage, meaning the use of information during training or testing that would not be available in a real forecast, is a common reason why ML studies overstate their results (Kaufman et al., 2012; Kapoor and Narayanan, 2023). Daily climate data are strongly autocorrelated, so splitting days at random puts almost identical days in both the training and the test set. The usual remedy is blocked cross-validation, where whole blocks of time, such as complete years, are held out and a buffer is left around them (Roberts et al., 2017). For the same reason, hyperparameters, calibration and decision thresholds must be chosen without looking at the test data; otherwise the test score itself becomes part of the fitting.

2.2 Review of previous studies

2.2.1 Marine heatwaves in the Bay of Bengal

Kumar et al. (2023) identified 107 MHW events in the Bay of Bengal between 1982 and 2021, about three a year, with an upward trend of 1.11 events per decade. The strongest event, in 2016, lasted 69 days, and the authors attributed most events to net heat flux and anticyclonic eddies. Gao et al. (2022) also found significant increases in MHW days and frequency over 1982–2020, and showed that ENSO dominates their year-to-year variation, with the North Atlantic Oscillation playing a smaller role. Lin et al. (2023) showed that El Niño raises the number of MHW days in a clearly seasonal way: in winter and spring the extra events occur in the southern bay, driven by stronger sunlight and a thick barrier layer, whereas in summer they shift to the

central bay, where surface heating and a deeper thermocline are important. Liang et al. (2024) linked autumn MHWs in the central bay to positive IOD events, which increase the net heat flux into the ocean by reducing latent heat loss and increasing shortwave radiation.

All of these studies describe and explain MHWs in the bay, but none of them tries to forecast them, and no ML forecasting study for the Bay of Bengal was found.

2.2.2 Machine-learning forecasts of marine heatwaves

Giamalaki et al. (2022) used random forests to predict whether an MHW would be present, and how severe it would be, in the northeast Pacific at weekly lead times, and reported 76% accuracy for presence or absence. Bonino et al. (2024) compared random forest, long short-term memory and convolutional neural networks for forecasting SST and MHW occurrence in 16 Mediterranean regions at short lead times, using a dynamical ocean forecast as the reference. Sun W. et al. (2023) combined a U-Net with a ConvLSTM to forecast MHWs in the South China Sea up to seven days ahead, and Sun D. et al. (2024) used a convolutional network to correct a dynamical subseasonal forecast of global MHWs. Parasyris et al. (2025) trained an attention U-Net to forecast Mediterranean MHW maps 3 and 7 days ahead. Mi et al. (2025) used ridge regression and random forest for summer MHWs in the South China Sea at weekly lead times, and Howard et al. (2026) developed a probabilistic U-Net for subseasonal MHW forecasts in the Indian Ocean. A broader review of ML methods for MHW prediction is given by Welandawe et al. (2025).

2.2.3 How previous studies benchmark against persistence

Table 2.1 summarises how these studies defined persistence. The information comes from the methods section of each paper, except for Mi et al. (2025), whose methods could not be read in full.

Table 2.1 Persistence benchmarks in recent ML forecasting studies of marine heatwaves.

Study	Region and model	Persistence benchmark	Five-day minimum in labels
Giamalaki et al. (2022)	Northeast Pacific; random forest	None; logistic regression is the only baseline	Yes
Bonino et al. (2024)	Mediterranean; RF, LSTM, CNN	None; compared with a dynamical forecast	Yes

Study	Region and model	Persistence benchmark	Five-day minimum in labels
Sun W. et al. (2023)	South China Sea; U-Net + ConvLSTM	None	Yes; gaps of 1 day joined
Sun D. et al. (2024)	Global; CNN correction of a dynamical forecast	Last observed state carried forward	Not stated
Parasyris et al. (2025)	Mediterranean; attention U-Net	MHW label map 3 or 7 days before the target	Yes
Mi et al. (2025)	South China Sea; ridge, RF	Used; definition not available	Not available
Howard et al. (2026)	Indian Ocean; probabilistic U-Net	Weekly SST anomaly carried forward, for SST scores only	No minimum duration

Several points stand out from this comparison. To begin with, three of the seven studies report MHW skill without any persistence reference at all, even though persistence is a strong competitor at lead times of a week or less. Parasyris et al. (2025) do use persistence, but their persistence forecast is the MHW label map from 3 or 7 days before the target date. Since their labels follow the five-day minimum, this map already contains information from the days after it was observed, which is exactly the label-based persistence examined in this thesis; their MHW maps are also used as model inputs. Finally, the studies define events differently (minimum duration, gap rule, daily or weekly units), so their scores refer to different targets and cannot be compared directly.

2.3 Research gap

Earlier studies either leave out a persistence reference or define it in a way that may use information from after the forecast date, and the size of this effect has never been measured, at any lead time or in any region. No study has separated real forecast skill from the simple fact that an event has already started on the forecast day. For the Bay of Bengal, there is no ML forecast of MHWs at all. This study addresses these three gaps.

CHAPTER 3
DATA AND METHODOLOGY

3.1 Data collection

The datasets used are listed in Table 3.1. All of them cover the period from 1 January 1982 to 31 December 2025, which is 16,071 days, except the second SST product used only for validation (Section 3.13), which ends on 31 December 2024, and the GODAS ocean reanalysis (Section 3.14), which was used for 1981–2024.

Table 3.1 Datasets used in the study.

Dataset	Variables	Resolution	Source
NOAA OISST v2.1	Daily mean SST	0.25°, daily	NOAA Physical Sciences Laboratory (Huang et al., 2021)
ERA5 single levels	2 m air temperature; 10 m u and v wind; surface latent and sensible heat flux; net shortwave and net longwave radiation	0.25°, hourly	Copernicus Climate Change Service (Hersbach et al., 2020)
Niño 3.4 index	Equatorial Pacific SST anomaly (1991–2020 base)	Monthly	NOAA Climate Prediction Center, from ERSST v5 (Huang et al., 2017)
Oceanic Niño Index (ONI)	Three-month running mean of the Niño 3.4 anomaly	Seasonal	NOAA Climate Prediction Center
Dipole Mode Index	Western minus south-eastern tropical Indian Ocean SST anomaly	Daily	Computed from OISST (Saji et al., 1999)
IOBW index	Tropical Indian Ocean SST anomaly, 40–100°E, 20°S–20°N	Daily	Computed from OISST (Yang et al., 2007)
RMM1, RMM2	Real-time Multivariate MJO index	Daily	Australian Bureau of Meteorology (Wheeler and Hendon, 2004)
ESA CCI/C3S SST analysis	Daily mean SST (validation only, 1982–2024)	0.05°, daily	Copernicus Marine Service (Embury et al., 2024)
NCEP GODAS	Potential temperature (5–45 m); isothermal-layer depth	1° × 1/3°, pentad	NOAA Climate Prediction Center (Behringer and Xue, 2004)

3.1.1 Sea surface temperature

Daily SST was taken from version 2.1 of the NOAA Optimum Interpolation SST analysis (OISST). This product combines satellite, ship and buoy observations on a global 0.25° grid (Reynolds et al., 2007; Huang et al., 2021). The yearly global files

were downloaded from the NOAA Physical Sciences Laboratory and cut to the study box and to the two comparison regions.

3.1.2 Atmospheric data

Hourly fields from the ERA5 reanalysis (Hersbach et al., 2020) were downloaded for 15–23°N, 85–93°E: 2 m air temperature, the two 10 m wind components, the surface latent and sensible heat fluxes, and the net shortwave and net longwave radiation at the surface. From these, eight daily variables were derived: 2 m air temperature (t2m), wind speed (wspd), the zonal and meridional wind (u10, v10), net shortwave radiation (swnet), latent heat flux (lhf), sensible heat flux (shf) and the net surface heat flux (qnet), which is the sum of the four flux terms. In ERA5 the fluxes are positive downwards, so a positive latent heat flux anomaly means that the ocean is losing less heat than usual.

3.1.3 Climate indices

Four climate indices were used to represent remote influences on the bay. ENSO is represented by the monthly Niño 3.4 SST anomaly published by the NOAA Climate Prediction Center, which is based on ERSST v5 (Huang et al., 2017) and uses a 1991–2020 base period. For the analysis of the previous year's ENSO (Section 3.15), the Oceanic Niño Index (ONI) of the same centre, a three-month running mean of the Niño 3.4 anomaly, was also used. The Dipole Mode Index (DMI) of Saji et al. (1999), which measures the IOD, was computed daily from OISST as the difference between the SST anomalies of the western box (50–70°E, 10°S–10°N) and the south-eastern box (90–110°E, 10°S–0°), with anomalies taken relative to a 1982–2021 daily climatology smoothed over 31 days. Over 528 months, the monthly means of this daily DMI correlate with the NOAA Physical Sciences Laboratory DMI based on HadISST (Rayner et al., 2003) at r = 0.82, and at r = 0.90 in September–November, when the IOD is strongest. The IOBW index is the area-mean SST anomaly over 40–100°E and 20°S–20°N (Yang et al., 2007), computed from OISST in the same way and then linearly detrended. Finally, the RMM1 and RMM2 indices describe the phase and strength of the MJO (Wheeler and Hendon, 2004).

3.1.4 Ocean reanalysis

Subsurface temperature was taken from the pentad (5-day mean) fields of the NCEP Global Ocean Data Assimilation System (GODAS; Behringer and Xue, 2004), distributed by the NOAA Climate Prediction Center. GODAS assimilates temperature profiles into an ocean model forced by the NCEP Reanalysis 2 and is provided on a 1° × 1/3° grid with 10 m vertical spacing in the upper ocean. Potential temperature at 5, 15, 25, 35 and 45 m and the depth of the bottom of the isothermal layer were extracted for 1981–2024.

3.2 Data processing

The regional SST series is the mean over the 805 ocean cells of the box, with each cell weighted by the cosine of its latitude. The same weighting was used for the Levantine Sea and the South China Sea, keeping only cells that are ocean throughout the record. The hourly ERA5 fluxes are stored as accumulations over one hour (J m^{-2}); they were converted to mean fluxes in W m^{-2} by dividing by 3600 s and then averaged to daily means over the box, which is equivalent to dividing the daily total by 86,400 s. The resulting values are physically reasonable: on average, net shortwave radiation is 203 W m^{-2}, net longwave radiation −61 W m^{-2}, latent heat flux −125 W m^{-2} and sensible heat flux −9 W m^{-2}, giving a small mean net heat gain of about 9 W m^{-2}. Each monthly Niño 3.4 value was assigned to every day of the following month, so that a forecast made on any day only uses a value that would already have been available.

Several checks were made before the analysis. There were no duplicate or missing days, and the ocean mask did not change over time. As a physical check, the daily net heat flux anomaly was compared with the day-to-day change in SST, and their cross-correlation peaked at lag 0, as expected if the flux drives the change in temperature.

The regional SST rises by 0.211 °C per decade over 1982–2025. A linear trend was fitted to the daily series and removed to obtain detrended SST, which forms the basis of the main target.

3.3 Marine heatwave detection

MHWs were detected with the method of Hobday et al. (2016), following the marineHeatWaves code used by Oliver et al. (2018). For each calendar day, the

climatological mean and the 90th-percentile threshold were calculated from the 1982–2021 baseline using all days within ±5 days of that date, and both were then smoothed with a 31-day moving average. The 1982–2021 baseline covers the first 40 complete years of the record, longer than the minimum of 30 years recommended by Hobday et al. (2016). Days with SST above the threshold were marked as exceedances, runs of at least five such days were kept as events, and events separated by two days or less were joined.

The detection was carried out twice. Applied to detrended SST, which gives the main target, it produced 109 events and 1192 MHW days. Applied to raw SST, the secondary target, it produced 133 events and 1652 MHW days. To make sure the labels were correct, a separate implementation was written for this study; it reproduced the original label file exactly, with no mismatched days and thresholds agreeing within 0.0003 °C.

3.4 Persistence benchmarks

A persistence forecast issued on day t for target day t + h simply predicts that the MHW state of day t will continue until day t + h. Three versions of this idea were compared (Fig. 3.1).

- **Label-based persistence** uses the official Hobday label of day t. Since that label depends on later days, this forecast relies on information that was not available on day t.
- **Honest persistence** predicts an MHW only if SST has been above the threshold on at least five consecutive days up to and including day t. It uses nothing after day t and serves as the persistence benchmark for RQ2.
- **Exceedance persistence** predicts an MHW whenever SST is above the threshold on day t, even if an event has not formed yet. It was added after the RQ2 results had been seen, as a stricter test, and it is reported as such.

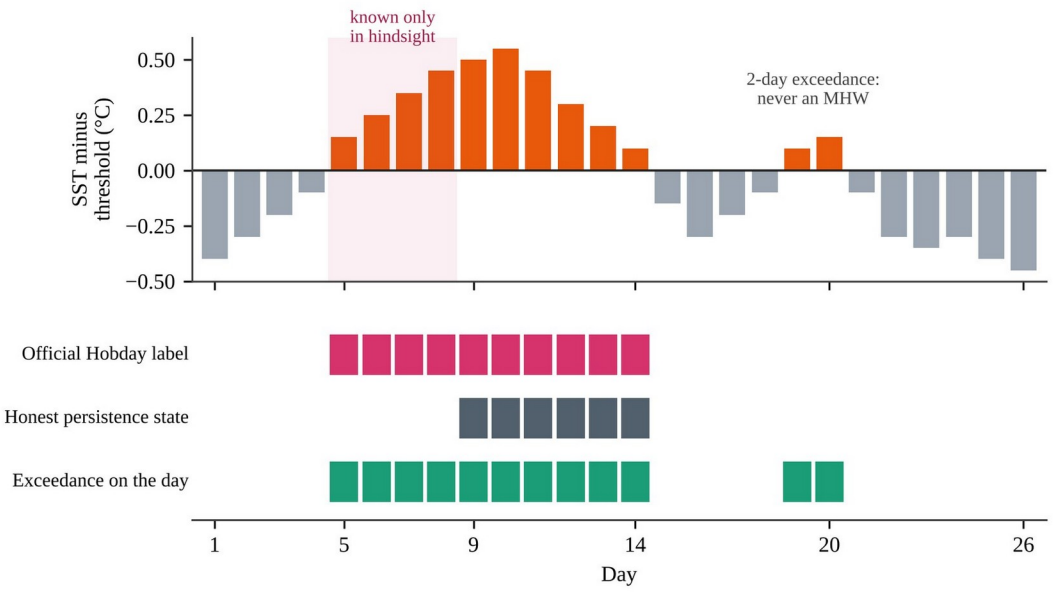


Figure 3.1 Schematic of the three persistence benchmarks for an idealised event. The first four days of the event are labelled as MHW days only once the fifth day has occurred, so they are known only in hindsight.

All three forecasts are scored against the official label of day t + h. The inflation caused by label-based persistence at lead h is defined as

$$Inflation(h) = MCC_{label}(h) - MCC_{honest}(h) \quad (3.1)$$

This difference was further split into the part caused by the five-day minimum, found by comparing with labels built without gap joining, and the part caused by gap joining. To express the inflation in days, the lead lost at lead h was calculated as the difference between h and the shorter lead at which honest persistence reaches the same MCC as label-based persistence does at h, using linear interpolation. The same analysis was repeated for the Levantine Sea and the South China Sea, each with its own MHW detection.

3.5 Predictors

For each forecast day t, 83 predictors were built using only data from day t or earlier (Table 3.2). SST anomalies were calculated relative to the fixed 1982–2021 Hobday climatology of the same series (the detrended series for the detrended target), and ERA5 anomalies relative to a fixed 1982–2021 day-of-year climatology smoothed with a 31-day window. The change predictors are simple differences of the anomaly, chg_k = anomaly(t) − anomaly(t − k).

Table 3.2 Predictor groups. All predictors use data on or before the forecast day.

Group	Number	Predictors
SST	17	Anomaly; distance to threshold; anomaly lagged 1, 3 and 7 days; running means over 3, 7, 14 and 30 days; 7-day standard deviation and maximum; changes over 1, 3 and 7 days; exceedance flag; length of the current exceedance run; honest MHW state
Calendar	2	Sine and cosine of the day of year
ERA5	56	For each of t2m, wspd, u10, v10, swnet, lhf, shf and qnet: anomaly; running means over 3, 7, 14 and 30 days; changes over 3 and 7 days
Climate indices	8	Niño 3.4, DMI, IOBW, RMM1, RMM2, RMM amplitude, 30-day means of DMI and IOBW

Three sets of predictors were compared: SST alone (the 17 SST predictors and the 2 calendar terms), SST + ERA5 (75 predictors) and, at the 7-day lead only, all 83 predictors.

3.6 Models

Three types of model were used (Table 3.3). Logistic regression is the simplest model that gives probabilities, so it was used as the reference ML model. Random forest was included because it can capture non-linear effects. XGBoost was tested at the 7-day lead but dropped afterwards, because it performed clearly worse than logistic regression. Class weights were used in all models to balance the small number of MHW days. The models were built with scikit-learn (Pedregosa et al., 2011) and XGBoost (Chen and Guestrin, 2016) in Python, using Google Colab.

Table 3.3 Models and hyperparameter grids.

Model	Fixed settings	Tuned hyperparameter
Logistic regression	Standardised predictors; balanced class weights	Penalty strength $C \in \{0.01, 0.1, 1\}$
Random forest	150 trees; $\sqrt{p}$ predictors per split; 50% bootstrap samples; balanced class weights	Minimum leaf size $\in \{10, 50\}$
XGBoost (7-day lead only)	300 trees; learning rate 0.05; subsampling 0.8; weight of positives = negatives/positives	Maximum depth $\in \{3, 5\}$

3.7 Evaluation design

The target at lead h is the official MHW label of day t + h. Every model was evaluated with leave-one-year-out (LOYO) cross-validation over the 44 years. In each

fold, the test set contains every forecast issued in the held-out year. The training set leaves out that year, every row whose target day falls in it (purging), and the first 30 days after it (an embargo), so that autocorrelated days next to the test year cannot inform the model. The lag-1 autocorrelation of the daily anomaly is 0.955, which for a first-order autoregressive process corresponds to an e-folding time of about 22 days, so a 30-day embargo covers the period over which neighbouring days remain strongly related.

All choices that depend on the data were made inside the training fold. An inner cross-validation grouped by year, with four folds for logistic regression and three for the tree models and the same purge and embargo, selected the hyperparameter value with the highest MCC. Isotonic regression fitted to the inner out-of-fold probabilities was then used to calibrate the scores (Zadrozny and Elkan, 2002), and the decision threshold was set where the calibrated inner probabilities gave the highest MCC. Finally, the model was refitted on the whole training fold and applied to the held-out year. The test year was never used to choose hyperparameters, calibration or thresholds. The MHW climatology, the threshold and the trend, however, were fixed once from the full record (Section 3.3), so the design limits leakage but is not fully free of it (Section 5.6).

Forecasts were evaluated at lead times of 1, 3, 5, 7, 10 and 14 days, with an additional 2-day lead in the leakage analysis of RQ1. Between 16,028 and 16,035 forecasts were available at each lead, and 7.4% of them had an MHW on the target day.

3.8 Verification metrics

The main score is the Matthews correlation coefficient:

$$MCC = (TP \cdot TN - FP \cdot FN) / \sqrt{[(TP + FP)(TP + FN)(TN + FP)(TN + FN)]} \quad (3.2)$$

where TP, FP, FN and TN are the numbers of hits, false alarms, misses and correct negatives. MCC lies between −1 and 1, and a value of 0 means no skill.

The quality of the forecast probabilities was measured with the Brier skill score:

$$BSS = 1 - BS / BS_{ref} \quad (3.3)$$

where BS is the mean squared difference between the forecast probability and the outcome. The reference score BS_{ref} comes from the monthly frequency of MHW days

in the training years. Persistence forecasts enter the Brier score as probabilities of 0 or 1.

Two more scores focus on new events. The new-event MCC is the MCC calculated only on forecast days when no MHW was known to be under way (honest MHW state of day t equal to 0), so it leaves out the days on which persistence is right almost by default. The onset hit rate is the fraction of event onset days that were correctly forecast. It has to be read with care, because it rises whenever the decision threshold is lowered, at the cost of more false alarms.

3.9 Statistical inference and decision rules

Uncertainty was estimated with a year-block bootstrap. The 44 years were resampled with replacement 2000 times, and each time the MCC was recalculated from the summed contingency counts of the selected years (Efron and Tibshirani, 1993; Künsch, 1989). Resampling whole years keeps the autocorrelation within each year intact. Differences between two forecasts were calculated on the same resamples, and a difference was only counted as significant if its entire 95% confidence interval (CI) was above zero. Because two forecasts tend to do well or badly in the same years, their separate CIs can overlap even when the paired difference is clearly positive, so significance was judged only from the paired difference.

The following decision rules were fixed before the results were seen:

- **RQ1** is supported if the inflation of label-based persistence has a CI above zero in the BoB and in at least one other region.
- **RQ2** is supported at a given lead if a model's MCC gain over honest persistence has a CI above zero.
- **RQ3** is supported if SST + ERA5 gives a higher new-event MCC than SST alone, with a CI above zero, at both the 5-day and the 7-day lead, for the same type of model.

3.10 Interpretation methods

Three further analyses were used to understand what the logistic SST model relies on. First, the model was refitted on all years with a fixed penalty (C = 0.1) at each lead, and its standardised coefficients were bootstrapped 200 times by year. Second, in a

drop-one-group test, the model was evaluated again with the same LOYO design after removing one group of predictors defined by time scale. The groups were the current state (anomaly, distance to the threshold, lags of 1 and 3 days, 3-day mean, exceedance flag, run length and honest MHW state), the tendency (changes over 1, 3 and 7 days), the background (lag of 7 days, means over 7, 14 and 30 days, and the 7-day maximum and standard deviation) and the season (the calendar terms). A loss with a CI above zero means that the group carries information that the other groups cannot replace. Third, three case studies were chosen by a rule decided before any forecast was looked at: the three detrended events with the largest cumulative intensity.

A fourth analysis, added after the main results were known, tested individual predictors rather than groups. Logistic regression with an L1 (lasso) penalty (Tibshirani, 1996), which sets the coefficients of uninformative predictors exactly to zero, was fitted with all 83 predictors in the LOYO design of Section 3.7. The penalty was chosen inside each training fold from $C \in \{0.0005, 0.001, 0.003, 0.01, 0.03\}$; at the 1-day lead, where every fold chose the largest value, the grid was extended to 0.1 and 0.3. The forecasts were compared with those of the logistic SST model over all days and on new-event days, using the paired year-block bootstrap of Section 3.9.

Because the 44 LOYO training sets share 43 of their years, selection frequencies over the folds mostly repeat the fit to the full record. Stability was therefore measured by refitting the model on 100 half-samples of the years, formed by 50 random splits of the 44 years into two complementary halves of 22 years (Meinshausen and Bühlmann, 2010; Shah and Samworth, 2013). At each lead the penalty was the value chosen most often in the LOYO folds, doubled because each half-sample contains half of the rows. A predictor was called stable if it was selected in at least 80% of the half-samples and kept the same sign in at least 95% of them. The expected number of falsely selected predictors among the stable ones was bounded by $q^2 / ((2\pi - 1)\, p)$ (Meinshausen and Bühlmann, 2010), where q is the mean number of predictors selected per fit, $\pi = 0.8$ and $p = 83$.

Two decision rules were fixed before this analysis was run. The L1 model adds skill only if its gain over the logistic SST model, in MCC or in new-event MCC, has a CI above zero at both 5 and 7 days. ERA5 or the climate indices carry a stable signal only if at least one of their predictors is stable at both 5 and 7 days.

3.11 Comparison regions

For the Levantine Sea and the South China Sea, daily box means were calculated from the same OISST files with the same weighting. Each series was detrended, MHWs were detected with the same method and baseline, and the leakage analysis described in Section 3.4 was repeated for both raw and detrended labels. No forecast models were trained for these regions.

3.12 Grid-cell analysis

Most published ML forecasts predict MHW maps rather than regional means, so the leakage analysis was also repeated for each of the 805 ocean cells in the study box. The daily SST of each cell was detrended separately (cell trends ranged from 0.18 to 0.24 °C per decade between the 10th and 90th percentiles). MHWs were detected with the same Hobday settings and 1982–2021 baseline, and label-based, honest and exceedance persistence were scored against each cell's own labels at the same lead times, with CIs from the same year-block bootstrap. As a check, the same code applied to the regional-mean series reproduced the original label file exactly (no mismatched days and thresholds within 0.0003 °C).

This analysis was added after the main results were available, but its decision rule was written down before it was run: the leakage holds at grid scale if at least 90% of cells have an inflation CI above zero at every lead from 1 to 7 days (detrended labels). All other statistics from the grid are only descriptive, because neighbouring cells are strongly correlated and cannot be treated as independent samples. No forecast models were trained for individual cells.

3.13 Validation with a second SST dataset

All forecasting results above rest on one SST product, so the main analyses were repeated with a second, largely independent one: the reprocessed SST analysis of the ESA Climate Change Initiative (CCI) and the Copernicus Climate Change Service (C3S), which is distributed by the Copernicus Marine Service (Embury et al., 2024; Copernicus Marine Service, 2025). It is a daily, gap-free analysis on a 0.05° grid, built from satellite observations alone (twenty infrared and two microwave radiometers) and with a high degree of independence from in situ measurements (Embury et al., 2024). OISST, in contrast, is built from infrared data of the AVHRR instruments

adjusted to ship, buoy and Argo observations (Huang et al., 2021). Both products use AVHRR infrared measurements, so they are not completely independent, but they differ in their other inputs, in their bias adjustment, grid and analysis method, which makes CCI/C3S a real check on OISST rather than a copy of it. ERA5 SST was not used for this purpose, because ERA5 does not produce its own SST analysis but takes it from external products (Hersbach et al., 2020).

The daily field (analysed_sst) was converted from kelvin to degrees Celsius and averaged over the ocean cells of the study box with cosine-of-latitude weights, as for OISST (20,679 cells, the same number in every year) for each day from 1 January 1982 to 31 December 2024, the last day available. To compare like with like, the OISST series was cut to the same 43 years, and the two series were then processed in exactly the same way. Each was detrended with its own linear trend, its own Hobday climatology and threshold were calculated for the 1982–2021 baseline, and MHWs were detected with the same settings as in Section 3.3. Only detrended labels were used.

Three analyses were repeated for both products: the leakage analysis of Section 3.4 at leads of 1, 2, 3, 5, 7, 10 and 14 days; honest and exceedance persistence at the six main leads; and the logistic SST model (the 17 SST predictors and 2 calendar terms of Table 3.2) with the full design of Section 3.7, including purging, the 30-day embargo, inner cross-validation of the penalty, isotonic calibration and the MCC-optimal threshold. The random forest and the ERA5 predictors were not repeated, because they added nothing with OISST (Sections 4.4 and 4.6). Confidence intervals came from the same year-block bootstrap, here with 43 years.

This analysis was added after the main results were known, but its decision rule was fixed before it was run. The OISST conclusions count as reproduced if all three of the following hold with CCI/C3S:

1. the inflation of label-based persistence has a CI above zero at every lead from 1 to 7 days;
2. the logistic SST model beats honest persistence, with a CI above zero, at every lead;
3. the logistic SST model is not significantly better than exceedance persistence at 1 and 3 days (the lower CI bound of its gain is zero or below).

The agreement between the products was described by the correlation and root-mean-square (RMS) difference of their detrended daily anomalies, the MCC between their two daily label series, and the share of OISST MHW days that are also MHW days in CCI/C3S. These statistics were also calculated by calendar month and for four periods (1982–1991, 1992–2002, 2003–2015 and 2016–2024). The anomaly correlation was calculated with one series shifted by up to three days in either direction, to make sure that the dates of the two products line up.

3.14 Upper-ocean heat memory test

Section 5.2 interprets the weak 10–14-day signal as possible heat memory of the upper ocean. To test this, four predictors were derived from GODAS over the 147 grid cells of the study box that are ocean at every level down to 45 m: the 0–50 m mean potential temperature anomaly, the isothermal-layer depth anomaly, and the change of each over 30 days. Box means were weighted by the cosine of latitude. Anomalies were taken relative to a 1982–2021 pentad climatology smoothed over three pentads and were then linearly detrended. Because an assimilation analysis can contain observations from after its nominal date, each pentad was used only from 14 days after its date. The test was repeated with a 28-day delay, and a gain that appeared with the shorter delay but disappeared with the longer one would have been treated as leakage. The four predictors were added to the logistic SST model of Section 3.13 and evaluated with the same design at leads of 7, 10 and 14 days, against detrended OISST and CCI/C3S labels for 1982–2024, on exactly the same forecast days for the models with and without GODAS. One pentad (27 September 2024) could not be retrieved and was filled by linear interpolation. In 91 of the 3,211 pentads, the isothermal-layer depth was undefined in up to 3 of the 147 cells, and the box mean was taken over the remaining cells.

This analysis was added after the main results were known, but its decision rule was fixed before it was run: heat memory is supported if adding GODAS raises the new-event MCC, with a CI above zero, at both 10 and 14 days with both SST products.

3.15 Previous-year ENSO and IOD

To test whether ENSO and the IOD in the previous year affect MHW activity in the study box, one value per year was formed for each index and for the MHWs. ENSO was represented by the ONI for November of the previous year to January (NDJ) and,

as a check, by the mean Niño 3.4 anomaly of the same three months; the two series are almost identical ($r = 0.997$). The IOD was represented by the mean daily DMI (Section 3.1.3) of September–November of the previous year, when the IOD is strongest. The responses were the number of MHW days and the number of events, counted by onset date, in February–December of each year from 1983 to 2025 (43 years). January was left out because it is part of the NDJ window. March–June, the season studied by Gao et al. (2022), and each calendar month were also examined. Detrended labels were used for the main test and raw labels as a secondary check.

Relationships were measured with Spearman's rank correlation, which is less affected by a few extreme years, with Pearson's correlation reported for comparison, and each correlation was recalculated with one year left out at a time. Because positive IOD events often occur together with El Niño, the IOD was also tested with a partial correlation controlling for Niño 3.4. Years were grouped by the previous-winter ONI into El Niño (≥ 0.5 °C), La Niña (< -0.5 °C) and neutral years.

The decision rules were fixed before the analysis was run. The ENSO effect is supported if the Spearman correlation between the ONI and detrended February–December MHW days is positive with $p < 0.05$ and stays positive when any single year is left out. The IOD has an effect of its own if its partial correlation has $p < 0.05$.

3.16 Software and computing

All data processing, statistical analysis and machine-learning work in this study was done in Python 3 on Google Colab, a cloud-based notebook environment that runs Python in the web browser. The raw data, the processed series and all model outputs were stored on Google Drive and read directly from the notebooks. Gridded data were read and averaged with xarray, daily series were handled with pandas and NumPy, and the logistic regression, random forest, isotonic calibration and cross-validation steps used scikit-learn (Pedregosa et al., 2011). The XGBoost models used the xgboost library (Chen and Guestrin, 2016), the GODAS GRIB files were decoded with the ecCodes library of ECMWF, correlations and their tests used SciPy, and the figures were drawn with Matplotlib and Cartopy. The main code is listed in Appendix C.

CHAPTER 4

RESULTS

4.1 Marine heatwave record of the northern Bay of Bengal

The regional-mean SST of the study box increased by 0.21 °C per decade between 1982 and 2025 (Fig. 4.1). Annual means stayed between about 27.5 and 28.0 °C until the late 1990s, but every year since 2020 has been above 28.5 °C. By construction the detrended anomaly has no long-term trend, whereas the raw anomaly has been mostly positive since 2015.

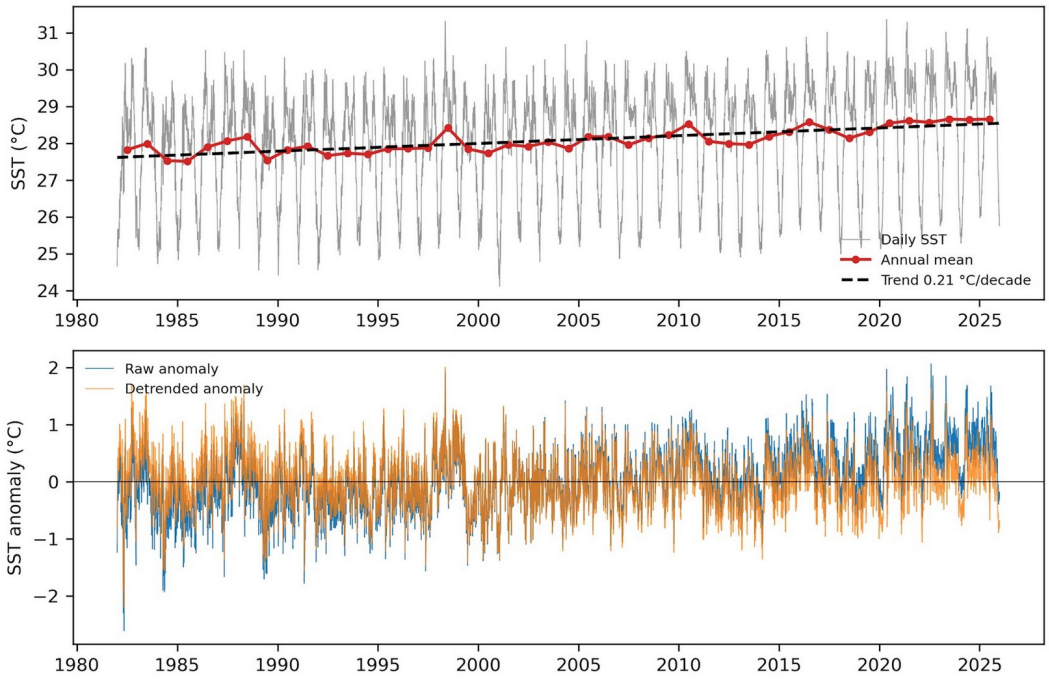


Figure 4.1 Regional-mean SST of the study box, 1982–2025: daily values, annual means and linear trend (top); raw and detrended daily anomalies (bottom).

Detection on detrended SST gave 109 events with 1192 MHW days, and detection on raw SST gave 133 events with 1652 MHW days (Table 4.1). Detrended events had a median duration of 8 days, and two thirds of them (67%) lasted 10 days or less. The two label types behave quite differently over time (Fig. 4.2). Detrended events are concentrated in a few years, most notably 1987–1988, 1998, 2010 and 2020, and show no long-term increase. Raw events, on the other hand, become steadily more common: since 2020 there have been more than 100 raw MHW days every year, with about 170 in 2024, because the warming trend pushes SST above a fixed threshold more and more often.

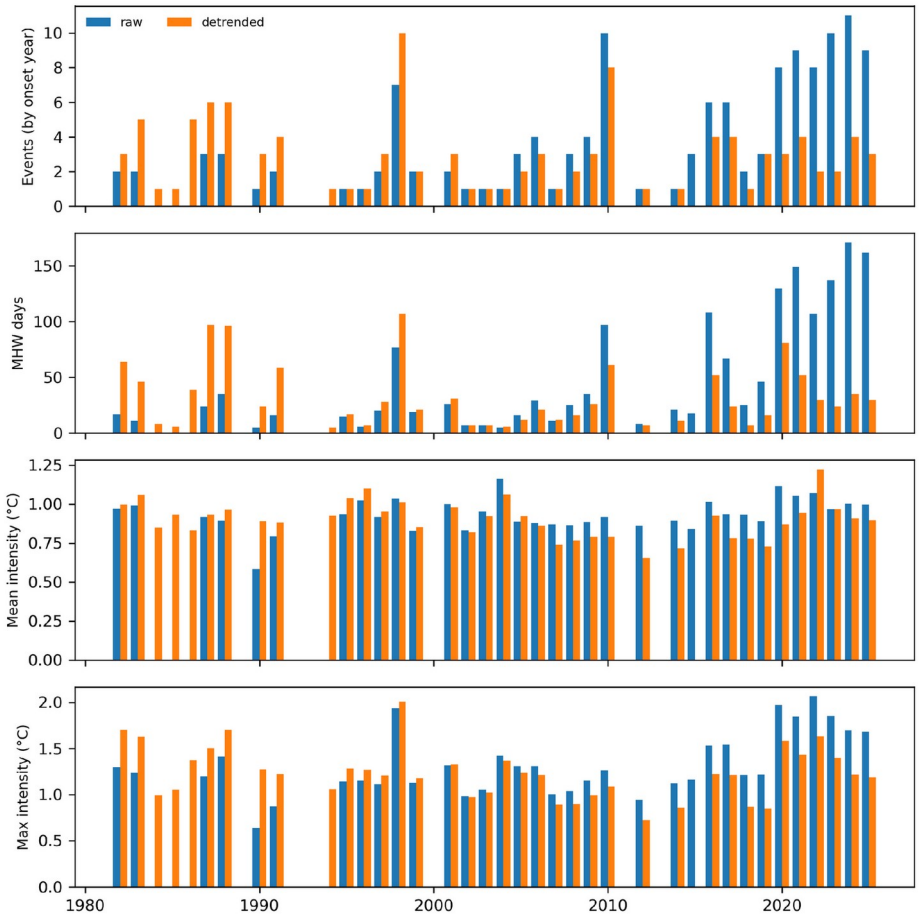


Figure 4.2 Annual MHW statistics for raw and detrended labels: number of events by onset year, MHW days, and mean and maximum intensity of events.

Detrended events most often begin in April–May and in September, and least often in January–March (Fig. 4.3). Raw events peak in June and November.

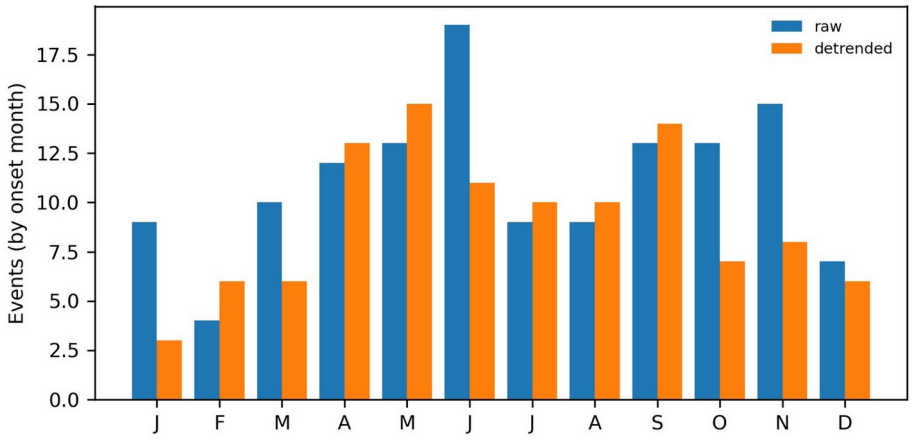


Figure 4.3 Number of MHW events by onset month for raw and detrended labels, 1982–2025.

One property of the labels is particularly important for forecasting: many MHW days cannot be recognised as such on the day itself. Of the 1192 detrended MHW days, 41.6% are labelled as MHW days only because later days completed the five-day minimum or closed a gap. For raw labels the figure is 35.2%. Table 4.1 gives the same statistics for the two comparison regions.

Table 4.1 MHW event statistics, 1982–2025, for the three regions and both label types.

Region	Labels	Events	MHW days	Median duration (days)	Events ≤ 10 days (%)	MHW days known only in hindsight (%)	SST trend (°C per decade)
Bay of Bengal	Detrended	109	1192	8	67.0	41.6	0.21
Bay of Bengal	Raw	133	1652	9	60.2	35.2	0.21
Levantine Sea	Detrended	96	1139	9	53.1	39.1	0.43
Levantine Sea	Raw	85	1818	11	44.7	22.8	0.43
South China Sea	Detrended	88	1231	10	56.8	30.9	0.22
South China Sea	Raw	97	1733	12	42.3	25.7	0.22

4.2 Atmospheric and oceanic precursors of MHW onset

Composites of standardised anomalies around the start of detrended events show a clear sequence of surface conditions (Fig. 4.4). Before onset the SST anomaly is already slightly positive, at about 0.5 standard deviations, and it rises sharply from roughly four days before the event begins. Wind speed is significantly below normal from 5 days before onset, the latent and sensible heat flux anomalies are positive (meaning less heat loss) from 8 and 9 days before, net shortwave radiation is above normal from 5 days before, and the net heat flux into the ocean is positive from 8 days before. After onset these anomalies reverse, as the warm water starts to lose heat. Among the climate indices, the IOBW index is significantly positive throughout the 30 days before onset, while the Niño 3.4 index is weakly positive until about 8 days before. On the yearly scale the link with ENSO is much clearer (Section 4.12). The DMI and the RMM indices show no significant anomaly before onset.

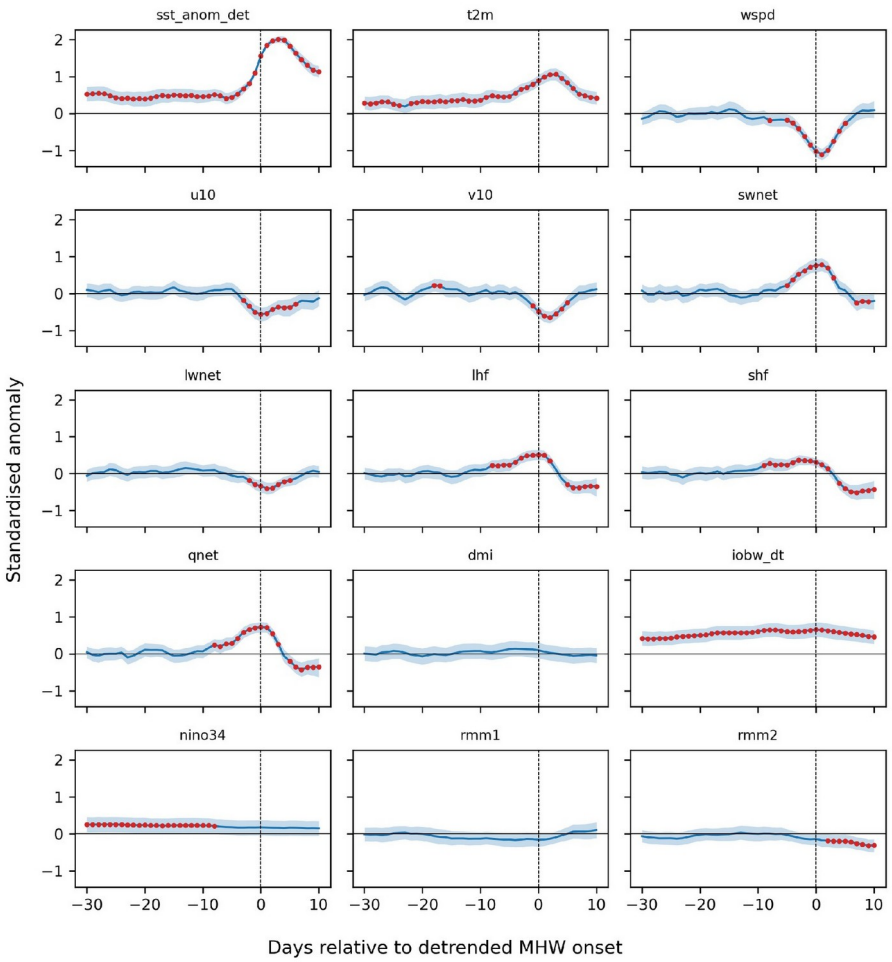


Figure 4.4 Composites of standardised anomalies from 30 days before to 10 days after the onset of detrended MHWs. Shading is the 95% bootstrap CI over events; red dots mark days on which the CI excludes zero.

These precursors are consistent with the heat-budget explanations of BoB MHWs given by Kumar et al. (2023) and Liang et al. (2024). They also suggested that ERA5 predictors might add forecast skill, which is what RQ3 tests.

4.3 Benchmark leakage (RQ1)

4.3.1 Regional mean and comparison regions

At every lead, label-based persistence scores much higher than honest persistence (Fig. 4.5). For detrended labels in the BoB, its MCC at a 1-day lead is 0.90, against 0.63 for honest persistence, and at 7 days the two values are 0.38 and 0.27. The inflation is +0.277 at 1 day, +0.266 at 3 days (61% of the honest score), +0.111 at 7 days and +0.036 at 14 days, and its CI is above zero at every lead (Table 4.2). Almost all of it, about 97%, comes from the five-day minimum, while joining gaps adds only about 0.01 MCC. The raw labels show the same behaviour (Fig. 4.5b).

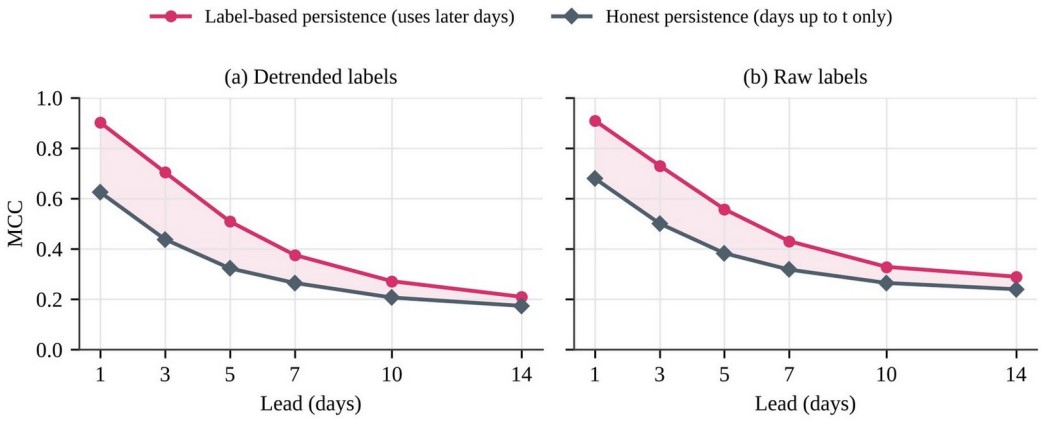


Figure 4.5 MCC of label-based and honest persistence by lead in the Bay of Bengal: (a) detrended labels, (b) raw labels. The shaded band is the inflation.

The two comparison regions give an equally clear result (Fig. 4.6, Table 4.2). For detrended labels the inflation at 1 day is +0.255 in the Levantine Sea and +0.194 in the South China Sea, and at 7 days it is +0.141 and +0.102. For raw labels the 1-day values are +0.140 and +0.160 (all three regions are listed in Table A.1). In all three regions, for both label types and at every lead, the CI lies above zero, so RQ1 is supported.

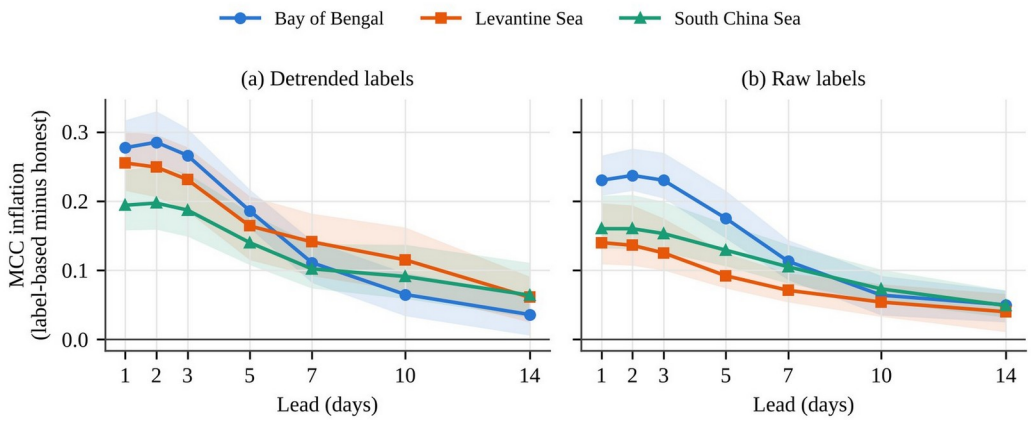


Figure 4.6 Inflation of label-based over honest persistence (MCC difference) by lead in three regions: (a) detrended labels, (b) raw labels. Shading is the 95% CI.

Table 4.2 Inflation of label-based over honest persistence (detrended labels), with 95% CIs, and the lead it effectively gains.

Lead (days)	Bay of Bengal	Levantine Sea	South China Sea	Lead lost, days (BoB / Lev. / SCS)
1	+0.277 (0.247 to 0.317)	+0.255 (0.215 to 0.301)	+0.194 (0.158 to 0.245)	more than the lead
2	+0.285 (0.254 to 0.330)	+0.249 (0.206 to 0.296)	+0.197 (0.159 to 0.251)	more than the lead

Lead (days)	Bay of Bengal	Levantine Sea	South China Sea	Lead lost, days (BoB / Lev. / SCS)
3	+0.266 (0.237 to 0.305)	+0.231 (0.185 to 0.278)	+0.187 (0.149 to 0.240)	more than the lead
5	+0.186 (0.161 to 0.217)	+0.164 (0.115 to 0.207)	+0.140 (0.108 to 0.183)	2.9 / 2.8 / 2.6
7	+0.111 (0.082 to 0.143)	+0.141 (0.092 to 0.182)	+0.102 (0.074 to 0.138)	2.9 / 3.0 / 2.8
10	+0.065 (0.034 to 0.096)	+0.115 (0.070 to 0.162)	+0.091 (0.059 to 0.137)	3.2 / 3.6 / 3.5
14	+0.036 (0.006 to 0.065)	+0.061 (0.025 to 0.091)	+0.064 (0.032 to 0.111)	4.1 / 4.6 / 4.3

When the inflation is converted into days, label-based persistence at 5–7 days performs about as well as honest persistence issued 3 days later, and at 14 days as well as one issued about 4–4.5 days later. These values are almost identical in the three regions. At 1–3 days, label-based persistence even beats honest persistence at a 1-day lead, so the lead lost is larger than the lead itself.

The size of the inflation at short leads follows the share of MHW days that can only be identified in hindsight (Table 4.1). Across the six combinations of region and label type, the 1-day inflation is ranked in exactly the same order as that share, from the detrended BoB (41.6%, +0.277) down to the raw Levantine Sea (22.8%, +0.140). At longer leads the Levantine Sea and the South China Sea show more inflation than the BoB, which fits with their longer events.

This has a direct consequence for evaluating models. At 7 days, label-based persistence (MCC 0.377) scores slightly higher than the best model in this study (0.369, Section 4.4), while honest persistence scores only 0.265. In other words, the error in the benchmark is as large as the skill the models add.

4.3.2 Grid-cell scale

For individual grid cells the leakage is larger than for the regional mean. When the same detection and persistence forecasts are applied to each of the 805 ocean cells, the median share of detrended MHW days that are only known in hindsight is 45.7%, compared with 41.6% for the regional mean. The median 1-day inflation across cells is +0.311 (10th–90th percentile 0.284 to 0.342), against +0.277 for the regional mean, and 96% of cells exceed the regional value (Table 4.3, Figs. 4.7 and 4.8). The gap

narrows as the lead increases: at 7 days the median cell value is +0.123 against +0.111, and at 10–14 days the cell and regional values are similar.

Table 4.3 Inflation of label-based over honest persistence in the 805 grid cells of the study box, compared with the regional mean.

Lead (days)	Detrended: regional mean	Detrended: cell median (10th–90th percentile)	Detrended: cells with CI > 0 (%)	Raw: regional mean	Raw: cell median (10th–90th percentile)	Raw: cells with CI > 0 (%)
1	+0.277	+0.311 (0.284 to 0.342)	100	+0.230	+0.259 (0.234 to 0.279)	100
2	+0.285	+0.316 (0.286 to 0.348)	100	+0.237	+0.260 (0.234 to 0.282)	100
3	+0.266	+0.294 (0.265 to 0.328)	100	+0.230	+0.244 (0.219 to 0.266)	100
5	+0.186	+0.197 (0.174 to 0.228)	100	+0.175	+0.171 (0.151 to 0.192)	100
7	+0.111	+0.123 (0.100 to 0.145)	100	+0.113	+0.113 (0.097 to 0.130)	100
10	+0.065	+0.067 (0.046 to 0.088)	99.3	+0.064	+0.069 (0.054 to 0.086)	100
14	+0.036	+0.034 (0.015 to 0.054)	67.7	+0.050	+0.041 (0.025 to 0.058)	87.6

The inflation is significant almost everywhere. For detrended labels, the CI is above zero in every cell at every lead from 1 to 7 days, in 99.3% of cells at 10 days and in 67.7% of cells at 14 days, so the rule set before this analysis (a CI above zero in at least 90% of cells at every lead from 1 to 7 days) is met. Raw labels give the same picture, with a CI above zero in every cell from 1 to 10 days. As for the regional mean, 95–98% of the inflation is caused by the five-day minimum.

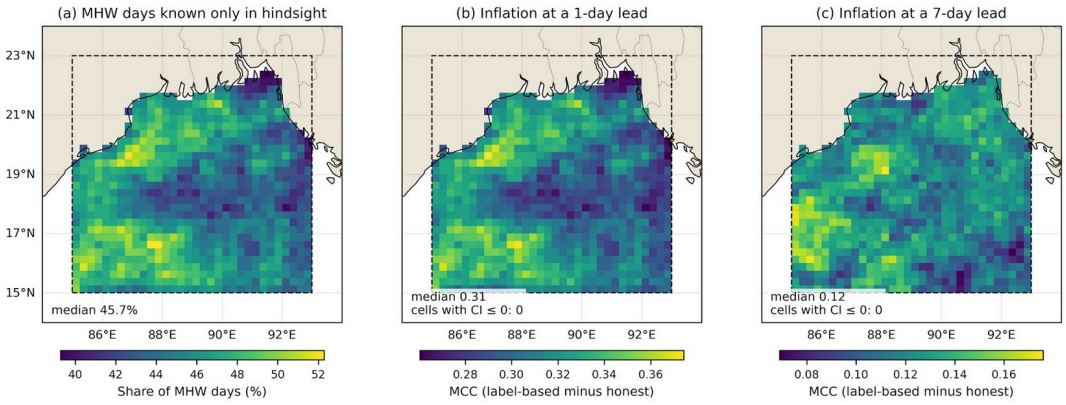


Figure 4.7 Grid-cell leakage for detrended labels in the study box (dashed): (a) share of MHW days known only in hindsight, (b) inflation at a 1-day lead, (c) inflation at a 7-day lead.

The median of each panel is shown at the lower left; no cell has a 95% CI that includes zero at either lead.

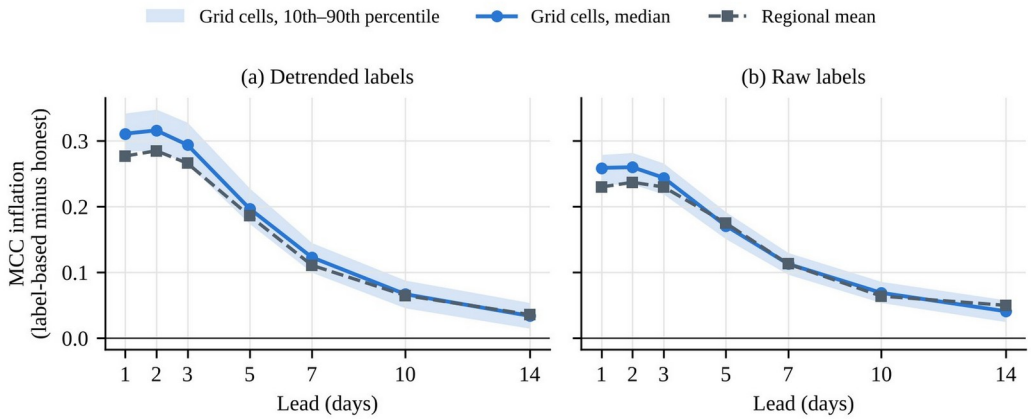


Figure 4.8 Inflation by lead in the 805 grid cells (median and 10th–90th percentile) compared with the regional mean: (a) detrended labels, (b) raw labels.

Across cells, the 1-day inflation follows the hindsight share almost perfectly (Spearman r = 1.00 for both label types, although this value is only descriptive because neighbouring cells are correlated). As a result, the two maps in Fig. 4.7a and b look almost the same. Both are highest in patches on the western side of the box, around 19–20°N, 86–88°E and 15–17°N, 85–88°E, and lowest in the north-east, near 21–22°N, 91–92°E. At 7 days the largest inflation is found in the west and south-west of the box (Fig. 4.7c). Exceedance persistence beats honest persistence by a median of 0.11–0.15 MCC per cell at 1–3 days, so the gap between the two honest baselines is also present at grid scale.

4.4 Forecast skill by lead (RQ2)

The skill of honest persistence and the four models at each lead is given in Table 4.4 and Fig. 4.9. Every model beats honest persistence at every lead. For the logistic SST model the gain is +0.129 to +0.194 MCC (95% CI) at 1 day, +0.047 to +0.145 at 7 days and +0.022 to +0.145 at 14 days. The only case without a significant gain is the random forest with SST predictors at 14 days (−0.006 to +0.121). RQ2 is therefore supported at every lead, although at 10–14 days the gains are small and their confidence intervals wide.

Table 4.4 MCC by lead for honest persistence and the four models (detrended labels).

Lead (days)	Honest persistence	Logistic, SST	Logistic, SST + ERA5	Random forest, SST	Random forest, SST + ERA5
1	0.625	0.783	0.779	0.782	0.777
3	0.438	0.616	0.620	0.602	0.608
5	0.323	0.458	0.468	0.445	0.464
7	0.265	0.361	0.365	0.346	0.369
10	0.207	0.287	0.264	0.267	0.288
14	0.174	0.257	0.241	0.230	0.242

Some further results are worth noting. The random forest never does better than logistic regression, and the four models stay within 0.02–0.03 MCC of each other at every lead. The models' probabilities also beat climatology at every lead (the BSS of the logistic SST model is 0.68 at 1 day, 0.18 at 7 days and 0.09 at 14 days; Table A.2), whereas honest persistence, which only gives yes-or-no forecasts, falls below climatology from 5 days onwards (BSS −0.10 at 5 days and −0.33 at 14 days). Skill of all kinds declines steadily with lead, and at 14 days the best MCC is only 0.26.

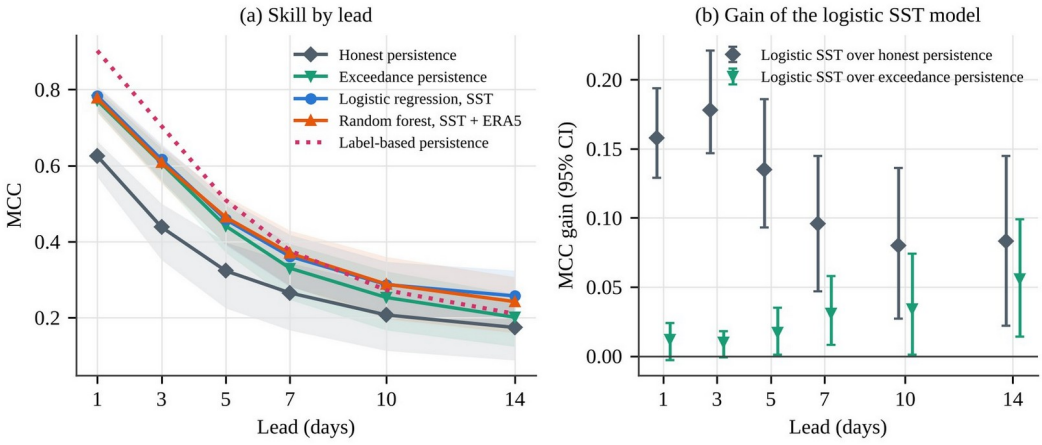


Figure 4.9 Forecast skill by lead for the detrended labels. (a) MCC by lead for the three persistence benchmarks and two models, with 95% CIs (shading). (b) MCC gain of the logistic SST model over honest and over exceedance persistence, with 95% CIs.

The choice of benchmark changes the conclusion (Fig. 4.9a). Measured against label-based persistence, the same models look worse than persistence at 1–5 days, about equal at 7 days and better only at 10–14 days. Measured against honest persistence, they win clearly at every lead.

4.5 A stricter baseline: exceedance persistence (post-hoc)

Exceedance persistence, which predicts an MHW whenever SST is above the threshold on the forecast day, was added after the RQ2 results had been seen and is reported as a post-hoc robustness test (Section 3.4). It turns out to be much stronger than honest persistence, with an MCC of 0.771 at 1 day compared with 0.625 (Table 4.5). At 1 and 3 days it does as well as the logistic SST model: the model's gain over it is −0.003 to +0.024 at 1 day and −0.001 to +0.018 at 3 days, neither of which is significant. The gain becomes significant at 5 days (+0.001 to +0.035) and rises to +0.014 to +0.099 at 14 days (Fig. 4.9b). The random forest with SST + ERA5 behaves in the same way, except at 10 days, where its gain is not significant (−0.003 to +0.068).

Table 4.5 Exceedance persistence and the gain of the logistic SST model over it, for detrended and raw labels (MCC, with 95% CIs for the gain).

Lead (days)	Exceedance persistence, detrended	Logistic gain, detrended	Exceedance persistence, raw	Logistic gain, raw
1	0.771	−0.003 to +0.024	0.808	−0.003 to +0.019
3	0.606	−0.001 to +0.018	0.650	−0.005 to +0.023
5	0.441	+0.001 to +0.035	0.503	−0.012 to +0.042
7	0.330	+0.008 to +0.058	0.396	+0.006 to +0.088
10	0.253	+0.001 to +0.074	0.308	+0.064 to +0.115
14	0.201	+0.014 to +0.099	0.273	+0.054 to +0.157

The same picture appears on new-event days, when no MHW was known to be under way on the forecast day. There the gain of the logistic SST model over exceedance persistence is not significant at 1–3 days, but it is significant from 5 to 14 days (+0.041 to +0.135 at 14 days).

With raw labels (Table 4.5) the short-lead result does not change, and there is no significant gain over exceedance persistence at 1, 3 or 5 days. The gain becomes significant from 7 days, and at 10–14 days it is about twice as large as for detrended labels (at 14 days the MCC is 0.383 against 0.273). For raw labels too, the logistic SST model beats honest persistence at every lead (gain at 1 day +0.117 to +0.152; Table A.3).

Put simply, the gain over honest persistence at 1–3 days comes entirely from the fact that SST is already above the threshold on the forecast day. Skill beyond that is

borderline at 5 days, clearest at 7 days, and small, at 0.02–0.06 MCC for detrended labels. At 10–14 days it is weak, and Section 4.9 shows that it depends on the SST product: the short-lead result holds with a second product, but the gain at 10–14 days does not.

4.6 ERA5 and new-event skill (RQ3)

Adding the 56 ERA5 predictors to the SST predictors did not improve the forecasts of new events at any lead (Table 4.6). For both logistic regression and random forest, the CI of the gain in new-event MCC includes zero at every lead, so the rule for RQ3 fixed in advance, which required a significant gain at both 5 and 7 days, is not met. The MCC over all days shows the same thing. At 10 and 14 days, logistic regression with ERA5 even scores a little lower than with SST alone (0.264 against 0.287 at 10 days), although the difference is not significant. The strongest penalty (C = 0.01) was chosen in nearly every fold, which suggests that the extra predictors mostly added noise.

Table 4.6 Gain from adding ERA5 to SST predictors (95% CIs) in MCC over all days and in new-event MCC.

Lead (days)	Logistic, MCC	Logistic, new-event MCC	Random forest, MCC	Random forest, new-event MCC
1	−0.024 to +0.014	−0.025 to +0.029	−0.019 to +0.007	−0.024 to +0.016
3	−0.012 to +0.022	−0.017 to +0.035	−0.013 to +0.034	−0.008 to +0.028
5	−0.016 to +0.037	−0.025 to +0.041	−0.003 to +0.047	−0.013 to +0.030
7	−0.027 to +0.034	−0.034 to +0.042	−0.001 to +0.046	−0.008 to +0.045
10	−0.074 to +0.025	−0.073 to +0.033	−0.008 to +0.052	−0.018 to +0.053
14	−0.061 to +0.038	−0.073 to +0.041	−0.018 to +0.050	−0.013 to +0.048

A wider comparison was carried out at the 7-day lead, including XGBoost and the full set of 83 predictors (Table 4.7). None of the model types gained significantly in MCC from either ERA5 or the climate indices, and XGBoost scored below logistic regression with every set of predictors (0.282–0.312 against 0.361–0.367). XGBoost with ERA5 did catch more onset days than XGBoost with SST alone (an onset hit rate of 0.42 against 0.25), but this comparison was exploratory, it came with lower overall skill, and it was not confirmed by the test fixed in advance above. RQ3 is therefore not supported. Section 4.10 tests the same question predictor by predictor.

Table 4.7 Seven-day comparison of all models and predictor sets (detrended labels): MCC with 95% CI and onset hit rate.

Model	Predictors	MCC (95% CI)	Onset hit rate
Honest persistence	—	0.265 (0.167 to 0.344)	0.09
Logistic regression	SST	0.361 (0.281 to 0.418)	0.17
Logistic regression	SST + ERA5	0.365 (0.289 to 0.427)	0.19
Logistic regression	All	0.367 (0.274 to 0.435)	0.26
Random forest	SST	0.346 (0.267 to 0.401)	0.15
Random forest	SST + ERA5	0.369 (0.283 to 0.429)	0.19
Random forest	All	0.360 (0.278 to 0.423)	0.19
XGBoost	SST	0.312 (0.241 to 0.364)	0.25
XGBoost	SST + ERA5	0.295 (0.227 to 0.349)	0.42
XGBoost	All	0.282 (0.217 to 0.334)	0.49

4.7 What the SST model uses

The standardised coefficients of the logistic SST model change in a systematic way with lead (Fig. 4.10). At 1 day the distance of SST from the threshold has the largest weight (1.48), followed by the 3-day mean anomaly (1.11) and the 1-day change (0.92). The weight on the 1-day change then falls steadily, reaching 0.21 at 14 days. The 30-day mean anomaly is the only predictor whose weight grows steadily with lead, from 0.41 at 1 day to 0.91 at 14 days, and it is significant at every lead. The 7-day maximum anomaly also gains weight at 10–14 days (0.79 and 0.72), and the 7-day standard deviation turns negative there (−0.23), which means that a steady warm anomaly is a better sign of a coming MHW than a fluctuating one.

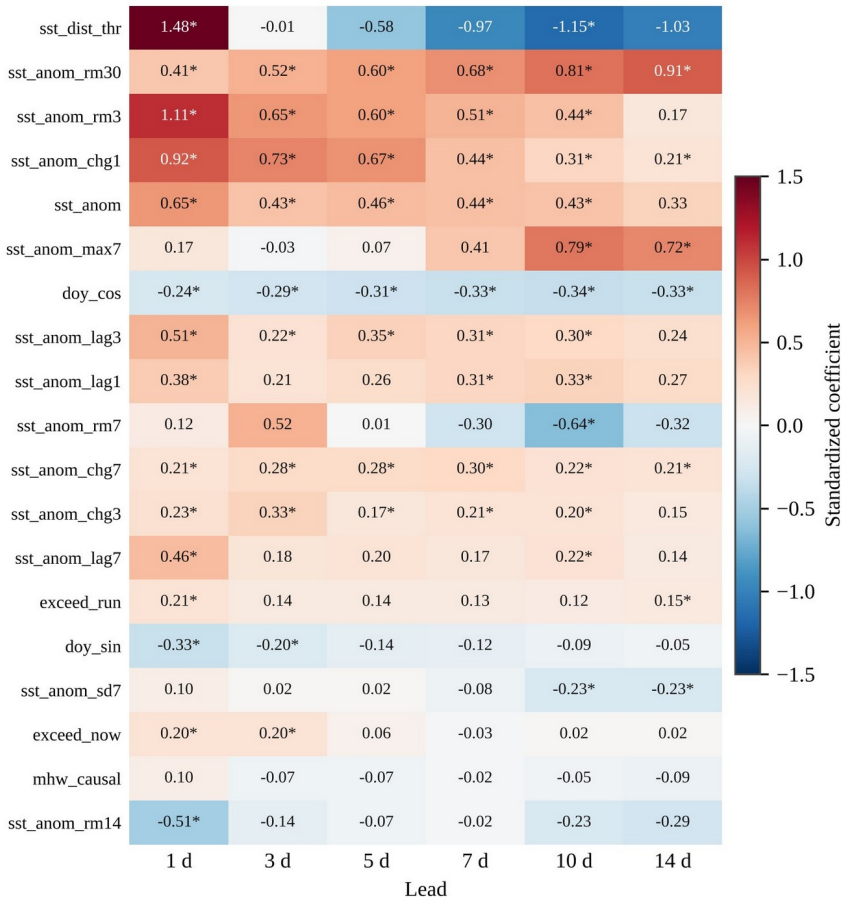


Figure 4.10 Standardised coefficients of the logistic SST model by lead. An asterisk marks coefficients whose 95% bootstrap CI excludes zero.

Figure 4.10 contains 114 intervals (19 predictors at 6 leads), so about six could exclude zero by chance alone. Only predictors that are significant at most leads, such as the 30-day mean anomaly, are interpreted.

Several predictors are close to being linear combinations of others, so individual coefficients have to be read carefully. The distance to the threshold differs from the anomaly only by a seasonal offset, and its sign changes from +1.48 at 1 day to about −1.0 at 7–14 days because, at long leads, the model replaces the distance to today's threshold with the anomaly itself and the season. Similarly, the negative weights on the 7- and 14-day means act as contrasts against the 30-day mean and the lags, rather than as genuinely negative effects.

The drop-one-group test avoids this problem by asking how much skill is lost when a whole group of predictors is removed (Table 4.8). Removing the current state costs a small but significant amount of skill at 1 day (+0.005 to +0.032) and at 3 days (+0.001 to +0.021). Removing the background group costs significant skill at 14 days, where

the MCC falls from 0.257 to 0.206 (loss +0.004 to +0.097) and the new-event MCC also falls (+0.014 to +0.124). At 10 days the background loss points the same way (0.287 to 0.263) but is not significant. Removing the season makes no difference at any lead.

Table 4.8 Drop-one-group test for the logistic SST model: MCC without the group and loss of MCC (95% CI). Bold losses are significant.

Lead (days)	Full model	Without current state	Without tendency	Without background	Without season
1	0.783	0.764 **(+0.005 to +0.032)**	0.784 (−0.009 to +0.004)	0.788 (−0.016 to +0.006)	0.783 (−0.010 to +0.009)
3	0.616	0.605 **(+0.001 to +0.021)**	0.614 (−0.007 to +0.008)	0.607 **(+0.001 to +0.017)**	0.616 (−0.009 to +0.008)
5	0.458	0.463 (−0.015 to +0.007)	0.459 (−0.007 to +0.006)	0.457 (−0.009 to +0.015)	0.462 (−0.020 to +0.015)
7	0.361	0.365 (−0.018 to +0.009)	0.369 (−0.019 to +0.002)	0.358 (−0.017 to +0.022)	0.367 (−0.031 to +0.016)
10	0.287	0.288 (−0.026 to +0.025)	0.279 (−0.001 to +0.019)	0.263 (−0.016 to +0.066)	0.292 (−0.030 to +0.022)
14	0.257	0.261 (−0.015 to +0.007)	0.258 (−0.012 to +0.009)	0.206 **(+0.004 to +0.097)**	0.259 (−0.025 to +0.023)

Removing the tendency group never reduces skill, but this result says nothing, because the tendency predictors are exact differences of the anomaly and its lags (Section 3.5). Those lags stay in the other groups, so a linear model can rebuild the tendency perfectly, and the test cannot show whether the recent trajectory of SST matters.

Table 4.8 contains 48 intervals (24 for MCC and 24 for new-event MCC), so one of them could be expected to exclude zero by chance alone. The small losses at 3 days are close to zero and may be such cases. The 14-day result for the background group is the clearest within OISST, since it is significant both overall and for new events and agrees with the rising coefficient of the 30-day mean. It should still be read as weak evidence: the intervals are wide, and, as Section 4.9 shows, the 14-day gain over persistence that this group supports is not reproduced with a second SST product.

4.8 Case studies

The three detrended events with the largest cumulative intensity were 19 September–7 November 1982 (50 days, 51.3 °C·days), 16 November 1987–30 January 1988 (76 days, 75.0 °C·days) and 25 June–13 August 2020 (50 days, 38.9 °C·days). Figure

4.11 and Table 4.9 compare the logistic SST model with exceedance and honest persistence for these events at 7 and 14 days.

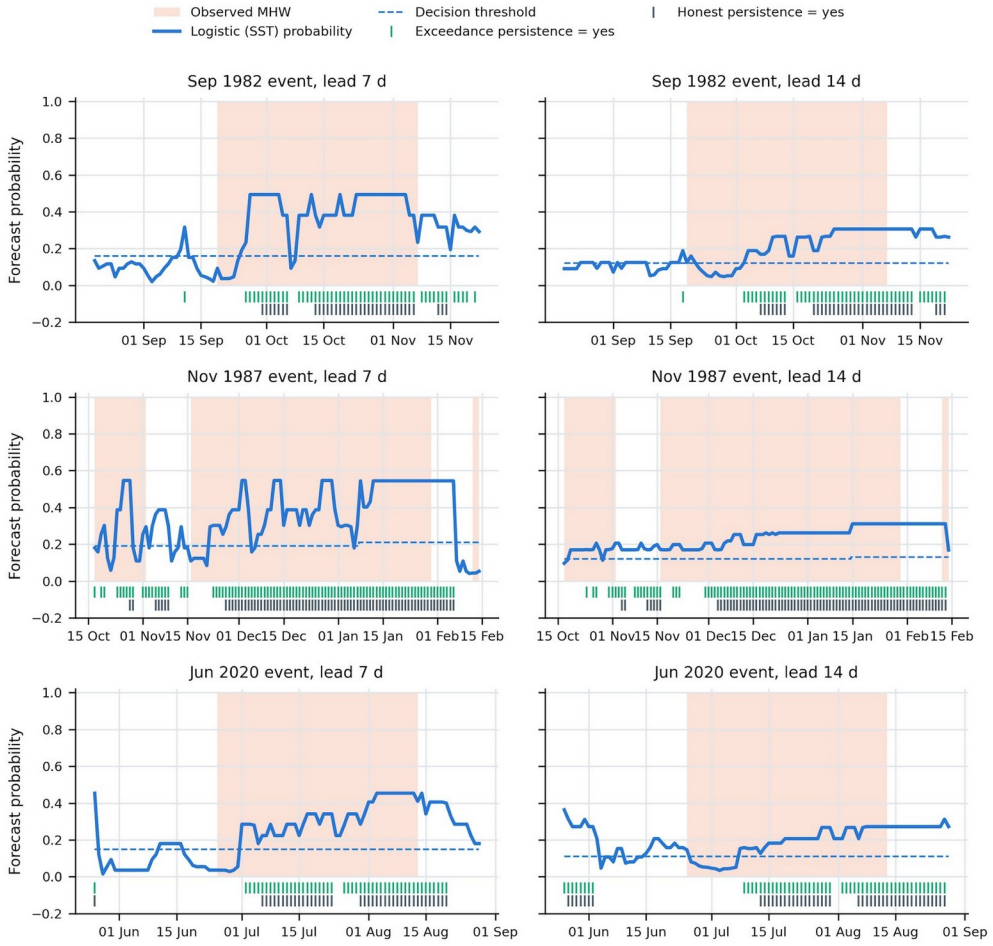


Figure 4.11 Forecasts for the three strongest detrended MHWs at 7-day (left) and 14-day (right) leads. Shading marks observed MHW days; the line is the calibrated probability of the logistic SST model with its decision threshold (dashed); ticks mark the days on which exceedance and honest persistence forecast an MHW. Forecasts are plotted at their target date.

Table 4.9 Case-study scores for the three strongest events: first event day caught (days after onset), share of event days forecast, and false-alarm days in the 30 days before onset.

Event	Lead (days)	Forecast	First day caught	Event days forecast (%)	False alarms before onset
Sep–Nov 1982	7	Logistic SST	6	84	2
Sep–Nov 1982	7	Exceedance persistence	7	80	1
Sep–Nov 1982	7	Honest persistence	11	64	0
Nov 1987–Jan 1988	7	Logistic SST	6	88	7
Nov 1987–Jan 1988	7	Exceedance persistence	7	91	10
Nov 1987–Jan 1988	7	Honest persistence	11	86	5

Event	Lead (days)	Forecast	First day caught	Event days forecast (%)	False alarms before onset
Jun–Aug 2020	7	Logistic SST	6	88	7
Jun–Aug 2020	7	Exceedance persistence	7	82	1
Jun–Aug 2020	7	Honest persistence	11	66	1
Sep–Nov 1982	14	Logistic SST	0	78	18
Sep–Nov 1982	14	Exceedance persistence	14	68	1
Sep–Nov 1982	14	Honest persistence	18	52	0
Nov 1987–Jan 1988	14	Logistic SST	0	100	13
Nov 1987–Jan 1988	14	Exceedance persistence	0	87	11
Nov 1987–Jan 1988	14	Honest persistence	0	78	6
Jun–Aug 2020	14	Logistic SST	0	76	21
Jun–Aug 2020	14	Exceedance persistence	14	68	8
Jun–Aug 2020	14	Honest persistence	18	52	7

At 7 days, the first event day that was caught is the same in all three events: day 6 for the model, day 7 for exceedance persistence and day 11 for honest persistence. This follows directly from how the forecasts work. Exceedance persistence can first flag the day that falls 7 days after the first exceeding day, and honest persistence must also wait for five exceeding days. The model is one day ahead of exceedance persistence, because it reacts to SST approaching the threshold. As a result, days 0–5 of the three strongest events in 44 years were missed by every forecast. Once an event was under way, though, all forecasts tracked it well (64–91% of event days).

At 14 days, the model does flag the first event day in all three events, but it also produces 13–21 false-alarm days in the 30 days before onset, compared with 1–11 for exceedance persistence. Its probability stays just above the threshold for weeks (Fig. 4.11), and in 2020 it even drops below the threshold during the first two weeks of the event. The 14-day forecasts are therefore broad warnings of a warm background rather than forecasts of when an event will start. All three forecasts caught the November 1987 onset at 14 days, but only because another MHW had just preceded it. Even inside these strong events, the calibrated 7-day probabilities rarely go above about 0.5.

4.9 Validation with a second SST dataset

4.9.1 Agreement between OISST and CCI/C3S

The two products give similar mean conditions but clearly different MHW records (Table 4.10). Over 1982–2024, CCI/C3S is on average 0.14 °C warmer than OISST, warms more slowly (0.126 against 0.207 °C per decade) and has fewer but longer events (91 against 105, with a median duration of 10 against 8 days). Its daily SST correlates with OISST at r = 0.970, but its detrended daily anomalies, which are what decide whether a day is an MHW day, correlate at only r = 0.751, with an RMS difference of 0.33 °C. The correlation is highest at zero lag (0.751, against 0.743 and 0.728 with a shift of one day), so the difference is not caused by misaligned dates. It also improves very little when the anomalies are averaged over 7 or 30 days (r = 0.775 and 0.783), which means that the products disagree about slow anomalies and not only about day-to-day noise. As a result, the two label series agree only moderately (MCC 0.457): just 52.5% of the OISST MHW days are also MHW days in CCI/C3S. This low agreement is largely what the anomaly correlation implies. For two Gaussian series correlated at r = 0.751, only about 51% of the days above the 90th percentile of one series are also above it in the other (ignoring the five-day minimum), and even at r = 0.95 the share would be only about 78%. Labels defined by a threshold therefore amplify small differences between the products.

Table 4.10 Detrended SST and MHW statistics of the two products over the common period 1982–2024.

Statistic	OISST	CCI/C3S
Mean SST (°C)	28.07	28.20
Linear trend (°C per decade)	0.207	0.126
Number of MHW events	105	91
Number of MHW days	1161	1284
Median event duration (days)	8	10
MHW days known only in hindsight (%)	40.9	32.2
Standard deviation of daily anomaly (°C)	0.476	0.447
Standard deviation of day-to-day change (°C)	0.143	0.109
Lag-1 autocorrelation of daily anomaly	0.955	0.970

The disagreement is not spread evenly through the year or the record (Fig. 4.12a and Table 4.11). From November to May the anomaly correlation is 0.77–0.88, but from

June to October it falls to 0.61–0.68. June is the extreme case: only 17 of the 102 OISST MHW days in June (17%) are also MHW days in CCI/C3S. Agreement also improves steadily over time, with the correlation rising from 0.69 in 1982–1991 to 0.86 in 2016–2024. In the first two periods CCI/C3S is 0.21 and 0.30 °C warmer than OISST, but after 2003 the mean difference is close to zero, which explains why its linear trend is smaller. Because each series is detrended separately, this difference also moves MHW days between decades: CCI/C3S has 422 MHW days in 1992–2002 against 223 for OISST, but 116 against 179 in 2003–2015. Finally, CCI/C3S is smoother from day to day than OISST (Table 4.10), which fits its longer events and smaller share of days known only in hindsight.

Table 4.11 Agreement between the two products by period: mean difference of raw SST (CCI/C3S minus OISST), correlation and RMS difference of detrended daily anomalies, and MHW days in each product and in both.

Period	Mean difference (°C)	Correlation	RMS difference (°C)	MHW days, OISST	MHW days, CCI/C3S	MHW days in both
1982–1991	+0.21	0.69	0.40	431	462	237
1992–2002	+0.30	0.73	0.36	223	422	148
2003–2015	+0.05	0.77	0.29	179	116	38
2016–2024	−0.01	0.86	0.24	328	284	187

4.9.2 Benchmark leakage and forecast skill with both products

Label-based persistence inflates skill with both products, and the inflation has a CI above zero at every lead from 1 to 14 days (Table 4.12, Fig. 4.12b). At 1–3 days the inflation is smaller with CCI/C3S (0.203 against 0.271 at 1 day), and its share of MHW days known only in hindsight is also smaller (32.2% against 40.9%). This is the same relationship that was found between the three regions and across the 805 grid cells (Section 4.3). From 5 days onwards the two curves are almost identical.

Table 4.12 Inflation of label-based over honest persistence with the two products, 1982–2024 (MCC, with 95% CIs).

Lead (days)	OISST	CCI/C3S
1	0.271 (0.241 to 0.309)	0.203 (0.176 to 0.241)
2	0.277 (0.245 to 0.321)	0.212 (0.183 to 0.255)
3	0.258 (0.230 to 0.298)	0.211 (0.180 to 0.256)
5	0.182 (0.158 to 0.213)	0.172 (0.143 to 0.215)

Lead (days)	OISST	CCI/C3S
7	0.110 (0.082 to 0.139)	0.118 (0.091 to 0.152)
10	0.063 (0.034 to 0.090)	0.064 (0.038 to 0.088)
14	0.039 (0.011 to 0.069)	0.045 (0.017 to 0.072)

The skill of the logistic SST model and the two honest benchmarks is compared in Table 4.13 and Fig. 4.12c–d. With both products the model beats honest persistence at every lead from 1 to 7 days. At 10 and 14 days, however, the two products part ways. With OISST the gain remains significant (+0.007 to +0.115 at 10 days and +0.035 to +0.150 at 14 days), but with CCI/C3S it disappears (−0.030 to +0.064 and −0.105 to +0.059). The model itself is not the reason, since its 14-day MCC is almost the same with both products (0.267 and 0.273). What changes is persistence: with CCI/C3S, honest persistence reaches 0.285 at 14 days against 0.174 with OISST, and exceedance persistence 0.301 against 0.203.

Table 4.13 MCC of honest persistence, exceedance persistence and the logistic SST model with the two products, 1982–2024, and the gain of the model over each benchmark (95% CIs).

Product	Lead (days)	Honest	Exceedance	Logistic SST	Gain over honest	Gain over exceedance
OISST	1	0.631	0.769	0.784	+0.124 to +0.186	+0.002 to +0.026
OISST	3	0.449	0.606	0.616	+0.133 to +0.209	+0.000 to +0.019
OISST	5	0.334	0.443	0.455	+0.082 to +0.171	−0.005 to +0.032
OISST	7	0.275	0.334	0.366	+0.046 to +0.140	+0.009 to +0.056
OISST	10	0.215	0.254	0.276	+0.007 to +0.115	−0.013 to +0.060
OISST	14	0.174	0.203	0.267	+0.035 to +0.150	+0.021 to +0.106
CCI/C3S	1	0.720	0.826	0.838	+0.092 to +0.145	−0.000 to +0.022
CCI/C3S	3	0.557	0.688	0.693	+0.102 to +0.176	−0.016 to +0.022
CCI/C3S	5	0.448	0.559	0.554	+0.072 to +0.148	−0.027 to +0.013
CCI/C3S	7	0.381	0.452	0.465	+0.038 to +0.128	−0.023 to +0.039
CCI/C3S	10	0.329	0.357	0.351	−0.030 to +0.064	−0.060 to +0.036

Product	Lead (days)	Honest	Exceedance	Logistic SST	Gain over honest	Gain over exceedance
CCI/C3S	14	0.285	0.301	0.273	−0.105 to +0.059	−0.133 to +0.049

Against exceedance persistence, the model shows no significant gain at any lead with CCI/C3S. With OISST the gains at 1 and 3 days now have lower bounds of +0.002 and +0.000, whereas over 1982–2025 they were −0.003 and −0.001 (Table 4.5). The estimates themselves hardly change (about +0.01 MCC), so whether they are called significant depends on a single year of data. What does hold in both products and both periods is the size of the gain: at 1–3 days the model adds at most about 0.025 MCC to exceedance persistence.

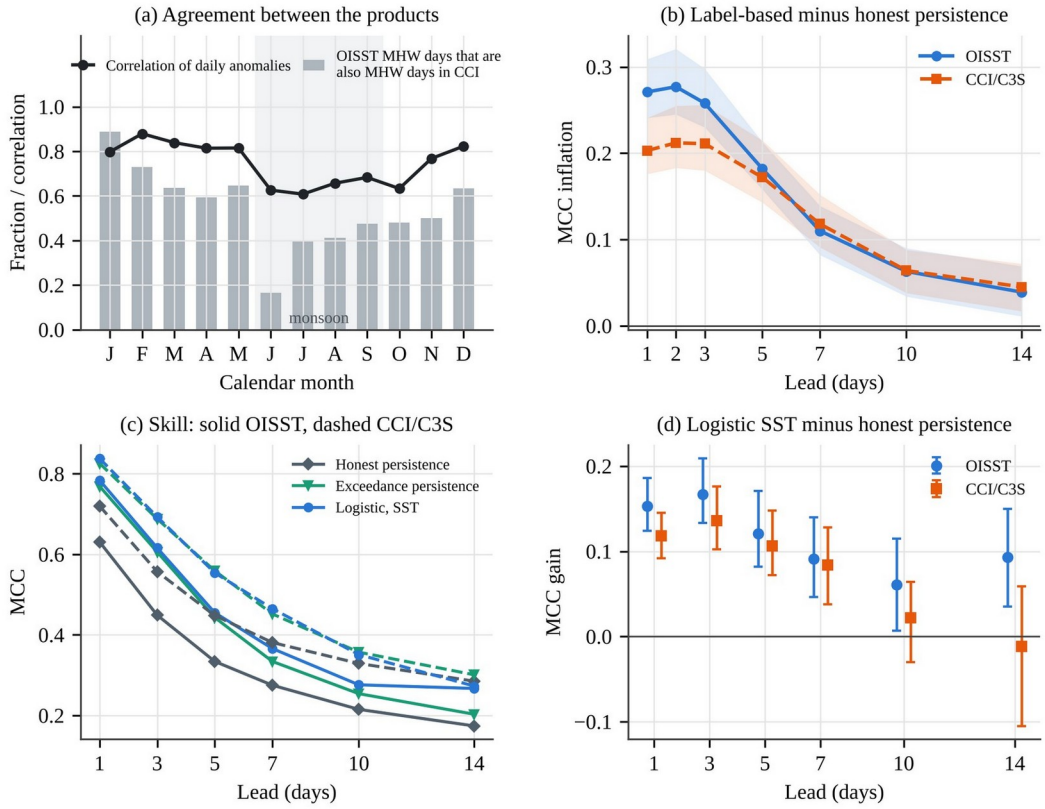


Figure 4.12 Comparison of OISST and CCI/C3S, 1982–2024. (a) Correlation of detrended daily anomalies and share of OISST MHW days that are also MHW days in CCI/C3S, by calendar month (June–September shaded). (b) Inflation of label-based over honest persistence with 95% CIs. (c) MCC of honest persistence, exceedance persistence and the logistic SST model (solid: OISST; dashed: CCI/C3S). (d) MCC gain of the logistic SST model over honest persistence with 95% CIs.

Applying the decision rule of Section 3.13, conditions 1 and 3 are met with CCI/C3S, but condition 2 is not, because the gain over honest persistence is not significant at 10

and 14 days. The OISST conclusions are therefore only partly reproduced. The benchmark inflation and the short-lead results hold with a second SST product, but the skill beyond persistence at 10–14 days does not.

4.10 Individual predictor selection (post-hoc)

Selecting individual predictors with the L1 penalty did not improve the forecasts (Table 4.14). With all 83 predictors, the MCC of the L1 model is within about 0.02 of the logistic SST model at every lead. Its gain has a CI that includes zero at every lead except 5 days, where it is marginally positive both over all days (+0.003 to +0.038) and on new-event days (+0.002 to +0.048). At 7 days the new-event MCC is unchanged (0.286 against 0.285). The rule fixed in advance, which required a significant gain at both 5 and 7 days, is therefore not met, in agreement with the result for the ERA5 group in Section 4.6.

Table 4.14 L1 logistic model with all 83 predictors compared with the logistic SST model (detrended labels, OISST): MCC over all days and on new-event days, with 95% CIs of the paired gain, and the size of the half-sample fits. Bold gains are significant.

Lead (days)	MCC, L1	MCC, SST	Gain (95% CI)	New-event MCC, L1	New-event MCC, SST	New-event gain (95% CI)	Predictors per half-sample fit	Bound on false selections
1	0.786	0.783	−0.010 to +0.018	0.666	0.659	−0.016 to +0.030	33.5	22.5
3	0.616	0.616	−0.019 to +0.020	0.532	0.525	−0.017 to +0.033	26.3	13.9
5	0.479	0.458	**+0.003 to +0.038**	0.407	0.383	**+0.002 to +0.048**	15.4	4.8
7	0.381	0.361	−0.024 to +0.056	0.286	0.285	−0.041 to +0.043	17.0	5.8
10	0.297	0.287	−0.045 to +0.057	0.235	0.238	−0.052 to +0.052	16.1	5.2
14	0.242	0.257	−0.069 to +0.032	0.189	0.214	−0.080 to +0.028	5.0	0.5

The half-sample frequencies (Fig. 4.13) separate the predictors more clearly than the LOYO folds. Six ERA5 and index predictors were selected in at least 80% of the LOYO folds at both 5 and 7 days, but three of them fell well below 80% in the half-samples: the air-temperature anomaly (0.59 and 0.61), the 7-day mean net heat flux (0.46 and 0.32) and the 7-day mean sensible heat flux (0.28 and 0.49). Three remained stable: the IOBW index, the net heat flux anomaly and the net shortwave

anomaly, so the rule for a stable non-SST signal is met. Among the SST predictors, the 30-day mean anomaly was stable at every lead (0.86–1.00), the anomaly itself from 3 to 14 days, the 1-day change at 1–7 days and the exceedance flag at 1–5 days, in line with the coefficients in Section 4.7.

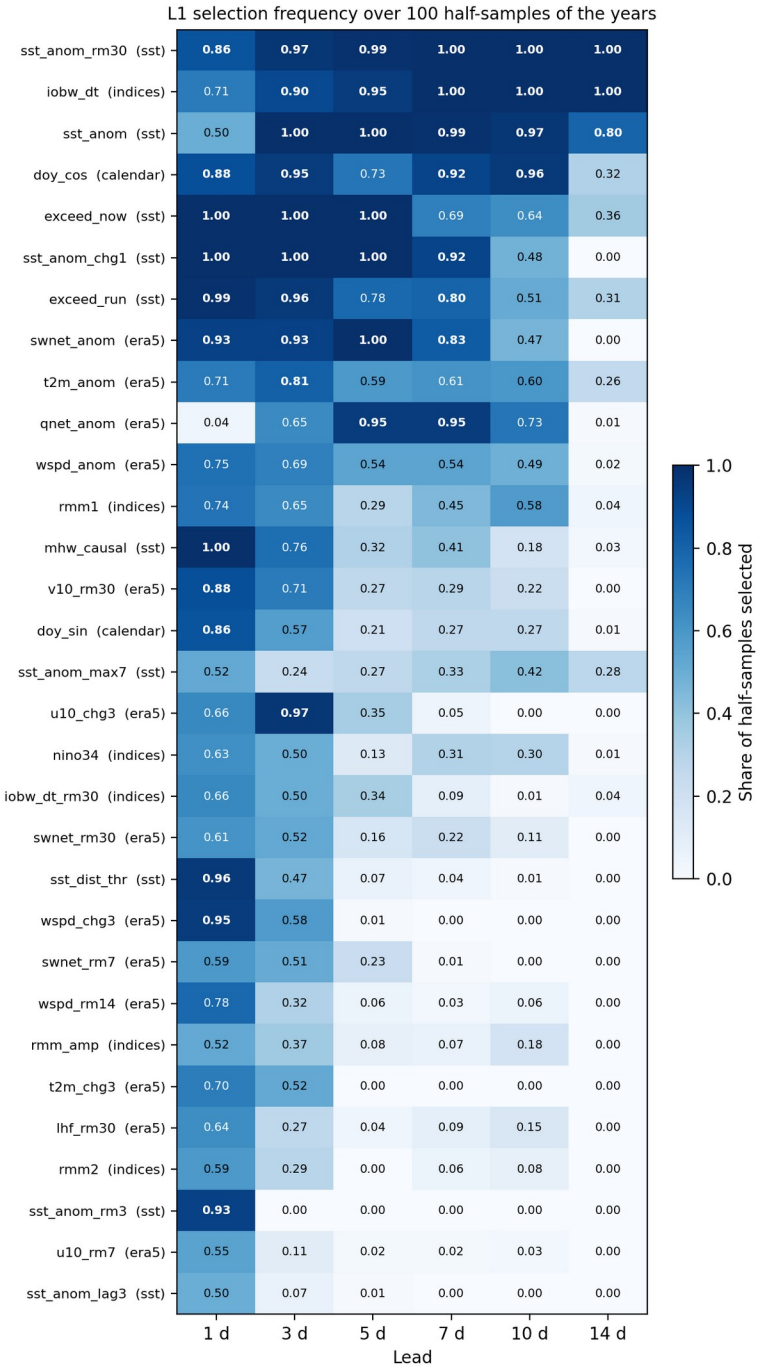


Figure 4.13 Share of 100 half-samples of the years in which each predictor was selected by the L1 logistic model, by lead (detrended labels, OISST). Only predictors selected in at least half of the half-samples at some lead are shown; bold values reach the stability threshold of 0.8.

The IOBW index is the most consistent non-SST predictor. It was selected in 90–100% of the half-samples at every lead from 3 to 14 days, always with a positive coefficient (median 0.21–0.36). At 14 days it is one of only three stable predictors, together with the 30-day mean and the current SST anomaly, and there each fit selects only five predictors on average, so the bound on false selections is 0.5. The net shortwave anomaly was stable at 1–7 days and the net heat flux anomaly only at 5–7 days, both with positive coefficients, meaning more heat entering the ocean. At 5 and 7 days, however, the bound on false selections (4.8 and 5.8) is close to the size of the stable sets (7 and 8 predictors), so the frequencies alone do not rule out chance there. The support for these two predictors comes from their consistent signs and their agreement with the onset composites (Section 4.2). At 1–3 days many predictors are selected per fit (26–34) and the bound is large (14–23), so stability at these leads is not interpreted. No DMI predictor reached 0.5 at any lead, and Niño 3.4 stayed below 0.35 beyond 3 days.

The IOBW index is computed from OISST over 20°S–20°N and 40–100°E (Section 3.1.3), which includes the southern part of the study box (15–20°N, 85–93°E), roughly 2% of the index area. Its signal is therefore mainly basin-scale, but not fully independent of the target SST.

4.11 Upper-ocean heat memory (post-hoc)

Adding the GODAS predictors did not improve the logistic SST model at any lead or with either product (Table 4.15, Fig. 4.14). With the 14-day delay, the MCC changed by between −0.015 and +0.007 and the new-event MCC by between −0.012 and +0.012, and every 95% CI included zero. At 10 and 14 days, where heat memory was expected to matter most, the new-event gains were +0.002 and −0.012 with OISST and +0.006 and +0.001 with CCI/C3S. Their upper CI bounds (+0.025 to +0.036) rule out gains larger than about 0.04 MCC. The largest gain, +0.012 in new-event MCC at 7 days with OISST (CI −0.002 to +0.026), was not significant and had the opposite sign with CCI/C3S (−0.009). The results with the 28-day delay were similar (Table A.4), so the decision rule of Section 3.14 is not met.

Table 4.15 MCC of the logistic SST model with and without the four GODAS predictors (14-day delay), 1982–2024, with the gain and its 95% CI.

Product	Lead (days)	MCC, SST	MCC, SST + GODAS	Gain (95% CI)	New-event MCC, SST	New-event MCC, SST + GODAS	New-event gain (95% CI)
OISST	7	0.366	0.372	+0.006 (−0.006 to +0.018)	0.278	0.290	+0.012 (−0.002 to +0.026)
OISST	10	0.275	0.282	+0.007 (−0.014 to +0.026)	0.222	0.224	+0.002 (−0.021 to +0.025)
OISST	14	0.268	0.253	−0.015 (−0.040 to +0.016)	0.222	0.210	−0.012 (−0.043 to +0.026)
CCI/C3S	7	0.465	0.455	−0.010 (−0.025 to +0.011)	0.337	0.328	−0.009 (−0.030 to +0.015)
CCI/C3S	10	0.352	0.354	+0.002 (−0.023 to +0.030)	0.216	0.222	+0.006 (−0.022 to +0.036)
CCI/C3S	14	0.272	0.274	+0.002 (−0.047 to +0.038)	0.177	0.178	+0.001 (−0.041 to +0.030)

The GODAS 0–50 m temperature anomaly was moderately correlated with the 30-day mean SST anomaly that the model already uses (r = 0.45 with OISST and 0.55 with CCI/C3S). The isothermal-layer depth anomaly was only weakly correlated with it (r = −0.23 and −0.14), but this independent part carried no useful information at 7–14 days.

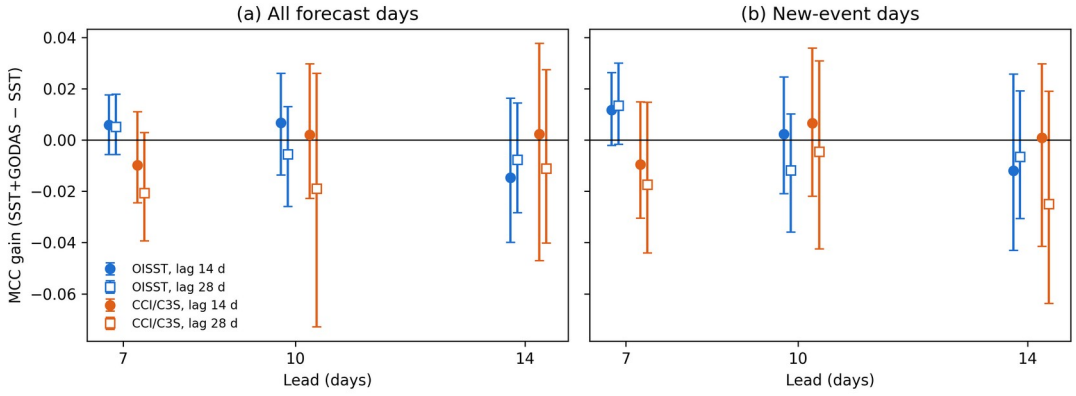


Figure 4.14 Change in MCC from adding the GODAS predictors to the logistic SST model, for (a) all forecast days and (b) new-event days. Filled markers: 14-day delay; open markers: 28-day delay. Bars are 95% year-block bootstrap CIs.

4.12 Previous-year ENSO and IOD (post-hoc)

MHW activity in the study box was clearly higher after El Niño winters (Fig. 4.15, Table 4.16). The Spearman correlation between the previous-winter ONI and detrended February–December MHW days was 0.40 (p = 0.008), and it stayed

between 0.36 and 0.46 when any one year was left out, so the decision rule of Section 3.15 is met. The correlation with the number of events was the same, and Niño 3.4 gave practically identical values. Years after an El Niño winter had on average 40.6 MHW days and 3.6 events, against 19.2 days and 2.1 events after neutral winters and 14.7 days and 1.5 events after La Niña (one-sided Mann–Whitney test, years after El Niño against all others, p = 0.014). The five years with the most MHW days (1998, 1987, 2020, 1988 and 2010) all followed an El Niño winter. The relationship is a tendency rather than a rule, however. The previous-winter ONI explains only about 16–26% of the year-to-year variance, and two years after El Niño, 1992 and 2015, had no MHW days at all.

Table 4.16 Spearman correlation (p-value) between previous-year climate indices and detrended MHW activity, 1983–2025.

Predictor (previous year)	MHW days, Feb–Dec	Events, Feb–Dec	MHW days, Mar–Jun	Events, Mar–Jun
ONI, Nov–Jan	0.40 (0.008)	0.40 (0.008)	0.47 (0.001)	0.55 (<0.001)
Niño 3.4, Nov–Jan	0.39 (0.009)	0.39 (0.011)	0.47 (0.001)	0.55 (<0.001)
DMI, Sep–Nov	0.26 (0.095)	0.18 (0.252)	0.31 (0.042)	0.37 (0.016)
DMI, Sep–Nov, partial on Niño 3.4	0.05 (0.776)	−0.06 (0.728)	0.06 (0.724)	0.08 (0.621)

The link was strongest in spring. For March–June, the correlations were 0.47 for MHW days and 0.55 for events (Pearson 0.53 and 0.62). By calendar month, the correlation was positive in every month from February to December. It was highest from March to July (0.25–0.37) and in November (0.33), and lowest in August, September and December (0.09–0.13) (Fig. 4.15b). With 11 monthly tests, these values are descriptive. With raw labels the correlations were weaker (0.24 for February–December MHW days, p = 0.11; Table A.5), because the number of raw MHW days is dominated by the warming trend rather than by year-to-year variability (Section 4.1).

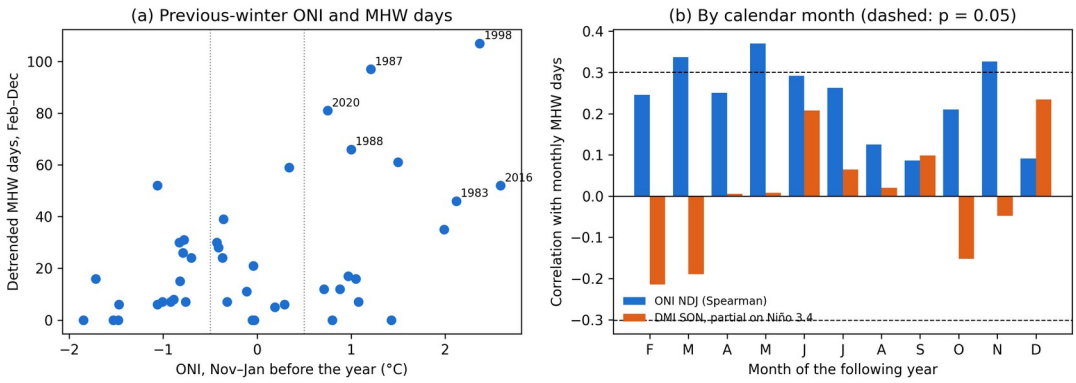


Figure 4.15 Previous-winter ENSO and MHW activity in the study box, 1983–2025: (a) previous-winter (November–January) ONI and detrended MHW days in February–December of the following year, 1983–2025; dotted lines mark ±0.5 °C. (b) Spearman correlation of monthly MHW days with the previous-winter ONI (blue), and partial correlation with the previous September–November DMI, controlling for Niño 3.4 (orange); dashed lines mark p = 0.05.

The IOD showed no effect of its own. The September–November DMI was positively correlated with MHW days in the following year (0.26, and 0.31 for March–June). However, it was also correlated with the previous-winter Niño 3.4 (0.57), and once ENSO was accounted for, the partial correlation fell to 0.05 ($p = 0.78$). No calendar month had a significant partial correlation. The apparent IOD signal is therefore the ENSO signal, carried by the positive IOD events that occurred together with El Niño.

CHAPTER 5
DISCUSSION

5.1 Why label-based persistence flatters persistence

The inflation found for RQ1 comes straight from the way the Hobday definition works. The first four days of every event are only labelled as MHW days once the fifth day has occurred, and gaps are only filled once an event resumes. A forecast that copies the label of the forecast day therefore "knows" that an event has started even when SST has been above the threshold for just one or two days, whereas honest persistence has to wait until the fifth day. Nearly all of the inflation (about 97%) comes from this minimum-duration rule, and gap joining contributes very little.

How large the effect is depends on how many MHW days can only be identified in hindsight. That share is highest where events are short, because the first four days then make up a larger part of each event. The detrended BoB series, with a median event length of 8 days and 41.6% of MHW days known only in hindsight, shows the largest short-lead inflation, while the raw Levantine Sea, with 11-day events and 22.8%, shows the smallest. At longer leads the order reverses, and the regions with longer events show more inflation, since a label that switched on early is still on further ahead. Because the effect appears in three seas with different climates and warming rates, and because it follows from the definition itself, it should be expected wherever Hobday labels are used as a persistence forecast.

The effect is even larger for single grid cells than for the regional mean. Averaging over a large box smooths out short, local exceedances, so regional events are longer and fewer of their days are known only in hindsight (41.6%, compared with a cell median of 45.7%). Across the 805 cells, the 1-day inflation follows the hindsight share almost exactly, which is the same relationship seen between regions. This matters in practice, because most ML studies forecast MHW maps at grid scale (Sun W. et al., 2023; Parasyris et al., 2025). For them, the relevant inflation is the grid-cell value, which in this study area has a median of about 0.31 at 1 day and 0.29 at 3 days, rather than the regional-mean value.

Expressed as lead time, label-based persistence at 5–7 days behaves like an honest forecast issued about 3 days later. For forecasts whose useful range is only about a week, that is a large error.

5.2 What the regional-mean models learn

The post-hoc comparison with exceedance persistence shows that, at 1–3 days, the models mostly detect events that have already started rather than predicting new ones. On the forecast day, SST above the threshold is a strong sign that the official label will soon be positive, whether or not five days have passed yet. Honest persistence ignores this signal by design, so any model that uses it will beat honest persistence easily. The coefficients agree with this: at 1 day the distance to the threshold and the 3-day mean anomaly dominate, and the drop-one-group test shows that only the current-state group is needed.

From about 7 days onwards the models add a small amount of information of a different kind. With OISST, the weight on the 30-day mean anomaly increases steadily with lead, removing the weekly-to-monthly background group costs about 20% of the skill at 14 days, and the case studies show long periods of raised probability before onset. At these leads the forecasts appear to signal that the ocean is primed for an MHW, rather than when the event will begin. This long-lead result, however, did not survive the change of SST product (Section 4.9). With CCI/C3S the model reaches the same 14-day MCC (0.273 against 0.267), but persistence is much stronger, and the gain over it disappears. The long-lead skill is therefore weak and depends on how persistent the anomalies of the chosen product are, not only on the ocean. A plausible interpretation of the background signal is the heat memory of the upper ocean, which accumulates surface heat fluxes over several weeks, so that a month of warm anomalies makes it more likely that a later fluctuation will cross the threshold. This interpretation was tested directly with GODAS (Section 4.11), and upper-ocean temperature and isothermal-layer depth added no skill at any lead with either product. About a quarter of the variance of the 0–50 m temperature anomaly is shared with the 30-day mean SST anomaly, and the rest carried no useful information at these leads. Whatever background signal exists is therefore already contained in the SST history, at least at the resolution of GODAS, and heat memory beyond what SST shows is not supported.

The same idea may explain why the ERA5 predictors added nothing, even though the composites in Section 4.2 show clear flux anomalies before onset. The fluxes change SST within a few days, so by the forecast day their combined effect is already part of the SST history that the model sees. The composite flux anomalies are also modest a week before onset (mostly below 0.5 standard deviations); on any single day they are weak signals against large day-to-day variability, even though they are significant when averaged over 109 events. Finally, the random forest did not beat logistic regression. For a smooth regional-mean series, the useful relationships seem to be close to linear, and the extra flexibility of the trees only adds variance.

The post-hoc L1 analysis (Section 4.10) refines this picture. When predictors are selected one by one, the IOBW index is retained at every lead from 3 to 14 days, and the net shortwave and net heat flux anomalies around 5–7 days, all with the signs expected from the onset composites. These predictors carry a consistent signal, yet they do not improve the forecasts significantly. The IOBW index changes slowly and describes whether the whole tropical Indian Ocean is warm, so it marks periods in which MHWs are more likely rather than the day an event begins, and the local 30-day mean anomaly already carries much of the same information. This agrees with the interpretation above: beyond a week, the useful information describes a warm background state, not the timing of onset. The link with basin-wide warming is consistent with earlier work on the Bay of Bengal (Gao et al., 2022; Lin et al., 2023), but here it rests on one SST product and was found after the main results.

5.3 Raw and detrended labels

For both label types the short-lead conclusion is the same: there is no skill beyond exceedance persistence at 1–3 days, and for raw labels not at 5 days either. At 10–14 days, however, the gain over exceedance persistence is roughly twice as large for raw labels. The most likely reason is the warming trend. With a fixed threshold, raw MHW days become far more common after 2015 (Fig. 4.2), so a model that sees the 30-day background is partly forecasting the warmer conditions of recent years rather than individual events. For this reason, the conclusions about event predictability are based on the detrended labels, and raw-label skill at long leads should not be presented as the ability to predict events.

5.4 Dependence on the SST product

The comparison with CCI/C3S shows that, in the northern Bay of Bengal, the MHW record itself depends strongly on the SST product. The two products agree on only about half of the MHW days, and the disagreement is concentrated in two places: the southwest monsoon and October, when cloud cover is heaviest and infrared retrievals are scarce, and the earliest decades, when fewer satellite sensors and in situ observations were available. Agreement is best after 2016. A similar pattern has been reported for the tropical Pacific by Chevillard et al. (2026), who compared six SST products and found that OISST detects more, shorter and more intense MHWs than the others, which they linked to its stronger variability on time scales of less than two weeks. They also found the lowest agreement in the western Pacific warm pool, where only 10–40% of MHW days were detected by all products, and differences that shrank over time. The same features appear here: OISST has more and shorter events and larger day-to-day changes than CCI/C3S, and the two products agree more closely in recent years.

This uncertainty in the labels puts the forecast results in perspective. On the same day, the labels of the two products agree with an MCC of only 0.457, which is lower than the MCC of a 3-day forecast scored against its own product (0.62–0.69). In other words, a forecast reproduces the labels of the product it was trained on better than two observational products reproduce each other. Much of this disagreement is expected from a threshold-based definition: with the observed anomaly correlation of 0.751, only about half of the exceedance days would coincide even if the two products differed only by random noise of this size (Section 4.9.1). Skill differences of 0.01–0.06 MCC between a model and a benchmark are small next to this, and a claim of skill beyond persistence is only meaningful if it is tested with more than one product.

Against this background, it matters which conclusions survived the change of product. The benchmark inflation did, at every lead, and its size changed in the way the mechanism predicts: less inflation where fewer days are known only in hindsight. This is expected, since the inflation follows from the Hobday definition and not from the details of any one dataset. The short-lead result also survived, because exceedance persistence uses the same SST as the labels and adjusts automatically to whatever product is used. The long-lead gain over persistence did not survive, because it depends on how strongly persistence itself performs, which in turn depends on the

smoothness of the product. The GODAS test (Section 4.11) points the same way: the long-lead gain with OISST is not explained by missing upper-ocean information, so it is more likely a property of the product than of the ocean.

5.5 Comparison with previous studies

Three of the seven studies reviewed in Section 2.2.3 report MHW skill without a persistence reference (Giamalaki et al., 2022; Sun W. et al., 2023; Bonino et al., 2024). The results of this thesis show why this is a problem: at lead times of a week or less, even a simple exceedance rule achieves most of the skill of a fitted model, so skill reported without such a reference cannot be credited to the model.

Parasyris et al. (2025) use the MHW label map from 3 or 7 days before the target as their persistence forecast, with labels that follow the five-day minimum. That benchmark is affected by the inflation measured here, and since their forecasts are maps, the grid-cell values in Section 4.3.2 are the relevant ones. Taken alone, an inflated benchmark would make a model look worse than it really is. However, because MHW maps are also used as inputs to their model, both the model and the benchmark may contain information from after the forecast date. Their scores therefore cannot be compared with forecasts that use only information available at the time, in either direction. This thesis does not re-evaluate their results; it only shows that the definition of the benchmark can, by itself, shift the MCC by 0.1–0.3 at such lead times.

Several studies also chose decision thresholds or input settings on the same data that they used to report skill, or estimated trends and climatologies from the full record. These issues are separate from the benchmark problem, but they point in the same direction: choices made during evaluation can change the reported skill as much as the choice of model. Based on these results, four practical recommendations can be made for MHW forecasting:

1. Report honest persistence and exceedance persistence alongside any model.
2. State how the minimum-duration rule is treated whenever labels are used in a benchmark or as an input.
3. Evaluate with blocked cross-validation, such as leave-one-year-out with purging and an embargo.

4. Choose hyperparameters, calibration and thresholds using the training data only.

The previous-year analysis (Section 4.12) agrees with physical studies of the bay. Gao et al. (2022) found that the December–February Niño 3.4 index of a decaying El Niño correlates with March–June MHW frequency in the northeastern bay at r = 0.53. The corresponding value here, for events in the study box, is 0.62, although the two studies filtered their series differently, and in both the link is concentrated in spring and early summer. Kumar et al. (2023) likewise attributed most MHW days in the bay to the decaying phase of El Niño. The link works through the warming of the whole tropical Indian Ocean that follows El Niño. This is why the IOBW index was positive throughout the 30 days before onset (Section 4.2) and was retained by the L1 selection (Section 4.10), and why the IOD has no effect once ENSO is accounted for. It also explains why Niño 3.4 added nothing to the 1–14-day forecasts. ENSO changes the chance of MHWs over a season, but by the time a forecast is made the resulting warmth is already part of the SST history, and ENSO carries no information about the week in which an event will start. The previous-winter ONI is therefore a candidate predictor for seasonal outlooks of MHW activity rather than for forecasts days ahead.

5.6 Limitations of the study

1. **Evaluation and operational limitations.** The climatology, threshold and detrending trend were estimated from the full record, and some robustness analyses (exceedance persistence, the grid-cell analysis, the second SST product, the L1 predictor selection, the GODAS test and the previous-year ENSO and IOD analysis) were added post-hoc; therefore, the framework is not fully operational in a strict real-time sense. The raw labels, which need no detrending, give the same short-lead result.
2. **Spatial and physical limitations.** Forecasting was based on a regional-mean SST series, which limits local-scale interpretation. Upper-ocean heat memory was tested only with GODAS, a coarse (1° × 1/3°) reanalysis whose top layer is relaxed to observed SST, whose salinity is synthetic and which had few subsurface observations in the Bay of Bengal before the Argo era. A null result

with GODAS therefore does not rule out that better subsurface data would add skill.

3. **Data and uncertainty limitations.** The analysis contains relatively few MHW events (109 detrended events in 44 years), many comparisons, and notable disagreement between OISST and CCI/C3S, which adds uncertainty to small skill differences and long-lead conclusions. The stable non-SST predictors found by the L1 selection were identified with OISST only, and the IOBW index is computed from the same product over a domain that includes part of the study box. The previous-year analysis rests on 43 years, of which 15 followed an El Niño winter.

CHAPTER 6
CONCLUSION

6.1 Major findings

This study set out to find how much of the skill of MHW forecasts in the northern Bay of Bengal is persistence and how much is genuine, at lead times of 1 to 14 days. The main findings are as follows.

1. **Label-based persistence uses information from after the forecast date (RQ1).** In the Bay of Bengal it raises the MCC of persistence by 0.28 at a 1-day lead and by 0.11 at 7 days, which at 5–7 days is equivalent to about three days of hindsight. The inflation is significant at every lead in the Bay of Bengal, the Levantine Sea and the South China Sea, for both detrended and raw labels, and it grows with the share of MHW days that can only be identified in hindsight. For individual grid cells it is larger still, with a median of 0.31 at 1 day across the 805 cells of the study area, and it is significant in every cell at leads of 1–7 days.
2. **For the regional-mean SST series, logistic regression and random forest beat honest persistence at every lead from 1 to 14 days (RQ2).** The gain is 0.13–0.18 MCC at 1–5 days and 0.08–0.10 at 7–14 days, with wide confidence intervals at 10–14 days.
3. **At 1–3 days, almost the whole gain comes from SST already being above the threshold.** In the post-hoc robustness test with exceedance persistence, the models add at most about 0.025 MCC there, for both detrended and raw labels and with both SST products.
4. **Skill beyond exceedance persistence is small, clearest at about 7 days, and weak and SST-product dependent at 10–14 days.** It amounts to 0.02–0.06 MCC overall. With OISST, the 14-day skill comes from the weekly-to-monthly background SST anomaly and gives broad warnings of a warm period rather than the timing of onset; in the three strongest events of the 44-year record, every forecast missed the first six event days at a 7-day lead.
5. **For the regional-mean series, ERA5 surface predictors do not improve the forecasts of new events, and random forest does not beat logistic**

regression (RQ3 not supported). A post-hoc L1 selection of individual predictors shows that the IOBW index, and at 5–7 days the net heat flux and net shortwave anomalies, carry a consistent signal with the expected signs. They do not significantly improve skill over SST predictors at 7 days or longer, and only marginally at 5 days.

6. **The benchmark and short-lead results hold with a second SST product, but the long-lead gain does not.** With the satellite-only CCI/C3S analysis, the inflation is significant at every lead and the model again beats honest persistence at 1–7 days, but its gain over persistence at 10–14 days disappears. The two products agree on only about half of the MHW days, mostly disagreeing during the monsoon and in the early decades.
7. **Upper-ocean predictors from GODAS do not add skill.** The 0–50 m temperature and the isothermal-layer depth did not improve the forecasts at 7–14 days with either SST product, and gains larger than about 0.04 MCC can be ruled out.
8. **MHWs are more frequent after El Niño winters, and the IOD has no effect of its own.** The previous-winter ONI correlates with the following year's detrended MHW days at 0.40 (0.47 in March–June), and years after El Niño have more than twice as many MHW days as other years. The apparent effect of the IOD disappears once ENSO is accounted for.

Taken together, these results show that, for regional-scale MHW forecasting in the northern Bay of Bengal, the choice of benchmark decides whether ML forecasts look useless, impressive or trivial. Against the label-based benchmark the models appear useless at 1–5 days, against honest persistence they appear strong, and against exceedance persistence they add almost nothing at 1–3 days and only weak skill beyond a week, which depends on the SST product.

6.2 Recommendations for future study

Several directions could build on this work. The evaluation could be repeated for individual grid points or sub-regions of the Bay of Bengal, where events are shorter and less coherent than in the regional mean. The upper-ocean test of this study could be repeated with a higher-resolution, Argo-era reanalysis such as GLORYS12 (from 1993), which resolves the shallow, salinity-stratified mixed layer of the northern bay

better than GODAS. The stable predictors found by the L1 selection, in particular the IOBW index, could be tested with CCI/C3S and with an IOBW index computed outside the study box, to check whether the basin-scale signal depends on the SST product or on the target region itself. The link between the previous-winter ONI and MHW activity could be developed into a seasonal outlook for the bay, combined with other seasonal predictors such as the North Atlantic Oscillation (Gao et al., 2022). In situ temperature records from moored buoys in the bay, such as those of the RAMA array (McPhaden et al., 2009), could be used to judge which SST product describes MHWs in the study area more accurately, and forecasts could be evaluated against several products at once. The climatology, threshold and trend could be estimated only from data before each forecast date, as an operational system would have to do. Subseasonal dynamical forecasts, for example from ECMWF, could be compared with honest and exceedance persistence using the same design. Finally, the audit of persistence definitions could be extended to more studies and regions, and published forecasts could be re-scored against honest baselines where the data allow. More generally, MHW forecasting studies should report honest and exceedance persistence, state how the minimum-duration rule is handled, evaluate their models with blocked designs that limit leakage, and check their main results with more than one SST product.

REFERENCES

1. Behringer, D.W., Xue, Y., 2004. Evaluation of the global ocean data assimilation system at NCEP: the Pacific Ocean. In: Eighth Symposium on Integrated Observing and Assimilation Systems for Atmosphere, Oceans, and Land Surface, AMS 84th Annual Meeting, Seattle, WA, Paper 2.3.
2. Bonino, G., Galimberti, G., Masina, S., McAdam, R., Clementi, E., 2024. Machine learning methods to predict sea surface temperature and marine heatwave occurrence: a case study of the Mediterranean Sea. Ocean Science 20, 417–432. https://doi.org/10.5194/os-20-417-2024
3. Breiman, L., 2001. Random forests. Machine Learning 45, 5–32.
4. Brier, G.W., 1950. Verification of forecasts expressed in terms of probability. Monthly Weather Review 78, 1–3.
5. Chen, T., Guestrin, C., 2016. XGBoost: a scalable tree boosting system. In: Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining. ACM, New York, pp. 785–794.
6. Chevillard, C., Le Gendre, R., Menkes, C., Izumo, T., Pagli, B., Van Wynsberge, S., Cravatte, S., 2026. Sensitivity of marine heatwaves metrics to SST products, focusing on the Tropical Pacific. Ocean Science 22, 1213–1236. https://doi.org/10.5194/os-22-1213-2026
7. Chicco, D., Jurman, G., 2020. The advantages of the Matthews correlation coefficient (MCC) over F1 score and accuracy in binary classification evaluation. BMC Genomics 21, 6.
8. Copernicus Marine Service, 2025. ESA SST CCI and C3S reprocessed sea surface temperature analyses (SST_GLO_SST_L4_REP_OBSERVATIONS_010_024). E.U. Copernicus Marine Service Information, Marine Data Store. https://doi.org/10.48670/moi-00169
9. Efron, B., Tibshirani, R.J., 1993. An Introduction to the Bootstrap. Chapman & Hall, New York.
10. Embury, O., Merchant, C.J., Good, S.A., Rayner, N.A., Høyer, J.L., Atkinson, C., Block, T., Alerskans, E., Pearson, K.J., Worsfold, M., McCarroll, N., Donlon, C., 2024. Satellite-based time-series of sea-surface temperature since 1980 for climate applications. Scientific Data 11. https://doi.org/10.1038/s41597-024-03147-w
11. Frölicher, T.L., Fischer, E.M., Gruber, N., 2018. Marine heatwaves under global warming. Nature 560, 360–364.

12. Gao, X., Li, G., Liu, J., Long, S.-M., 2022. The trend and interannual variability of marine heatwaves over the Bay of Bengal. Atmosphere 13, 469. https://doi.org/10.3390/atmos13030469

13. Giamalaki, K., Beaulieu, C., Prochaska, J.X., 2022. Assessing predictability of marine heatwaves with random forests. Geophysical Research Letters 49, e2022GL099069. https://doi.org/10.1029/2022GL099069

14. Hastie, T., Tibshirani, R., Friedman, J., 2009. The Elements of Statistical Learning, 2nd ed. Springer, New York.

15. Hersbach, H., Bell, B., Berrisford, P., Hirahara, S., Horányi, A., Muñoz-Sabater, J., et al., 2020. The ERA5 global reanalysis. Quarterly Journal of the Royal Meteorological Society 146, 1999–2049.

16. Hobday, A.J., Alexander, L.V., Perkins, S.E., Smale, D.A., Straub, S.C., Oliver, E.C.J., Benthuysen, J.A., Burrows, M.T., Donat, M.G., Feng, M., Holbrook, N.J., Moore, P.J., Scannell, H.A., Sen Gupta, A., Wernberg, T., 2016. A hierarchical approach to defining marine heatwaves. Progress in Oceanography 141, 227–238.

17. Hobday, A.J., Oliver, E.C.J., Sen Gupta, A., Benthuysen, J.A., Burrows, M.T., Donat, M.G., Holbrook, N.J., Moore, P.J., Thomsen, M.S., Wernberg, T., Smale, D.A., 2018. Categorizing and naming marine heatwaves. Oceanography 31(2), 162–173.

18. Holbrook, N.J., Scannell, H.A., Sen Gupta, A., Benthuysen, J.A., Feng, M., Oliver, E.C.J., et al., 2019. A global assessment of marine heatwaves and their drivers. Nature Communications 10, 2624.

19. Holbrook, N.J., Sen Gupta, A., Oliver, E.C.J., Hobday, A.J., Benthuysen, J.A., Scannell, H.A., Smale, D.A., Wernberg, T., 2020. Keeping pace with marine heatwaves. Nature Reviews Earth & Environment 1, 482–493.

20. Howard, L., Subramanian, A.C., Nadimpalli, J.R., Giglio, D., Hoteit, I., 2026. Skillful subseasonal Indian Ocean marine heatwave forecasts using a neural network. Environmental Data Science 5, e6. https://doi.org/10.1017/eds.2026.10033

21. Huang, B., Thorne, P.W., Banzon, V.F., Boyer, T., Chepurin, G., Lawrimore, J.H., Menne, M.J., Smith, T.M., Vose, R.S., Zhang, H.-M., 2017. Extended Reconstructed Sea Surface Temperature, Version 5 (ERSSTv5): upgrades, validations, and intercomparisons. Journal of Climate 30, 8179–8205.

22. Huang, B., Liu, C., Banzon, V., Freeman, E., Graham, G., Hankins, B., Smith, T., Zhang, H.-M., 2021. Improvements of the Daily Optimum Interpolation Sea Surface Temperature (DOISST) Version 2.1. Journal of Climate 34, 2923–2939.

23. Jacox, M.G., Alexander, M.A., Amaya, D., Becker, E., Bograd, S.J., et al., 2022. Global seasonal forecasts of marine heatwaves. Nature 604, 486–490.

24. Jolliffe, I.T., Stephenson, D.B. (Eds.), 2012. Forecast Verification: A Practitioner's Guide in Atmospheric Science, 2nd ed. Wiley-Blackwell, Chichester.

25. Kapoor, S., Narayanan, A., 2023. Leakage and the reproducibility crisis in machine-learning-based science. Patterns 4(9), 100804. https://doi.org/10.1016/j.patter.2023.100804

26. Kaufman, S., Rosset, S., Perlich, C., Stitelman, O., 2012. Leakage in data mining: formulation, detection, and avoidance. ACM Transactions on Knowledge Discovery from Data 6(4), 15.

27. Kumar, S., Chakraborty, A., Chandrakar, R., Kumar, A., Sadhukhan, B., Roy Chowdhury, R., 2023. Analysis of marine heatwaves over the Bay of Bengal during 1982–2021. Scientific Reports 13, 14235. https://doi.org/10.1038/s41598-023-39884-y

28. Künsch, H.R., 1989. The jackknife and the bootstrap for general stationary observations. The Annals of Statistics 17, 1217–1241.

29. Liang, K., Qiu, Y., Lin, X., Lin, W., Ni, X., He, Y., 2024. An increase in autumn marine heatwaves caused by the Indian Ocean Dipole in the Bay of Bengal. Journal of Climate 37(17), 4523–4539. https://doi.org/10.1175/JCLI-D-23-0541.1

30. Lin, X., Qiu, Y., Wang, J., Teng, H., Ni, X., Liang, K., 2023. Seasonal diversity of El Niño-induced marine heatwave increases in the Bay of Bengal. Geophysical Research Letters 50, e2022GL100807. https://doi.org/10.1029/2022GL100807

31. Matthews, B.W., 1975. Comparison of the predicted and observed secondary structure of T4 phage lysozyme. Biochimica et Biophysica Acta 405, 442–451.

32. McPhaden, M.J., Meyers, G., Ando, K., Masumoto, Y., Murty, V.S.N., Ravichandran, M., Syamsudin, F., Vialard, J., Yu, L., Yu, W., 2009. RAMA: the Research Moored Array for African–Asian–Australian Monsoon Analysis and Prediction. Bulletin of the American Meteorological Society 90, 459–480.

33. Meinshausen, N., Bühlmann, P., 2010. Stability selection. Journal of the Royal Statistical Society: Series B (Statistical Methodology) 72, 417–473. https://doi.org/10.1111/j.1467-9868.2010.00740.x

34. Mi, C., Yi, A., Xue, J., Dong, C., Shan, H., 2025. Forecasting summer marine heatwaves in the South China Sea using explainable machine learning models. Deep Sea Research Part I: Oceanographic Research Papers 218, 104457. https://doi.org/10.1016/j.dsr.2025.104457

35. Oliver, E.C.J., Benthuysen, J.A., Darmaraki, S., Donat, M.G., Hobday, A.J., Holbrook, N.J., Schlegel, R.W., Sen Gupta, A., 2021. Marine heatwaves. Annual Review of Marine Science 13, 313–342.

36. Oliver, E.C.J., Donat, M.G., Burrows, M.T., Moore, P.J., Smale, D.A., Alexander, L.V., et al., 2018. Longer and more frequent marine heatwaves over the past century. Nature Communications 9, 1324.

37. Parasyris, A., Metheniti, V., Kampanis, N., Darmaraki, S., 2025. Marine heatwaves in the Mediterranean Sea: a convolutional neural network study for extreme event prediction. Ocean Science 21, 897–912. https://doi.org/10.5194/os-21-897-2025

38. Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., et al., 2011. Scikit-learn: machine learning in Python. Journal of Machine Learning Research 12, 2825–2830.

39. Rayner, N.A., Parker, D.E., Horton, E.B., Folland, C.K., Alexander, L.V., Rowell, D.P., Kent, E.C., Kaplan, A., 2003. Global analyses of sea surface temperature, sea ice, and night marine air temperature since the late nineteenth century. Journal of Geophysical Research 108(D14), 4407.

40. Reynolds, R.W., Smith, T.M., Liu, C., Chelton, D.B., Casey, K.S., Schlax, M.G., 2007. Daily high-resolution-blended analyses for sea surface temperature. Journal of Climate 20, 5473–5496.

41. Roberts, D.R., Bahn, V., Ciuti, S., Boyce, M.S., Elith, J., Guillera-Arroita, G., et al., 2017. Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. Ecography 40, 913–929.

42. Saji, N.H., Goswami, B.N., Vinayachandran, P.N., Yamagata, T., 1999. A dipole mode in the tropical Indian Ocean. Nature 401, 360–363.

43. Shah, R.D., Samworth, R.J., 2013. Variable selection with error control: another look at stability selection. Journal of the Royal Statistical Society: Series B (Statistical Methodology) 75, 55–80. https://doi.org/10.1111/j.1467-9868.2011.01034.x

44. Smith, K.E., Burrows, M.T., Hobday, A.J., Sen Gupta, A., Moore, P.J., et al., 2021. Socioeconomic impacts of marine heatwaves: global issues and opportunities. Science 374(6566), eabj3593.

45. Sun, D., Jing, Z., Liu, H., 2024. Deep learning improves sub-seasonal marine heatwave forecast. Environmental Research Letters 19, 064035.

46. Sun, W., Zhou, S., Yang, J., Gao, X., Ji, J., Dong, C., 2023. Artificial intelligence forecasting of marine heatwaves in the South China Sea using a combined U-Net and ConvLSTM system. Remote Sensing 15(16), 4068. https://doi.org/10.3390/rs15164068

47. Tibshirani, R., 1996. Regression shrinkage and selection via the lasso. Journal of the Royal Statistical Society: Series B (Methodological) 58, 267–288. https://doi.org/10.1111/j.2517-6161.1996.tb02080.x

48. Welandawe, S., Priyadarshana, Y.H.P.P., Senanayake, N., Silva, E.N.S., 2025. Machine learning techniques for marine heatwave prediction: a comprehensive review. Intelligent Marine Technology and Systems 3, 28. https://doi.org/10.1007/s44295-025-00076-1

49. Wheeler, M.C., Hendon, H.H., 2004. An all-season real-time multivariate MJO index: development of an index for monitoring and prediction. Monthly Weather Review 132, 1917–1932.

50. Wilks, D.S., 2011. Statistical Methods in the Atmospheric Sciences, 3rd ed. Academic Press, Oxford.

51. Yang, J., Liu, Q., Xie, S.-P., Liu, Z., Wu, L., 2007. Impact of the Indian Ocean SST basin mode on the Asian summer monsoon. Geophysical Research Letters 34, L02708.

52. Zadrozny, B., Elkan, C., 2002. Transforming classifier scores into accurate multiclass probability estimates. In: Proceedings of the 8th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining. ACM, New York, pp. 694–699.

APPENDICES

APPENDIX A — Supplementary tables

Table A.1 Inflation of label-based over honest persistence for raw labels (MCC difference, 95% CI), and lead lost in days.

Lead (days)	Bay of Bengal	Levantine Sea	South China Sea	Lead lost, days (BoB / Lev. / SCS)
1	+0.230 (0.208 to 0.266)	+0.140 (0.109 to 0.197)	+0.160 (0.132 to 0.208)	more than the lead
2	+0.237 (0.215 to 0.276)	+0.136 (0.107 to 0.194)	+0.160 (0.131 to 0.209)	more than the lead
3	+0.230 (0.204 to 0.270)	+0.125 (0.100 to 0.175)	+0.153 (0.126 to 0.199)	more than the lead
5	+0.175 (0.146 to 0.215)	+0.092 (0.074 to 0.125)	+0.129 (0.106 to 0.167)	2.7 / 2.5 / 2.6
7	+0.113 (0.086 to 0.144)	+0.071 (0.054 to 0.096)	+0.105 (0.081 to 0.137)	2.8 / 2.5 / 2.7
10	+0.064 (0.035 to 0.092)	+0.054 (0.033 to 0.080)	+0.073 (0.054 to 0.101)	3.3 / 3.0 / 3.3
14	+0.050 (0.025 to 0.071)	+0.040 (0.011 to 0.066)	+0.049 (0.032 to 0.071)	5.4 / 3.3 / 4.4

Table A.2 Skill of honest persistence and the logistic SST model by lead (detrended labels): MCC with 95% CI, Brier skill score and new-event MCC.

Lead (days)	Honest persistence MCC	Logistic SST MCC	Gain over honest persistence	BSS, persistence / logistic	New-event MCC, logistic
1	0.625 (0.570 to 0.664)	0.783 (0.747 to 0.810)	+0.129 to +0.194	0.361 / 0.679	0.659
3	0.438 (0.352 to 0.500)	0.616 (0.561 to 0.657)	+0.147 to +0.221	0.074 / 0.449	0.525
5	0.323 (0.225 to 0.398)	0.458 (0.391 to 0.511)	+0.093 to +0.186	−0.101 / 0.281	0.383
7	0.265 (0.167 to 0.344)	0.361 (0.281 to 0.418)	+0.047 to +0.145	−0.191 / 0.180	0.285
10	0.207 (0.113 to 0.284)	0.287 (0.210 to 0.347)	+0.027 to +0.136	−0.280 / 0.114	0.238
14	0.174 (0.088 to 0.242)	0.257 (0.174 to 0.324)	+0.022 to +0.145	−0.328 / 0.086	0.214

Table A.3 Raw-label comparison of honest persistence, exceedance persistence and the logistic SST model (MCC).

Lead (days)	Honest persistence	Exceedance persistence	Logistic SST	Logistic gain over honest (95% CI)
1	0.680	0.808	0.815	+0.117 to +0.152
3	0.501	0.650	0.660	+0.134 to +0.187
5	0.383	0.503	0.518	+0.087 to +0.192
7	0.318	0.396	0.444	+0.066 to +0.187
10	0.265	0.308	0.400	+0.088 to +0.174
14	0.240	0.273	0.383	+0.071 to +0.209

Table A.4 MCC of the logistic SST model with and without the four GODAS predictors with a 28-day delay, 1982–2024, with the gain and its 95% CI.

Product	Lead (days)	MCC, SST	MCC, SST + GODAS	Gain (95% CI)	New-event MCC, SST	New-event MCC, SST + GODAS	New-event gain (95% CI)
OISST	7	0.366	0.371	+0.005 (−0.006 to +0.018)	0.278	0.291	+0.013 (−0.002 to +0.030)
OISST	10	0.275	0.270	−0.005 (−0.026 to +0.013)	0.222	0.210	−0.012 (−0.036 to +0.010)
OISST	14	0.268	0.260	−0.008 (−0.028 to +0.014)	0.222	0.216	−0.006 (−0.031 to +0.019)
CCI/C3S	7	0.465	0.444	−0.021 (−0.039 to +0.003)	0.337	0.320	−0.017 (−0.044 to +0.015)
CCI/C3S	10	0.352	0.333	−0.019 (−0.073 to +0.026)	0.216	0.211	−0.005 (−0.042 to +0.031)
CCI/C3S	14	0.272	0.260	−0.012 (−0.040 to +0.027)	0.177	0.152	−0.025 (−0.064 to +0.019)

Table A.5 Spearman correlation (p-value) between previous-year climate indices and raw-label MHW activity, 1983–2025.

Predictor (previous year)	MHW days, Feb–Dec	Events, Feb–Dec	MHW days, Mar–Jun	Events, Mar–Jun
ONI, Nov–Jan	0.24 (0.114)	0.28 (0.064)	0.30 (0.051)	0.30 (0.052)
Niño 3.4, Nov–Jan	0.22 (0.150)	0.27 (0.083)	0.29 (0.061)	0.29 (0.056)
DMI, Sep–Nov	0.19 (0.221)	0.14 (0.386)	0.26 (0.088)	0.25 (0.101)
DMI, Sep–Nov, partial on Niño 3.4	0.08 (0.620)	−0.02 (0.892)	0.13 (0.429)	0.11 (0.489)

APPENDIX B — Data and code availability

All analyses were run in Python on Google Colab, and the main code is listed in Appendix C. The processed data and all forecast outputs are stored in the project's Google Drive folder (MHW_Project_Data/processed). The main files are the daily predictor and label table (predictor_table_1982_2025.csv), the feature tables, the out-of-fold forecasts for each lead (folders oof_p5, oof_lead1 to oof_lead14, oof_l1 and godas/oof), and the result tables listed in Table B.1.

Table B.1 Result files behind the tables and figures of Chapter 4.

Result	File
MHW event statistics, three regions	wp1_event_stats_3regions.csv
Benchmark inflation by lead	wp1_leakage_curve_bob.csv, wp1_leakage_curve_3regions.csv
Benchmark inflation per grid cell	wp1c_grid_leakage_summary.csv, wp1c_grid_leakage_cells.csv
Skill by lead, all models	wp2_skill_by_lead_det.csv
Exceedance persistence, detrended and raw	wp2_exceedance_persistence_det.csv, rob1_raw_hierarchy.csv
Seven-day comparison of all models	p5_results_det.csv
Coefficients and drop-one-group test	wp5_logit_sst_coefficients.csv, wp5_drop_group_det.csv
Case studies	wp5_case_studies.csv
Comparison of OISST and CCI/C3S	v2_dataset_comparison.csv, v2_leakage_both.csv, v2_skill_both.csv, v2_diag_month.csv, v2_diag_period.csv
Individual predictor selection, skill	l1_skill_with_new_event.csv, l1_skill_summary.csv
Individual predictor selection, stability	l1_halfsample_selection_frequency_by_lead.csv, l1_selection_frequency_by_lead.csv
Upper-ocean heat memory (GODAS)	godas/godas_box_pentad.csv, wp6_godas_heat_memory.csv
Previous-year ENSO and IOD	wp7_enso_iod_lag.csv, wp7_enso_iod_monthly.csv

APPENDIX C — Main processing code

The listings below are the main code cells that produced the results of this thesis, taken from the Google Colab notebooks used for the study. Cells that only drew figures or printed checks were left out, and working notes in the comments were removed. File paths refer to the project folder on Google Drive, and the cells are listed in the order in which they were run.

C.1 Regional SST and MHW labels

```
from google.colab import drive
drive.mount('/content/drive')

import glob, os, re, warnings
import numpy as np
import pandas as pd
import xarray as xr
import matplotlib.pyplot as plt
warnings.filterwarnings("ignore")

# CONFIG
OISST_GLOB = "/content/drive/MyDrive/Colab Notebooks/oisst_daily.nc"
USER_LABELS_CSV = "/content/drive/MyDrive/Colab Notebooks/daily_mhw_detection_raw_sst.csv"
OUT_DIR = "/content/drive/MyDrive/MHW/mhw_check"
LAT = (15.0, 23.0)
LON = (85.0, 93.0)
BASELINE = (1982, 2021)
START = "1982-01-01"
#
os.makedirs(OUT_DIR, exist_ok=True)

def regional_mean_from_file(path):
    ds = xr.open_dataset(path)                    # mask_and_scale=True -> fill values become NaN
    var = "sst" if "sst" in ds else [v for v in ds.data_vars if ds[v].ndim >= 3][0]
    da = ds[var]
    for dim in ("zlev", "lev", "depth"):
        if dim in da.dims:
            da = da.isel({dim: 0}, drop=True)
    latn = "lat" if "lat" in da.dims else "latitude"
    lonn = "lon" if "lon" in da.dims else "longitude"
    da = da.sortby(latn).sortby(lonn)
    box = da.sel({latn: slice(*LAT), lonn: slice(*LON)}).load()
    w = np.cos(np.deg2rad(box[latn]))
    mean = box.weighted(w).mean((latn, lonn), skipna=True)
    nvalid = box.notnull().sum((latn, lonn))
    df = pd.DataFrame({
        "date": pd.to_datetime(box["time"].values).floor("D"),
        "sst": mean.values.astype(float),
        "n_valid": nvalid.values.astype(int),
        "n_lat": box.sizes[latn], "n_lon": box.sizes[lonn],
        "file": os.path.basename(path),
        "units": str(da.attrs.get("units", "?")),
    })
    ds.close()
    return df

files = sorted(glob.glob(OISST_GLOB, recursive=True))
print(f"{len(files)} files found")
cache = os.path.join(OUT_DIR, "regional_sst_with_provenance.csv")
if os.path.exists(cache):
    raw = pd.read_csv(cache, parse_dates=["date"])
    print("loaded cache:", cache, "(delete it to rebuild)")
else:
    parts = []
    for k, f in enumerate(files):
        parts.append(regional_mean_from_file(f))
        if k % 500 == 0:
            print(f" {k}/{len(files)} {os.path.basename(f)}")
    raw = pd.concat(parts, ignore_index=True)
    raw.to_csv(cache, index=False)
raw["prelim"] = raw["file"].str.contains("prelim", case=False)

problems = []
def report(ok, msg):
    print(("PASS " if ok else "FAIL   ") + msg)
    if not ok:
        problems.append(msg)

raw = raw[raw["date"] >= START].sort_values(["date", "prelim"]).reset_index(drop=True)

print("Grid cells in box:", raw["n_lat"].iloc[0], "x", raw["n_lon"].iloc[0],
      "| units attribute:", list(raw["units"].unique()))

report(raw["n_lat"].nunique() == 1 and raw["n_lon"].nunique() == 1,
       "every file gives the same box size")

dup = raw[raw.duplicated("date", keep=False)]
if len(dup):
    spread = dup.groupby("date")["sst"].agg(lambda s: s.max() - s.min())
    print(f" {dup['date'].nunique()} dates appear more than once; "
          f"max value difference between copies = {spread.max():.3f} C")
    print(dup.head(10).to_string())
report(len(dup) == 0, "no duplicated dates")
# keep one copy per date, final over preliminary
```

```
raw = raw.drop_duplicates("date", keep="first").set_index("date")

full = pd.date_range(raw.index.min(), raw.index.max(), freq="D")
missing = full.difference(raw.index)
if len(missing):
    print(" missing dates (first 20):", list(missing[:20].date))
report(len(missing) == 0, f"continuous daily record {full[0].date()} -> {full[-1].date()} "
       f"({len(full)} days, {len(missing)} missing)")

mode_n = raw["n_valid"].mode()[0]
bad_n = raw[raw["n_valid"] != mode_n]
report(len(bad_n) == 0, f"same number of ocean cells every day ({mode_n}); "
       f"{len(bad_n)} days differ")
if len(bad_n):
    print(bad_n[["n_valid", "file"]].head(10).to_string())

report(raw["sst"].between(20, 34).all(),
       f"SST range plausible in deg C (min {raw['sst'].min():.2f}, max {raw['sst'].max():.2f})")
report(raw["sst"].notna().all(), f"no NaN regional means ({raw['sst'].isna().sum()} NaN)")
report(raw.index.year.min() <= BASELINE[0] and raw.index.year.max() >= BASELINE[1],
       f"record covers the {BASELINE[0]}-{BASELINE[1]} baseline")
if raw["prelim"].any():
    print(f" NOTE: {raw['prelim'].sum()} days still come from PRELIMINARY files "
          f"({raw.index[raw['prelim']].min().date()} onward). Fine at the end of the record
    only.")

# repair only what is safe to repair: reindex to a continuous calendar (gaps stay NaN;
# the detector fills NaN with climatology, exactly as Oliver's code does)
sst = raw["sst"].reindex(full)
print("\nIntegrity:", "ALL PASS" if not problems else f"{len(problems)} FAIL(s) — see above")

def _runs(x):
    out, i, n = [], 0, len(x)
    while i < n:
        if x[i]:
            j = i
            while j + 1 < n and x[j + 1]:
                 j += 1
            out.append([i, j]); i = j + 1
        else:
            i += 1
    return out

def _doy366(dates):
    ref = pd.date_range("2000-01-01", "2000-12-31")
    lut = {(d.month, d.day): i + 1 for i, d in enumerate(ref)}
    return np.array([lut[(d.month, d.day)] for d in dates])

def _periodic_runavg(x, w):
    N = len(x)
    return np.convolve(np.concatenate([x, x, x]), np.ones(w) / w, mode="same")[N:2 * N]

def climatology(dates, temp, base, pctile=90, half_window=5, smooth=31, pool=True):
    doy = _doy366(dates)
    yrs = dates.year.to_numpy()
    b0 = np.where(yrs == base[0])[0][0]
    b1 = np.where(yrs == base[1])[0][-1]
    thr = np.full(366, np.nan); sea = np.full(366, np.nan)
    for d in range(1, 367):
        if d == 60:
            continue
        tt0 = np.where(doy[b0:b1 + 1] == d)[0]
        hw = half_window if pool else 0
        tt = (b0 + tt0[:, None] + np.arange(-hw, hw + 1)[None, :]).ravel()
        tt = tt[(tt >= 0) & (tt < len(temp))]
        thr[d - 1] = np.nanpercentile(temp[tt], pctile)
        sea[d - 1] = np.nanmean(temp[tt])
    thr[59] = 0.5 * (thr[58] + thr[60]); sea[59] = 0.5 * (sea[58] + sea[60])
    thr = _periodic_runavg(thr, smooth); sea = _periodic_runavg(sea, smooth)
    return sea[doy - 1], thr[doy - 1]

def detect(dates, temp, base, min_dur=5, max_gap=2, order="filter_then_join",
           pool=True, smooth=31):
    temp = np.asarray(temp, float)
    seas, thresh = climatology(dates, temp, base, pool=pool, smooth=smooth)
    filled = np.where(np.isnan(temp), seas, temp)
    runs = _runs(filled > thresh)
    def join(rr):
        out = []
        for r in rr:
            if out and r[0] - out[-1][1] - 1 <= max_gap:
                 out[-1][1] = r[1]
            else:
                 out.append(list(r))
        return out
    if order == "filter_then_join":         # Hobday / Oliver (correct)
        ev = join([r for r in runs if r[1] - r[0] + 1 >= min_dur])
```

```
    elif order == "join_then_filter":
        ev = [r for r in join(runs) if r[1] - r[0] + 1 >= min_dur]
    elif order == "no_join":
        ev = [r for r in runs if r[1] - r[0] + 1 >= min_dur]
    mhw = np.zeros(len(temp), int)
    rows = []
    for s, e in ev:
        mhw[s:e + 1] = 1
        a = filled[s:e + 1] - seas[s:e + 1]
        rows.append(dict(start=dates[s].date(), end=dates[e].date(), duration_days=e - s + 1,
                         peak_date=dates[s + int(np.argmax(a))].date(),
                         max_intensity_C=round(float(a.max()), 3),
                         mean_intensity_C=round(float(a.mean()), 3),
                         cumulative_intensity_Cdays=round(float(a.sum()), 3)))
    daily = pd.DataFrame(dict(sst=temp, seas=seas, thresh=thresh, ssta=temp - seas, mhw=mhw),
                         index=dates)
    return daily, pd.DataFrame(rows)

dates = sst.index
vals = sst.to_numpy()

# Main analysis: raw SST
d_raw, ev_raw = detect(dates, vals, BASELINE)

# Sensitivity: linearly detrended SST (trend pivoted at baseline mid-point, so baseline-era values
    barely move)
t = np.arange(len(vals), dtype=float)
ok = ~np.isnan(vals)
slope = np.polyfit(t[ok], vals[ok], 1)[0]
pivot = np.argmin(np.abs(dates - pd.Timestamp(f"{(BASELINE[0] + BASELINE[1] + 1) // 2}-01-01")))
vals_dt = vals - slope * (t - pivot)
d_dt, ev_dt = detect(dates, vals_dt, BASELINE)
print(f"Linear SST trend in box: {slope * 3652.5:.3f} C/decade")

def yearly(daily, ev):
    y = pd.DataFrame(index=sorted(set(daily.index.year)))
    y["events (by start year)"] = (pd.Series(pd.to_datetime(ev["start"]).dt.year).value_counts()
                                   if len(ev) else pd.Series(dtype=int))
    y["MHW days"] = daily["mhw"].groupby(daily.index.year).sum()
    m = daily["mhw"] == 1
    y["mean intensity C"] = daily.loc[m, "ssta"].groupby(daily.index[m].year).mean().round(2)
    y["max intensity C"] = daily.loc[m, "ssta"].groupby(daily.index[m].year).max().round(2)
    return y.fillna(0)

y_raw, y_dt = yearly(d_raw, ev_raw), yearly(d_dt, ev_dt)
table = pd.concat({"RAW": y_raw, "DETRENDED": y_dt}, axis=1)
pd.set_option("display.width", 200)
print(table.to_string())
print(f"\nRAW: {len(ev_raw)} events, {d_raw.mhw.sum()} MHW days "
      f"({100 * d_raw.mhw.mean():.1f}% of all days)")
print(f"DETRENDED: {len(ev_dt)} events, {d_dt.mhw.sum()} MHW days")
first, last = y_raw.loc[:2001, "MHW days"].mean(), y_raw.loc[2002:, "MHW days"].mean()
print(f"Mean MHW days/yr: 1982-2001 = {first:.0f}, 2002-end = {last:.0f}")

out = pd.DataFrame({"sst": d_raw["sst"], "clim_mean": d_raw["seas"], "thresh90": d_raw["thresh"],
                    "ssta": d_raw["ssta"], "mhw_raw": d_raw["mhw"],
                    "sst_detrended": d_dt["sst"], "thresh90_detrended": d_dt["thresh"],
                    "mhw_detrended": d_dt["mhw"]}, index=dates)
out.index.name = "date"
out.to_csv(os.path.join(OUT_DIR, "mhw_daily_labels.csv"))
ev_raw.to_csv(os.path.join(OUT_DIR, "mhw_events_raw.csv"), index=False)
ev_dt.to_csv(os.path.join(OUT_DIR, "mhw_events_detrended.csv"), index=False)
table.to_csv(os.path.join(OUT_DIR, "mhw_yearly_stats.csv"))
print("Saved to", OUT_DIR)

print("\nTop 10 events (raw) by cumulative intensity:")
print(ev_raw.sort_values("cumulative_intensity_Cdays",
    ascending=False).head(10).to_string(index=False))
```

C.2 ERA5 daily fields and regional means

```
from google.colab import drive; drive.mount('/content/drive')
import xarray as xr, numpy as np, pandas as pd, os, glob
BASE   = '/content/drive/MyDrive/Colab Notebooks/MHW_Project_Data/raw'
INST = sorted(glob.glob(f'{BASE}/ERA5/temp_u10_v10/ERA5_t2m_u10_v10_*.nc'))
ACC = sorted(glob.glob(f'{BASE}/ERA5/Flux/ERA5_surface_flux_*.nc'))
print(len(INST), len(ACC))
print(xr.open_dataset(INST[0])); print(xr.open_dataset(ACC[0]))

PROC = '/content/drive/MyDrive/Colab Notebooks/MHW_Project_Data/processed'
OUT = f'{PROC}/era5_daily'; os.makedirs(OUT, exist_ok=True)
FLUX = [('ssr','swnet'), ('str','lwnet'), ('slhf','lhf'), ('sshf','shf')]
```

```
def op(f, first_only=False):
    ds = xr.open_dataset(f).drop_vars(['expver','number'], errors='ignore')
    if first_only:
        ds = ds.isel(valid_time=[0])
    if 'time' in ds.dims:                                  # collapse the 12 monthly slices
        v0 = list(ds.data_vars)[0]
        chk = ds[v0].isel(latitude=16, longitude=16).notnull().sum('time')
        assert int(chk.min()) == 1 and int(chk.max()) == 1, f'unexpected structure in {f}'
        ds = ds.max('time', skipna=True)
    return ds.rename(valid_time='time').sortby('time')

yr_of = lambda f: pd.Timestamp(xr.open_dataset(f).valid_time.values[0]).year   # reads metadata
    only
inst = {yr_of(f): f for f in INST}
acc = {yr_of(f): f for f in ACC}
assert sorted(inst) == sorted(acc) == list(range(1982, 2026))

for yr in range(1982, 2026):
    out = f'{OUT}/era5_daily_{yr}.nc'
    if os.path.exists(out): continue
    a = op(inst[yr]).sel(time=str(yr))
    d = xr.Dataset()
    d['t2m'] = a.t2m.resample(time='1D').mean() - 273.15
    d['u10'] = a.u10.resample(time='1D').mean()
    d['v10'] = a.v10.resample(time='1D').mean()
    d['wspd'] = np.hypot(a.u10, a.v10).resample(time='1D').mean()
    b = op(acc[yr])
    if yr + 1 in acc:
        b = xr.concat([b, op(acc[yr + 1], first_only=True)], dim='time')
    b = b.assign_coords(time=b.time - np.timedelta64(1, 'h')).sel(time=str(yr))
    for src, dst in FLUX:
        d[dst] = b[src].resample(time='1D').mean() / 3600
    d['qnet'] = d.swnet + d.lwnet + d.lhf + d.shf
    d.to_netcdf(out)
    print(yr, 'inst hours:', a.time.size, 'flux hours:', b.time.size,
          'NaN t2m:', int(d.t2m.isnull().sum()), 'NaN qnet:', int(d.qnet.isnull().sum()))

ds = xr.open_mfdataset(sorted(glob.glob(f'{OUT}/era5_daily_*.nc')), combine='by_coords').load()
units = {'t2m':'degC','u10':'m s-1','v10':'m s-1','wspd':'m s-1',
         'swnet':'W m-2','lwnet':'W m-2','lhf':'W m-2','shf':'W m-2','qnet':'W m-2'}
for v, u in units.items(): ds[v].attrs['units'] = u
ds.attrs = {'source':'ERA5 hourly single levels, Copernicus CDS',
            'domain':'15-23N, 85-93E',
            'processing':'daily means 00-23 UTC; fluxes from hourly accumulations /3600; '
                         'positive = downward (into ocean); wspd = daily mean of hourly speed; '
                         '2025-12-31 fluxes from 23 h'}
assert ds.time.size == 16071
enc = {v: {'zlib': True, 'complevel': 4, 'dtype': 'float32'} for v in ds.data_vars}
ds.to_netcdf(f'{PROC}/era5_daily_1982_2025_BoB.nc', encoding=enc)
print(ds)

OISST_DAY = '/content/drive/MyDrive/Colab Notebooks/oisst_daily.nc'
sst0 = xr.open_dataset(OISST_DAY).sst
sst0 = sst0.isel(time=0) if 'time' in sst0.dims else sst0
sst0 = sst0.squeeze(drop=True).sel(lat=slice(15, 23), lon=slice(85, 93))
ocean = sst0.notnull()
assert int(ocean.sum()) == 805, int(ocean.sum())

era = xr.open_dataset(f'{PROC}/era5_daily_1982_2025_BoB.nc').sortby('latitude')
g = era.interp(latitude=ocean.lat.values, longitude=ocean.lon.values) \
       .rename(latitude='lat', longitude='lon').where(ocean)
assert int(g.t2m.isel(time=0).notnull().sum()) == 805

w = np.cos(np.deg2rad(ocean.lat))
reg = g.weighted(w).mean(('lat','lon')).to_dataframe()[list(units)]
reg.index = pd.to_datetime(reg.index).normalize()
reg.to_csv(f'{PROC}/era5_regional_daily.csv')

# ---- checks ----
print(reg.shape, 'NaN:', int(reg.isna().sum().sum()))
print(reg.describe().round(2))
print(reg[['swnet','lwnet','lhf','shf','qnet']].groupby(reg.index.month).mean().round(1))
```

C.3 Climate indices and the daily predictor table

```
url = 'https://www.cpc.ncep.noaa.gov/data/indices/ersst5.nino.mth.91-20.ascii'
raw = pd.read_csv(url, sep=r'\s+')
per = pd.to_datetime(dict(year=raw.YR, month=raw.MON, day=1)).dt.to_period('M')
n34 = pd.Series(raw.iloc[:, -1].values, index=per)          # last column = Niño-3.4 anomaly
n34.index = n34.index + 1                                    # month m becomes usable in month m+1
days = pd.date_range('1982-01-01', '2025-12-31', freq='D')
nino34 = pd.Series(n34.reindex(days.to_period('M')).values, index=days, name='nino34')
print(nino34.isna().sum(), nino34['1997-12':'1998-02'].resample('MS').first())
```

```
import subprocess
IOD = f'{PROC}/io_boxes'; os.makedirs(IOD, exist_ok=True)
TMP = '/content/oisst_tmp'; os.makedirs(TMP, exist_ok=True)
DL = 'https://downloads.psl.noaa.gov/Datasets/noaa.oisst.v2.highres/sst.day.mean.{}.nc'

def boxmean(da, s, n, w, e):
    b = da.sel(lat=slice(s, n), lon=slice(w, e))
    return b.weighted(np.cos(np.deg2rad(b.lat))).mean(('lat','lon')).values

for yr in range(1982, 2026):
    out = f'{IOD}/io_{yr}.csv'
    if os.path.exists(out): continue
    f = f'{TMP}/sst_{yr}.nc'
    r = subprocess.run(['wget', '-q', '-c', '--tries=10', '--waitretry=60', '-O', f,
    DL.format(yr)])
    if r.returncode != 0:
        raise RuntimeError(f'{yr}: download failed, wget code {r.returncode}')
    with xr.open_dataset(f, engine='netcdf4') as ds:
        sst = ds.sst.sel(lat=slice(-20, 20), lon=slice(40, 110)).load()
    pd.DataFrame({'wtio': boxmean(sst, -10, 10, 50, 70),
                  'seio': boxmean(sst, -10, 0, 90, 110),
                  'iobw': boxmean(sst, -20, 20, 40, 100)},
                 index=pd.to_datetime(sst.time.values)).to_csv(out)
    os.remove(f)
    print(yr, sst.time.size)

io = pd.concat([pd.read_csv(f, index_col=0, parse_dates=True)
                for f in sorted(glob.glob(f'{IOD}/io_*.csv'))])
def anom(s):                                    # 1982–2021 daily climatology, 31-day smoothed
    doy = s.index.dayofyear - (s.index.is_leap_year & (s.index.month > 2))
    clim = s[s.index.year <= 2021].groupby(doy[s.index.year <= 2021]).mean()
    c = pd.concat([clim.iloc[-15:], clim, clim.iloc[:15]]).rolling(31,
    center=True).mean().iloc[15:-15]
    c.index = clim.index
    return s - c.reindex(doy).values
dmi = (anom(io.wtio) - anom(io.seio)).rename('dmi')
ia   = anom(io.iobw)
t    = np.arange(len(ia))
iobw = (ia - np.polyval(np.polyfit(t, ia.values, 1), t)).rename('iobw_dt')   # detrended
print(len(dmi), dmi.isna().sum(), dmi['1997-10':'1997-11'].mean().round(2), dmi['2019-10':'2019-
    11'].mean().round(2))

import subprocess
RMM = f'{PROC}/rmm.74toRealtime.txt'
if not os.path.exists(RMM) or os.path.getsize(RMM) < 1_000_000:
    subprocess.run(['wget', '-q', '-U', 'Mozilla/5.0', '-O', RMM,
                    'https://www.bom.gov.au/clim_data/IDCKGEM000/rmm.74toRealtime.txt'])
print('file size:', os.path.getsize(RMM) if os.path.exists(RMM) else 'missing')

r = pd.read_csv(RMM, sep=r'\s+', skiprows=2, header=None, usecols=range(7),
                names=['y','m','d','rmm1','rmm2','phase','amp'])
r.index = pd.to_datetime(dict(year=r.y, month=r.m, day=r.d))
rmm = r.loc['1982':'2025', ['rmm1','rmm2']]
print('last date in file:', r.index.max().date())
print(len(rmm), (rmm.abs() > 100).sum().to_dict())

# 1. save Cell G results (still in memory)
#pd.concat([dmi, iobw], axis=1).to_csv(f'{PROC}/io_indices_daily.csv')

# 2. load every piece

era   = pd.read_csv(f'{PROC}/era5_regional_daily.csv', index_col=0, parse_dates=True)
io    = pd.read_csv(f'{PROC}/io_indices_daily.csv',   index_col=0, parse_dates=True)

raw  = pd.read_csv('https://www.cpc.ncep.noaa.gov/data/indices/ersst5.nino.mth.91-20.ascii',
    sep=r'\s+')
n34 = pd.Series(raw.iloc[:, -1].values,
                index=pd.to_datetime(dict(year=raw.YR, month=raw.MON, day=1)).dt.to_period('M') +
    1)
days = pd.date_range('1982-01-01', '2025-12-31', freq='D')
nino = pd.Series(n34.reindex(days.to_period('M')).values, index=days, name='nino34')

r    = pd.read_csv(f'{PROC}/rmm.74toRealtime.txt', sep=r'\s+', skiprows=2, header=None,
    usecols=range(7),
                  names=['y','m','d','rmm1','rmm2','phase','amp'])
r.index = pd.to_datetime(dict(year=r.y, month=r.m, day=r.d))

hits = glob.glob('/content/drive/MyDrive/MHW/mhw_check/mhw_daily_labels.csv', recursive=True)
print('labels file:', hits)
lab = pd.read_csv(hits[0], index_col=0, parse_dates=True)
print('label columns:', list(lab.columns))

# 3. merge on date
for d in (era, io, lab): d.index = pd.to_datetime(d.index).normalize()
df = pd.concat([era, io, nino, r[['rmm1','rmm2']], lab], axis=1).loc['1982-01-01':'2025-12-31']
df.to_csv(f'{PROC}/predictor_table_1982_2025.csv')
```

```
# 4. checks
print(df.shape, '| NaN per column:', df.isna().sum()[df.isna().sum() > 0].to_dict())
print(df.head(3).T)
```

C.4 Persistence benchmarks and benchmark leakage

```
from google.colab import drive; drive.mount('/content/drive')
import os, numpy as np, pandas as pd
import matplotlib.pyplot as plt

PROC = '/content/drive/MyDrive/Colab Notebooks/MHW_Project_Data/processed'
FIG = f'{PROC}/figures'; os.makedirs(FIG, exist_ok=True)
df = pd.read_csv(f'{PROC}/predictor_table_1982_2025.csv', index_col=0, parse_dates=True)

def doy366(idx):
    d = idx.dayofyear.values.copy()
    d[(~idx.is_leap_year) & (idx.month > 2)] += 1           # 1 Mar is always day 61
    return d

def hobday_clim(s, base=(1982, 2021), win=5, smooth=31, pct=90):
    """Hobday et al. (2016) climatological mean and percentile threshold (as in Oliver's
    marineHeatWaves)."""
    v, d = s.values, doy366(s.index)
    inb = (s.index.year >= base[0]) & (s.index.year <= base[1])
    m, p = np.full(367, np.nan), np.full(367, np.nan)
    for k in range(1, 367):
        if k == 60: continue
        idx = np.where((d == k) & inb)[0]
        pool = np.unique(np.clip((idx[:, None] + np.arange(-win, win + 1)).ravel(), 0, len(v) -
    1))
        x = v[pool[inb[pool]]]
        m[k], p[k] = np.nanmean(x), np.nanpercentile(x, pct)
    out = []
    for c in (m, p):
        c[60] = (c[59] + c[61]) / 2
        c = c[1:]
        c = np.convolve(np.r_[c[-(smooth // 2):], c, c[:smooth // 2]], np.ones(smooth) / smooth,
    'valid')
        out.append(pd.Series(c[d - 1], index=s.index))
    return out                                              # clim_mean, threshold

def runs(b):
    """Start/end indices of consecutive True runs."""
    b = np.asarray(b, bool).astype(int)
    st = np.where(np.diff(np.r_[0, b]) == 1)[0]; en = np.where(np.diff(np.r_[b, 0]) == -1)[0]
    return st, en

def detect(sst, thr, min_dur=5, max_gap=2):
    """causal - knowable on day t: SST above threshold for >= min_dur consecutive days up to t
       nojoin - Hobday label with the minimum-duration rule only
       label - full Hobday label: qualifying runs, then gaps <= max_gap days joined"""
    exc = (sst > thr).values
    n = len(exc)
    nojoin = np.zeros(n, int); causal = np.zeros(n, int)

    st, en = runs(exc)
    keep = [(s, e) for s, e in zip(st, en) if e - s + 1 >= min_dur]
    for s, e in keep:
        nojoin[s:e + 1] = 1
        causal[s + min_dur - 1:e + 1] = 1
    label = nojoin.copy()
    for (s1, e1), (s2, e2) in zip(keep[:-1], keep[1:]):
        if s2 - e1 - 1 <= max_gap: label[e1 + 1:s2] = 1
    f = lambda a: pd.Series(a, index=sst.index)
    return f(causal), f(nojoin), f(label)

# verification: rebuild the labels from SST alone (must match mhw_daily_labels.csv)
CHK = {}
for kind, s_col, t_col, l_col in (('raw', 'sst', 'thresh90', 'mhw_raw'),
                                  ('det', 'sst_detrended', 'thresh90_detrended',
    'mhw_detrended')):
    clim, thr = hobday_clim(df[s_col])
    c, nj, lab = detect(df[s_col], thr)
    CHK[kind] = dict(thr_diff=float((thr - df[t_col]).abs().max()), days=int(lab.sum()),
                     days_file=int(df[l_col].sum()), mismatched_days=int((lab !=
    df[l_col]).sum()))
print(pd.DataFrame(CHK).T)

LEADS = [1, 2, 3, 5, 7, 10, 14]
B = 2000

def mcc_from(tp, fp, fn, tn):
```

```
    tp, fp, fn, tn = (np.asarray(v, float) for v in (tp, fp, fn, tn))
    den = np.sqrt((tp + fp) * (tp + fn) * (tn + fp) * (tn + fn))
    return np.where(den > 0, (tp * tn - fp * fn) / np.where(den > 0, den, 1), 0.0)

def year_counts(pred, truth, years, yrs):
    g = pd.DataFrame({'tp': (pred == 1) & (truth == 1), 'fp': (pred == 1) & (truth == 0),
                      'fn': (pred == 0) & (truth == 1), 'tn': (pred == 0) & (truth == 0), 'year':
    years}).groupby('year').sum()
    return g.reindex(yrs, fill_value=0).values

def leakage_curve(sst, thr, region, kind):
    """Persistence forecasts for day t+h issued on day t, scored against the official Hobday label
    on t+h."""
    causal, nojoin, label = detect(sst, thr)
    yrs = sorted(sst.index.year.unique())
    idx = np.random.default_rng(0).integers(0, len(yrs), (B, len(yrs)))
    rows = []
    for h in LEADS:
        y = label.shift(-h).dropna().astype(int)
        years = y.index.year
        r = {'region': region, 'target': kind, 'lead': h}
        boot = {}
        for nm, s in (('label', label), ('nojoin', nojoin), ('causal', causal)):
            p = s.reindex(y.index).values
            c = year_counts(p, y.values, years, yrs)
            r[f'MCC_{nm}'] = float(mcc_from(*c.sum(0)))
            boot[nm] = mcc_from(*c[idx].sum(1).T)
        for a, b, nm in (('label', 'causal', 'inflation_total'), ('nojoin', 'causal',
    'from_min_duration'),
                         ('label', 'nojoin', 'from_gap_joining')):
            d = boot[a] - boot[b]
            r[nm] = r[f'MCC_{a}'] - r[f'MCC_{b}']
            r[f'{nm}_lo'], r[f'{nm}_hi'] = np.percentile(d, 2.5), np.percentile(d, 97.5)
        r['inflation_pct'] = 100 * r['inflation_total'] / r['MCC_causal'] if r['MCC_causal'] > 0
    else np.nan
        rows.append(r)
    # event statistics that explain the size of the leakage
    st, en = runs(label.values.astype(bool))
    dur = en - st + 1
    stj, enj = runs(nojoin.values.astype(bool))
    stats = {'region': region, 'target': kind, 'events': len(st), 'MHW_days': int(label.sum()),
             'median_duration': float(np.median(dur)), 'pct_events_<=10d': 100 * np.mean(dur <=
    10),
             'pct_events_joined_across_gap': 100 * (1 - len(st) / max(len(stj), 1)) if len(stj)
    else np.nan,
             'pct_MHW_days_known_only_in_hindsight': 100 * float(((label == 1) & (causal ==
    0)).sum() / label.sum())}
    return pd.DataFrame(rows), stats

curves, stats = [], []
for kind, s_col in (('det', 'sst_detrended'), ('raw', 'sst')):
    _, thr = hobday_clim(df[s_col])
    c, s = leakage_curve(df[s_col], thr, 'Bay of Bengal', kind)
    curves.append(c); stats.append(s)

CUR = pd.concat(curves); STA = pd.DataFrame(stats)
CUR.to_csv(f'{PROC}/wp1_leakage_curve_bob.csv', index=False);
    STA.to_csv(f'{PROC}/wp1_event_stats_bob.csv', index=False)
pd.set_option('display.width', 220)
show = ['target', 'lead', 'MCC_label', 'MCC_nojoin', 'MCC_causal', 'inflation_total',
    'inflation_total_lo',
        'inflation_total_hi', 'inflation_pct', 'from_min_duration', 'from_gap_joining']
print(CUR[show].round(3).to_string(index=False))
print(); print(STA.round(1).to_string(index=False))
```

C.5 Predictors

```
# P3-1 setup, fixed references, feature builder
from google.colab import drive; drive.mount('/content/drive')
import os, json, numpy as np, pandas as pd

PROC = '/content/drive/MyDrive/Colab Notebooks/MHW_Project_Data/processed'
df = pd.read_csv(f'{PROC}/predictor_table_1982_2025.csv', index_col=0, parse_dates=True)
H = 7                                                    # forecast lead (days)

def doy366(idx):
    d = idx.dayofyear.values.copy()
    d[(~idx.is_leap_year) & (idx.month > 2)] += 1
    return d

def hobday_clim(s, base=(1982, 2021), win=5, smooth=31):
    v, d = s.values, doy366(s.index)
    inb = (s.index.year >= base[0]) & (s.index.year <= base[1])
```

```
    clim = np.full(367, np.nan)
    for k in range(1, 367):
        if k == 60: continue
        idx = np.where((d == k) & inb)[0]
        pool = np.unique(np.clip((idx[:, None] + np.arange(-win, win + 1)).ravel(), 0, len(v) -
    1))
        clim[k] = np.nanmean(v[pool[inb[pool]]])
    clim[60] = (clim[59] + clim[61]) / 2
    c = clim[1:]
    c = np.convolve(np.r_[c[-(smooth // 2):], c, c[:smooth // 2]], np.ones(smooth) / smooth,
    'valid')
    return pd.Series(c[d - 1], index=s.index)

def doy_clim(s):                                          # fixed 1982-2021 day-of-year mean,
    31-day smoothed
    base = s[s.index.year <= 2021]
    c = base.groupby(doy366(base.index)).mean()
    c = pd.Series(np.convolve(np.r_[c.values[-15:], c.values, c.values[:15]], np.ones(31) / 31,
    'valid'), index=c.index)
    return pd.Series(c.reindex(doy366(s.index)).values, index=s.index)

# ---- fixed reference series (calendar functions of the 1982-2021 baseline; the same for every
    fold) ----
ERA = ['t2m', 'wspd', 'u10', 'v10', 'swnet', 'lhf', 'shf', 'qnet']
REF = pd.DataFrame(index=df.index)
REF['clim_raw'], REF['thr_raw'] = df.clim_mean, df.thresh90
REF['clim_det'] = hobday_clim(df.sst_detrended)
REF['thr_det'] = df.thresh90_detrended
for v in ERA: REF[f'clim_{v}'] = doy_clim(df[v])

def run_length(b):                                        # consecutive True days ending at t
    (causal)
    g = (~b).cumsum()
    return b.astype(int).groupby(g).cumsum()

def build_features(d, ref, kind):
    """kind = 'det' or 'raw'. Uses only rows <= t (rolling / shift / diff on the past) plus fixed
    references."""
    sst = d.sst_detrended if kind == 'det' else d.sst
    clim, thr = ref[f'clim_{kind}'].loc[d.index], ref[f'thr_{kind}'].loc[d.index]
    a, dist = sst - clim, sst - thr
    X = pd.DataFrame(index=d.index)
    # --- SST group ---
    X['sst_anom'], X['sst_dist_thr'] = a, dist
    for k in (1, 3, 7):        X[f'sst_anom_lag{k}'] = a.shift(k)
    for w in (3, 7, 14, 30):   X[f'sst_anom_rm{w}'] = a.rolling(w).mean()
    X['sst_anom_sd7'], X['sst_anom_max7'] = a.rolling(7).std(), a.rolling(7).max()
    for k in (1, 3, 7):        X[f'sst_anom_chg{k}'] = a - a.shift(k)
    exc = dist > 0
    X['exceed_now'] = exc.astype(int)
    X['exceed_run'] = run_length(exc)
    X['mhw_causal'] = (X.exceed_run >= 5).astype(int)        # MHW state knowable on day t

    # --- ERA5 group (anomalies from the fixed 1982-2021 climatology) ---
    for v in ERA:
        x = d[v] - ref[f'clim_{v}'].loc[d.index]
        X[f'{v}_anom'] = x
        for w in (3, 7, 14, 30): X[f'{v}_rm{w}'] = x.rolling(w).mean()
        for k in (3, 7):          X[f'{v}_chg{k}'] = x - x.shift(k)
    # --- climate-index group ---
    for v in ('nino34', 'dmi', 'iobw_dt', 'rmm1', 'rmm2'): X[v] = d[v]
    X['dmi_rm30'], X['iobw_dt_rm30'] = d.dmi.rolling(30).mean(), d.iobw_dt.rolling(30).mean()
    X['rmm_amp'] = np.hypot(d.rmm1, d.rmm2)
    # --- calendar group ---
    doy = d.index.dayofyear
    X['doy_sin'], X['doy_cos'] = np.sin(2 * np.pi * doy / 365.25), np.cos(2 * np.pi * doy /
    365.25)
    return X

GROUPS = lambda cols: {
    'sst':      [c for c in cols if c.startswith(('sst_', 'exceed_', 'mhw_causal'))],
    'era5':     [c for c in cols if c.split('_')[0] in ERA],
    'indices': [c for c in cols if c.startswith(('nino34', 'dmi', 'iobw', 'rmm'))],
    'calendar': ['doy_sin', 'doy_cos']}

# P3-2 targets, tables, saving
TGT = {'det': 'mhw_detrended', 'raw': 'mhw_raw'}
tables = {}
for kind, lab in TGT.items():
    X = build_features(df, REF, kind)
    y = df[lab].shift(-H).rename('y')                     # label on day t+7
    T = pd.concat([X, y], axis=1)
    T['year'] = T.index.year                              # year of the forecast day t (for
    fold assignment)
    T['target_year'] = (T.index + pd.Timedelta(days=H)).year # year the target falls in (for
    purging)
    T = T.dropna()
```

```
    T['y'] = T.y.astype(int)
    tables[kind] = T
    T.to_csv(f'{PROC}/features_{kind}_t+{H}.csv')
    print(f'{kind}: {T.shape[0]} rows x {X.shape[1]} features | {T.index[0].date()} -> {T.index[-
    1].date()} '
          f'| MHW share {T.y.mean():.1%}')
groups = GROUPS(list(X.columns))
json.dump(groups, open(f'{PROC}/feature_groups.json', 'w'), indent=1)
print({g: len(c) for g, c in groups.items()}, '| total', sum(len(c) for c in groups.values()))
assert sum(len(c) for c in groups.values()) == X.shape[1], 'a feature is missing from the groups'
```

C.6 Evaluation framework

```
from google.colab import drive; drive.mount('/content/drive')
import os, json, time, numpy as np, pandas as pd
from sklearn.pipeline import make_pipeline
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.isotonic import IsotonicRegression
from sklearn.metrics import average_precision_score

PROC = '/content/drive/MyDrive/Colab Notebooks/MHW_Project_Data/processed'
OOF = f'{PROC}/oof'; os.makedirs(OOF, exist_ok=True)
H = 7
groups = json.load(open(f'{PROC}/feature_groups.json'))
T = {k: pd.read_csv(f'{PROC}/features_{k}_t+{H}.csv', index_col=0, parse_dates=True) for k in
    ('det', 'raw')}
YEARS = sorted(T['det'].year.unique())

def train_mask(tab, test_years):
    """Drop test years, rows whose t+7 target is in a test year (purge), first 30 days after a
    test year (embargo)."""
    m = ~tab.year.isin(test_years) & ~tab.target_year.isin(test_years)
    after = tab.year.isin([y + 1 for y in test_years]) & (tab.index.dayofyear <= 30)
    return m & ~after

def mcc_from(tp, fp, fn, tn):
    tp, fp, fn, tn = (np.asarray(v, dtype=float) for v in (tp, fp, fn, tn))
    den = np.sqrt((tp + fp) * (tp + fn) * (tn + fp) * (tn + fn))
    return np.where(den > 0, (tp * tn - fp * fn) / np.where(den > 0, den, 1), 0.0)

def best_threshold(y, p):
    th = np.unique(np.r_[np.arange(0.02, 0.99, 0.01), p])       # includes the calibrated step
    values
    th = th[(th > 0) & (th < 1)]
    P = p[None, :] >= th[:, None]

    tp = (P & (y == 1)).sum(1); fp = (P & (y == 0)).sum(1)
    fn = ((~P) & (y == 1)).sum(1); tn = ((~P) & (y == 0)).sum(1)
    m = mcc_from(tp, fp, fn, tn)
    return th[np.argmax(m)], m.max()

def loyo(kind, grid, cols, name, inner_folds=4, verbose=True):
    """Leave-one-year-out. Inside each training fold only:
    1) inner year-grouped CV picks the hyper-parameters (max MCC),
    2) isotonic calibration is fitted on the inner out-of-fold probabilities,
    3) the threshold is chosen on the calibrated out-of-fold probabilities.
    grid = list of (label, model_factory)."""
    tab, rows, t0 = T[kind], [], time.time()
    for y in YEARS:
        tr, te = tab[train_mask(tab, [y])], tab[tab.year == y]
        chunks = np.array_split(np.array(sorted(tr.year.unique())), inner_folds)
        best = None
        for lab, make in grid:
            oof = pd.Series(np.nan, index=tr.index)
            for ch in chunks:
                itr, iva = tr[train_mask(tr, list(ch))], tr[tr.year.isin(ch)]
                oof.loc[iva.index] = make().fit(itr[cols], itr.y).predict_proba(iva[cols])[:, 1]
            ok = oof.notna()
            score = best_threshold(tr.y[ok].values, oof[ok].values)[1]
            if best is None or score > best[0]: best = (score, lab, make, oof[ok], tr.y[ok])
        _, lab, make, oof, yo = best
        iso = IsotonicRegression(out_of_bounds='clip', y_min=0, y_max=1).fit(oof.values,
    yo.values)
        th, _ = best_threshold(yo.values, iso.predict(oof.values))
        p = iso.predict(make().fit(tr[cols], tr.y).predict_proba(te[cols])[:, 1])
        rows.append(pd.DataFrame({'year': y, 'y': te.y.values, 'prob': p, 'pred': (p >=
    th).astype(int),
                                  'thr': th, 'params': lab}, index=te.index))
    out = pd.concat(rows); out.to_csv(f'{OOF}/{name}_{kind}.csv')
    if verbose: print(f'{name} [{kind}] done in {time.time() - t0:.0f}s | params chosen: '
                      f'{out.groupby("year").params.first().value_counts().to_dict()}')
    return out
```

```
def persistence(kind):
    """Honest persistence: MHW state knowable on day t (SST above threshold >= 5 consecutive
    days)."""
    tab = T[kind]; p = tab.mhw_causal.astype(float)
    out = pd.DataFrame({'year': tab.year, 'y': tab.y, 'prob': p, 'pred': p.astype(int), 'thr':
    0.5}, index=tab.index)
    out.to_csv(f'{OOF}/persistence_{kind}.csv'); return out

def climatology(kind):
    """Brier-score reference only: P(MHW) by calendar month from the training years; no yes/no
    forecast."""
    tab, rows = T[kind], []
    for y in YEARS:
        tr, te = tab[train_mask(tab, [y])], tab[tab.year == y]
        pm = tr.y.groupby(tr.index.month).mean()
        rows.append(pd.DataFrame({'year': y, 'y': te.y.values, 'prob':
    pm.reindex(te.index.month).values,
                                  'pred': np.nan, 'thr': np.nan}, index=te.index))
    out = pd.concat(rows); out.to_csv(f'{OOF}/climatology_{kind}.csv'); return out

def onset_rows(o):
    """Forecast days whose target day t+7 is the first day of an MHW event."""
    prev = o.y.shift(1); consecutive = (o.index.to_series().diff().dt.days == 1).values
    return (o.y == 1) & (prev == 0) & consecutive

def metrics(o, clim):
    y, p = o.y.values, o.prob.values
    r = {'PR_AUC': average_precision_score(y, p) if y.any() else np.nan,
         'BSS': 1 - np.mean((p - y) ** 2) / np.mean((clim.prob.reindex(o.index).values - y) ** 2)}
    if o.pred.notna().all():
        pr = o.pred.values.astype(int)
        tp = int(((pr == 1) & (y == 1)).sum()); fp = int(((pr == 1) & (y == 0)).sum())
        fn = int(((pr == 0) & (y == 1)).sum()); tn = int(((pr == 0) & (y == 0)).sum())
        on = onset_rows(o)
        r.update(MCC=float(mcc_from(tp, fp, fn, tn)), F1=2 * tp / max(2 * tp + fp + fn, 1),
                 CSI=tp / max(tp + fp + fn, 1), onset_hit=float(o.pred[on].mean()),
    n_onsets=int(on.sum()),
                 TP=tp, FP=fp, FN=fn, TN=tn)
    return r

def year_counts(o):
    g = o.assign(tp=(o.pred == 1) & (o.y == 1), fp=(o.pred == 1) & (o.y == 0),
                 fn=(o.pred == 0) & (o.y == 1), tn=(o.pred == 0) & (o.y == 0)).groupby('year')
    return g[['tp', 'fp', 'fn', 'tn']].sum().reindex(YEARS).values

def bootstrap(o, refs, B=2000, seed=0):
    """Resample whole years (same resamples for every model): MCC CI and paired MCC-gain CI vs
    each reference."""
    idx = np.random.default_rng(seed).integers(0, len(YEARS), (B, len(YEARS)))
    m = mcc_from(*year_counts(o)[idx].sum(1).T)
    out = {'MCC_lo': np.percentile(m, 2.5), 'MCC_hi': np.percentile(m, 97.5)}
    for rn, ro in refs.items():
        d = m - mcc_from(*year_counts(ro)[idx].sum(1).T)
        out[f'gain_vs_{rn}_lo'], out[f'gain_vs_{rn}_hi'] = np.percentile(d, 2.5), np.percentile(d,
    97.5)
    return out

def report(results, refs=('persistence', 'logit_sst'), years=None):
    """refs: honest persistence (plan rule) and the SST-only model (research question)."""
    rows = {}
    clim = results['climatology'] if years is None else results['climatology']
    [results['climatology'].year.isin(years)]
    for name, o in results.items():
        oo = o if years is None else o[o.year.isin(years)]
        r = metrics(oo, clim)
        if years is None and name != 'climatology':
            r.update(bootstrap(oo, {rn: results[rn] for rn in refs if rn in results and rn !=
    name}))
        rows[name] = r
    return pd.DataFrame(rows).T

OOF = f'{PROC}/oof_p5'; os.makedirs(OOF, exist_ok=True)

def best_threshold(y, p):
    """Threshold with the highest MCC. Candidates: a 0.01 grid plus the distinct calibrated values
    (or 300 upper quantiles when the probabilities are continuous)."""
    u = np.unique(p)
    extra = u if len(u) <= 500 else np.quantile(p, np.linspace(0.5, 1, 301))
    th = np.unique(np.r_[np.arange(0.01, 1.0, 0.01), extra])
    th = th[(th > 0) & (th < 1)]
    P = p[None, :] >= th[:, None]
    tp = (P & (y == 1)).sum(1); fp = (P & (y == 0)).sum(1)
    fn = ((~P) & (y == 1)).sum(1); tn = ((~P) & (y == 0)).sum(1)
    m = mcc_from(tp, fp, fn, tn)
    return th[np.argmax(m)], m.max()
```

C.7 Models, skill by lead and exceedance persistence

```
# P5-1
import os, numpy as np, pandas as pd
from sklearn.ensemble import RandomForestClassifier
from xgboost import XGBClassifier

class XGBW:
    """XGBoost with class weighting computed from each training set (scale_pos_weight =
    negatives / positives)."""
    def __init__(self, **kw): self.kw = kw
    def fit(self, X, y):
        spw = float((y == 0).sum()) / max(float((y == 1).sum()), 1.0)
        self.m = XGBClassifier(scale_pos_weight=spw, subsample=0.8, colsample_bytree=0.8,
    min_child_weight=5,
                               tree_method='hist', n_jobs=-1, random_state=0,
    eval_metric='logloss', **self.kw)
        self.m.fit(X, y); return self
    def predict_proba(self, X): return self.m.predict_proba(X)

logit = lambda C: (lambda: make_pipeline(StandardScaler(),
                           LogisticRegression(C=C, class_weight='balanced', max_iter=3000)))
rf = lambda leaf: (lambda: RandomForestClassifier(n_estimators=150, max_features='sqrt',
    min_samples_leaf=leaf, max_samples=0.5,
                                                  class_weight='balanced_subsample', n_jobs=-1,
    random_state=0))
xgb = lambda d: (lambda: XGBW(max_depth=d, learning_rate=0.05, n_estimators=300))
GRIDS = {'logit': [(f'C={C}', logit(C)) for C in (0.01, 0.1, 1.0)],
         'rf':    [(f'leaf={l}', rf(l)) for l in (10, 50)],
         'xgb':   [(f'depth={d}', xgb(d)) for d in (3, 5)]}
FSETS = {'sst': groups['sst'] + groups['calendar'],
         'era5': groups['sst'] + groups['calendar'] + groups['era5'],
         'all': groups['sst'] + groups['calendar'] + groups['era5'] + groups['indices']}

def run(kind, model, fset):
    """Resumable: a finished run is read back from the OOF folder instead of being recomputed."""
    name = f'{model}_{fset}'
    f = f'{OOF}/{name}_{kind}.csv'

    if os.path.exists(f):
        return pd.read_csv(f, index_col=0, parse_dates=True)
    return loyo(kind, GRIDS[model], FSETS[fset], name, inner_folds=4 if model == 'logit' else 3)

print('P5-1 definitions loaded')

# WP2-1 skill at leads 1, 3, 5, 10, 14 days
import os, inspect, numpy as np, pandas as pd, matplotlib.pyplot as plt
FIG = f'{PROC}/figures'; os.makedirs(FIG, exist_ok=True)

LEADS2 = [1, 3, 5, 10, 14]
MODELS = [('logit', 'sst'), ('logit', 'era5'), ('rf', 'sst'), ('rf', 'era5')]
feat = pd.read_csv(f'{PROC}/features_det_t+7.csv', index_col=0,
    parse_dates=True).drop(columns=['y', 'target_year'])
lab_det = pd.read_csv(f'{PROC}/predictor_table_1982_2025.csv', index_col=0,
    parse_dates=True).mhw_detrended

def table_for(h):
    """Same features on day t (built only from days <= t); target = detrended MHW label on day
    t+h."""
    y = lab_det.shift(-h).reindex(feat.index)
    tb = feat.assign(y=y, target_year=(feat.index +
    pd.Timedelta(days=h)).year).dropna(subset=['y'])
    tb['y'] = tb.y.astype(int)
    return tb

RES = {}
for h in LEADS2:
    OOF = f'{PROC}/oof_lead{h}'; os.makedirs(OOF, exist_ok=True)      #
    loyo/persistence/climatology write here
    T = {'det': table_for(h)}
    print(f'--- lead {h} d: {len(T["det"])} rows, MHW share {T["det"].y.mean():.1%}')
    R = {'persistence': persistence('det'), 'climatology': climatology('det')}
    for m, s in MODELS:
         R[f'{m}_{s}'] = run('det', m, s)
    RES[h] = R

# lead 7: runs of the same code, read back from file
RES[7] = {n: pd.read_csv(f'{PROC}/oof_p5/{n}_det.csv', index_col=0, parse_dates=True)
          for n in ['persistence', 'climatology'] + [f'{m}_{s}' for m, s in MODELS]}

# WP2-2 / WP3 skill-by-lead table: occurrence skill and new-event skill
LALL = sorted(RES)
idx = np.random.default_rng(0).integers(0, len(YEARS), (2000, len(YEARS)))
ci = lambda d: f'{np.percentile(d, 2.5):+.3f} to {np.percentile(d, 97.5):+.3f}'
quiet_all = feat.mhw_causal == 0                                  # no MHW knowable on day t
rows = []
```

```
for h in LALL:
    R = RES[h]
    names = [n for n in R if n != 'climatology']
    bm = {n: mcc_from(*year_counts(R[n])[idx].sum(1).T) for n in names}
    sub = {n: R[n][quiet_all.reindex(R[n].index).fillna(False).values] for n in names if n !=
    'persistence'}
    bq = {n: mcc_from(*year_counts(o)[idx].sum(1).T) for n, o in sub.items()}
    for n in names:
        r = metrics(R[n], R['climatology'])
        row = {'lead': h, 'model': n, 'MCC': round(r['MCC'], 3),
               'MCC 95% CI': f'{np.percentile(bm[n], 2.5):.3f} to {np.percentile(bm[n],
    97.5):.3f}',
               'MCC_lo': np.percentile(bm[n], 2.5), 'MCC_hi': np.percentile(bm[n], 97.5),
               'BSS': round(r['BSS'], 3), 'onset hit': round(r['onset_hit'], 3)}
        if n != 'persistence':
            m, s = n.split('_')
            o = sub[n]; y, p = o.y.values, o.pred.values.astype(int)
            row['gain vs persistence'] = ci(bm[n] - bm['persistence'])
            row['new-event MCC'] = round(float(mcc_from(((p == 1) & (y == 1)).sum(), ((p == 1) &
    (y == 0)).sum(),
                                                        ((p == 0) & (y == 1)).sum(), ((p == 0) &
    (y == 0)).sum())), 3)
            if s == 'era5':
                row['ERA5 gain (MCC)'] = ci(bm[n] - bm[f'{m}_sst'])
                row['ERA5 gain (new-event MCC)'] = ci(bq[n] - bq[f'{m}_sst'])
        rows.append(row)
SK = pd.DataFrame(rows); SK.to_csv(f'{PROC}/wp2_skill_by_lead_det.csv', index=False)
pd.set_option('display.width', 250); pd.set_option('display.max_columns', 20)
for h in LALL:
    print(f'\n===== lead {h} d =====')
    print(SK[SK.lead == h].drop(columns=['lead', 'MCC_lo',
    'MCC_hi']).set_index('model').fillna('').to_string())

# WP2-4 diagnostic: exceedance persistence (SST above threshold on day t)
pt = pd.read_csv(f'{PROC}/predictor_table_1982_2025.csv', index_col=0, parse_dates=True)

exc = (pt.sst_detrended > pt.thresh90_detrended).astype(int)
bmcc = lambda o: mcc_from(*year_counts(o)[idx].sum(1).T)
pmcc = lambda o: float(mcc_from(*year_counts(o).sum(0)))
rows = []
for h in LALL:
    R = RES[h]
    E = R['persistence'].copy(); E['pred'] = exc.reindex(E.index).values.astype(int)
    q = quiet_all.reindex(E.index).fillna(False).values
    bE, bEq = bmcc(E), bmcc(E[q])
    row = {'lead': h, 'honest persistence': round(pmcc(R['persistence']), 3),
           'exceedance persistence': round(pmcc(E), 3),
           'exc. pers. 95% CI': f'{np.percentile(bE, 2.5):.3f} to {np.percentile(bE, 97.5):.3f}',
           'exc. pers. new-event MCC': round(pmcc(E[q]), 3)}
    for n in ('logit_sst', 'rf_era5'):
        row[f'{n} gain vs exc. pers.'] = ci(bmcc(R[n]) - bE)
        row[f'{n} new-event gain vs exc. pers.'] = ci(bmcc(R[n][q]) - bEq)
    rows.append(row)
EX = pd.DataFrame(rows); EX.to_csv(f'{PROC}/wp2_exceedance_persistence_det.csv', index=False)
print(EX.set_index('lead').to_string())
```

C.8 Coefficients and drop-one-group test

```
# WP5-1 what the SST-only model uses: standardized logistic coefficients by lead
from sklearn.pipeline import make_pipeline
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
COLS = FSETS['sst']
LEADS_ALL = [1, 3, 5, 7, 10, 14]
rng = np.random.default_rng(0)
coef, sig = {}, {}
for h in LEADS_ALL:
    tb = table_for(h).dropna(subset=COLS)
    X, y, yr = tb[COLS].values, tb.y.values, tb.index.year.values
    fit = lambda i: make_pipeline(StandardScaler(), LogisticRegression(C=0.1,
    class_weight='balanced', max_iter=3000)
                                  ).fit(X[i], y[i])[-1].coef_[0]
    pos = {u: np.where(yr == u)[0] for u in np.unique(yr)}
    ys = list(pos)
    coef[h] = fit(np.arange(len(y)))
    boot = np.array([fit(np.concatenate([pos[u] for u in rng.choice(ys, len(ys))])) for _ in
    range(200)])
    lo, hi = np.percentile(boot, [2.5, 97.5], axis=0)
    sig[h] = (lo > 0) | (hi < 0)
    print(f'lead {h} d done')

CO = pd.DataFrame(coef, index=COLS); SG = pd.DataFrame(sig, index=COLS)
CO.to_csv(f'{PROC}/wp5_logit_sst_coefficients.csv');
```

```
    SG.to_csv(f'{PROC}/wp5_logit_sst_coef_significant.csv')
show = CO.round(2).astype(str) + np.where(SG, '*', ' ')
print('\nStandardized coefficients (* = 95% bootstrap CI excludes 0), sorted by |coef| at 7 d')
print(show.loc[CO[7].abs().sort_values(ascending=False).index].to_string())

# WP5-1b which time scale carries the skill: drop-one-group test (logit_sst, same honest LOYO)
GRP = {'current state': ['sst_anom', 'sst_dist_thr', 'sst_anom_lag1', 'sst_anom_lag3',
    'sst_anom_rm3',
                         'exceed_now', 'exceed_run', 'mhw_causal'],
       'tendency':      ['sst_anom_chg1', 'sst_anom_chg3', 'sst_anom_chg7'],
       'background':    ['sst_anom_lag7', 'sst_anom_rm7', 'sst_anom_rm14', 'sst_anom_rm30',
    'sst_anom_max7', 'sst_anom_sd7'],
       'season':        ['doy_sin', 'doy_cos']}
assert sorted(sum(GRP.values(), [])) == sorted(FSETS['sst']), 'groups must cover the SST-model
    features exactly'

DROP = {}
for h in LALL:
    OOF = f'{PROC}/oof_lead{h}'; os.makedirs(OOF, exist_ok=True)
    T = {'det': table_for(h)}
    for g, gcols in GRP.items():
        name = f'logit_sst_no_{g.replace(" ", "_")}'
        f = f'{OOF}/{name}_det.csv'
        DROP[(h, g)] = (pd.read_csv(f, index_col=0, parse_dates=True) if os.path.exists(f) else
                        loyo('det', GRIDS['logit'], [c for c in FSETS['sst'] if c not in gcols],
    name, inner_folds=4))

bmcc = lambda o: mcc_from(*year_counts(o)[idx].sum(1).T)
pmcc = lambda o: float(mcc_from(*year_counts(o).sum(0)))
qmask = lambda o: quiet_all.reindex(o.index).fillna(False).values
rows = []
for h in LALL:
    full = RES[h]['logit_sst']

    bF, bFq = bmcc(full), bmcc(full[qmask(full)])
    for g in GRP:
        d = DROP[(h, g)]
        loss, lossq = bF - bmcc(d), bFq - bmcc(d[qmask(d)])
        rows.append({'lead': h, 'group dropped': g, 'MCC full': round(pmcc(full), 3), 'MCC
    without': round(pmcc(d), 3),
                     'MCC loss 95% CI': ci(loss), 'sig': '*' if np.percentile(loss, 2.5) > 0 else
    '',
                     'new-event MCC loss 95% CI': ci(lossq), 'sig new-event': '*' if
    np.percentile(lossq, 2.5) > 0 else ''})
DG = pd.DataFrame(rows); DG.to_csv(f'{PROC}/wp5_drop_group_det.csv', index=False)
for g in GRP:
    print(f'\n===== dropped: {g} =====')
    print(DG[DG['group dropped'] == g].drop(columns='group
    dropped').set_index('lead').to_string())
```

C.9 Grid-cell analysis

```
# WP1c-1 gridded leakage: setup, detection on every ocean cell
import os, glob, numpy as np, pandas as pd, xarray as xr, matplotlib.pyplot as plt
PROC = os.environ.get('PROC', '/content/drive/MyDrive/Colab Notebooks/MHW_Project_Data/processed')
ROOT = os.environ.get('ROOT', '/content/drive/MyDrive')
FIG = f'{PROC}/figures'; os.makedirs(FIG, exist_ok=True)
LEADS = [1, 2, 3, 5, 7, 10, 14]
B = 2000

hits = glob.glob(f'{ROOT}/**/oisst_box_grid.nc', recursive=True); print(hits)
ds = xr.open_dataset(hits[0])
ds = ds.rename({k: v for k, v in {'latitude': 'lat', 'longitude': 'lon'}.items() if k in
    ds.dims})
sst = (ds['sst'] if 'sst' in ds else ds[list(ds.data_vars)[0]]).squeeze(drop=True)
sst = sst.sel(time=slice('1982-01-01', '2025-12-31')).sortby('lat').load()
A   = sst.values.astype('float64')                                    # time x lat x lon
T   = pd.DatetimeIndex(sst.time.values)
ocean = np.isfinite(A).all(0)
X   = A[:, ocean]                                                     # time x cells
LAT, LON = np.meshgrid(sst.lat.values, sst.lon.values, indexing='ij')
clat, clon = LAT[ocean], LON[ocean]
tc = np.arange(len(T)) - (len(T) - 1) / 2
slope = (X * tc[:, None]).sum(0) / (tc ** 2).sum()                    # per-cell linear trend (per
    day)
Xd = X - slope[None, :] * tc[:, None]                                 # detrended, same form as
the
    box mean
print(f'{X.shape[0]} days x {X.shape[1]} ocean cells | cell trends {np.percentile(slope*3652.5,
    10):.2f} to '
      f'{np.percentile(slope*3652.5, 90):.2f} degC/decade (10th-90th pct)')

def doy366(idx):
```

```
    d = idx.dayofyear.values.copy(); d[(~idx.is_leap_year) & (idx.month > 2)] += 1
    return d

def hobday_thr(M, T, base=(1982, 2021), win=5, smooth=31, pct=90):
    """Hobday et al. (2016) 90th-percentile threshold for every column of M (time x cells), as in
    WP1."""
    d, inb = doy366(T), (T.year >= base[0]) & (T.year <= base[1])
    c = np.full((367, M.shape[1]), np.nan)
    for k in range(1, 367):
        if k == 60: continue
        idx = np.where((d == k) & inb)[0]
        pool = np.unique(np.clip((idx[:, None] + np.arange(-win, win + 1)).ravel(), 0, len(T) -
    1))
        c[k] = np.nanpercentile(M[pool[inb[pool]]], pct, axis=0)
    c[60] = (c[59] + c[61]) / 2
    c = c[1:]; h = smooth // 2
    c = np.concatenate([c[-h:], c, c[:h]])
    cs = np.cumsum(np.vstack([np.zeros((1, c.shape[1])), c]), axis=0)
    c = (cs[smooth:] - cs[:-smooth]) / smooth                          # 31-day running mean,
    wrapped
    return c[d - 1]

def detect_cols(E, min_dur=5, max_gap=2):
    """Per column: causal (knowable on day t), nojoin (min-duration only), label (full Hobday)."""
    causal = np.zeros(E.shape, np.int8); nojoin = np.zeros(E.shape, np.int8); label =
    np.zeros(E.shape, np.int8)
    for j in range(E.shape[1]):
        b = np.diff(np.r_[0, E[:, j].astype(np.int8), 0])
        st, en = np.where(b == 1)[0], np.where(b == -1)[0] - 1
        keep = en - st + 1 >= min_dur; st, en = st[keep], en[keep]
        for s, e in zip(st, en):
            nojoin[s:e + 1, j] = 1; causal[s + min_dur - 1:e + 1, j] = 1

        label[:, j] = nojoin[:, j]
        for k in range(len(st) - 1):
            if st[k + 1] - en[k] - 1 <= max_gap: label[en[k] + 1:st[k + 1], j] = 1
    return causal, nojoin, label

# verification: the grid code applied to the box-mean series must reproduce the label file
df = pd.read_csv(f'{PROC}/predictor_table_1982_2025.csv', index_col=0, parse_dates=True)
for kind, s_col, t_col, l_col in (('raw', 'sst', 'thresh90', 'mhw_raw'), ('det', 'sst_detrended',
    'thresh90_detrended', 'mhw_detrended')):
    th = hobday_thr(df[[s_col]].values, df.index)
    _, _, lab = detect_cols(df[[s_col]].values > th)
    print(f'check {kind}: max threshold diff {np.abs(th[:, 0] - df[t_col].values).max():.4f} degC
    | '
          f'mismatched label days {int((lab[:, 0] != df[l_col].values).sum())}')

LAB = {}
for kind, M in (('det', Xd), ('raw', X)):
    th = hobday_thr(M, T)
    exc = M > th
    c, nj, lab = detect_cols(exc)
    LAB[kind] = dict(exc=exc.astype(np.int8), causal=c, nojoin=nj, label=lab)
    print(f'{kind}: median MHW days per cell {int(np.median(lab.sum(0)))}, '
          f'median hindsight share {np.median(((lab == 1) & (c == 0)).sum(0) / lab.sum(0)):.1%}')

# WP1c-2 leakage per cell, by lead, with year-block bootstrap
def mcc(tp, fp, fn, tn):
    tp, fp, fn, tn = (np.asarray(v, float) for v in (tp, fp, fn, tn))
    den = np.sqrt((tp + fp) * (tp + fn) * (tn + fp) * (tn + fn))
    return np.where(den > 0, (tp * tn - fp * fn) / np.where(den > 0, den, 1), 0.0)

yrs = np.unique(T.year)
rng = np.random.default_rng(0); idx = rng.integers(0, len(yrs), (B, len(yrs)))

def year_counts(p, y, tyear):
    """Contingency counts per target year and cell -> array (4, years, cells)."""
    starts = np.r_[0, np.where(np.diff(tyear) != 0)[0] + 1]
    out = []
    for a in ((p == 1) & (y == 1), (p == 1) & (y == 0), (p == 0) & (y == 1), (p == 0) & (y == 0)):
        out.append(np.add.reduceat(a.astype(np.int32), starts, axis=0))
    return np.stack(out)

def boot_mcc(c, chunk=250):
    """Year-block bootstrap MCC per cell, in chunks to keep memory low."""
    return np.vstack([mcc(*c[:, idx[k:k + chunk]].sum(2)) for k in range(0, B, chunk)])

rows, CELL = [], {}
box = pd.read_csv(f'{PROC}/wp1_leakage_curve_bob.csv')
for kind in ('det', 'raw'):
    L = LAB[kind]
    hind = ((L['label'] == 1) & (L['causal'] == 0)).sum(0) / L['label'].sum(0)
    for h in LEADS:
        y = L['label'][h:]; tyear = T.year.values[h:]
        cnt = {n: year_counts(L[n][:-h], y, tyear) for n in ('label', 'nojoin', 'causal', 'exc')}
        pt = {n: mcc(*c.sum(1)) for n, c in cnt.items()}                    # point MCC per
```

```
    cell
        bt = {n: boot_mcc(c) for n, c in cnt.items() if n in ('label', 'causal')}                #
    B x cells
        d   = bt['label'] - bt['causal']
        lo, hi = np.percentile(d, 2.5, 0), np.percentile(d, 97.5, 0)
        infl = pt['label'] - pt['causal']
        CELL[(kind, h)] = dict(infl=infl, lo=lo, hi=hi, gap_exc=pt['exc'] - pt['causal'],
    hind=hind,
                               mcc_label=pt['label'], mcc_causal=pt['causal'])
        bm = box[(box.target == kind) & (box.lead == h)].inflation_total.values[0]
        rows.append({'labels': kind, 'lead': h, 'box-mean inflation': round(bm, 3),
                     'cell inflation median': round(float(np.median(infl)), 3),
                     'cell inflation 10th-90th pct': f'{np.percentile(infl, 10):.3f} to
    {np.percentile(infl, 90):.3f}',
                     'cells with CI > 0 (%)': round(100 * float(np.mean(lo > 0)), 1),
                     'cells where cell > box (%)': round(100 * float(np.mean(infl > bm)), 1),
                     'share from 5-day minimum (median)': round(float(np.median((pt['nojoin'] -
    pt['causal']) / np.where(infl != 0, infl, np.nan))), 2),
                     'exceedance minus honest (median)': round(float(np.median(pt['exc'] -
    pt['causal'])), 3)})
GR = pd.DataFrame(rows); GR.to_csv(f'{PROC}/wp1c_grid_leakage_summary.csv', index=False)
pd.DataFrame({'lat': clat, 'lon': clon, **{f'{k}_{h}_{v}': CELL[(k, h)][v] for (k, h) in CELL for
    v in ('infl', 'lo', 'hi', 'gap_exc')},
              'hind_det': CELL[('det', 1)]['hind'], 'hind_raw': CELL[('raw', 1)]
    ['hind']}).to_csv(f'{PROC}/wp1c_grid_leakage_cells.csv', index=False)
pd.set_option('display.width', 250); pd.set_option('display.max_columns', 20)
print(GR.to_string(index=False))

for kind in ('det', 'raw'):
    c1 = CELL[(kind, 1)]
    r = pd.Series(c1['hind']).corr(pd.Series(c1['infl']), method='spearman')
    print(f'{kind}: Spearman r across cells, hindsight share vs 1-day inflation = {r:.2f}
    (descriptive; cells are spatially correlated)')
print('RULE (detrended): grid-scale leakage holds if >= 90% of cells have CI > 0 at every lead 1-7
    d ->',
      'MET' if all(GR[(GR.labels == 'det') & (GR.lead <= 7)]['cells with CI > 0 (%)'] >= 90) else
    'NOT MET')
```

C.10 Second SST dataset

```
# V1-0 Copernicus Marine login + list the datasets in the CCI/C3S product
!pip -q install copernicusmarine
import copernicusmarine
copernicusmarine.login()
info = copernicusmarine.describe(product_id='SST_GLO_SST_L4_REP_OBSERVATIONS_010_024')
IDS = [d.dataset_id for p in info.products for d in p.datasets]
print(IDS)
import xarray as xr; print('zarr' in xr.backends.list_engines())

# V1-1 CCI/C3S daily SST -> study-box mean, 1982-2024
import os, numpy as np, pandas as pd, xarray as xr
PROC = '/content/drive/MyDrive/Colab Notebooks/MHW_Project_Data/processed'
OUTD = f'{PROC}/cci_sst'; os.makedirs(OUTD, exist_ok=True)
TMP = '/content/cci_tmp'; os.makedirs(TMP, exist_ok=True)
for yr in range(1982, 2025):
    out = f'{OUTD}/cci_box_{yr}.csv'
    if os.path.exists(out): continue
    f = None
    for ds_id in IDS:
        try:
            copernicusmarine.subset(dataset_id=ds_id, variables=['analysed_sst'],
                                    minimum_longitude=85, maximum_longitude=93,
    minimum_latitude=15, maximum_latitude=23,
                                    start_datetime=f'{yr}-01-01T00:00:00', end_datetime=f'{yr}-
12-
    31T23:59:59',
                                    output_directory=TMP, output_filename=f'cci_{yr}.nc',
    overwrite=True)
            f = f'{TMP}/cci_{yr}.nc'; break
        except Exception as e:
            print(yr, ds_id, 'not available:', str(e)[:100])
    if f is None: raise RuntimeError(f'{yr}: no dataset in {IDS} covers this year')
    with xr.open_dataset(f) as ds:
        v = ds['analysed_sst'].squeeze(drop=True).load()
    if float(v.mean()) > 200: v = v - 273.15                        # Kelvin -> degC
    lat = 'latitude' if 'latitude' in v.dims else 'lat'; lon = 'longitude' if 'longitude' in
    v.dims else 'lon'
    ocean = np.isfinite(v).all('time')
    w = np.cos(np.deg2rad(v[lat])) * ocean
    s = ((v.where(ocean) * w).sum((lat, lon)) / w.sum()).to_series()
    s.index = pd.to_datetime(s.index).normalize()
    pd.DataFrame({'sst': s, 'ocean_cells': int(ocean.sum())}).to_csv(out)
    os.remove(f); print(yr, len(s), 'days |', int(ocean.sum()), 'ocean cells')
```

```
# V2-1 OISST vs CCI/C3S: same processing on both, 1982-2024
import os, glob, time, numpy as np, pandas as pd, matplotlib.pyplot as plt, matplotlib.dates as
    mdates
from sklearn.pipeline import make_pipeline
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.isotonic import IsotonicRegression
PROC = os.environ.get('PROC', '/content/drive/MyDrive/Colab Notebooks/MHW_Project_Data/processed')
FIG = f'{PROC}/figures'; os.makedirs(FIG, exist_ok=True)
END = '2024-12-31'
df   = pd.read_csv(f'{PROC}/predictor_table_1982_2025.csv', index_col=0, parse_dates=True)
cci = pd.concat([pd.read_csv(f, index_col=0, parse_dates=True) for f in
    sorted(glob.glob(f'{PROC}/cci_sst/cci_box_*.csv'))]).sst
S    = pd.DataFrame({'OISST': df.sst[:END], 'CCI': cci}).dropna()
assert len(S) == len(pd.date_range('1982-01-01', END)), f'{len(S)} days: a CCI year is missing'
YEARS = sorted(S.index.year.unique()); B = 2000
rng = np.random.default_rng(0); BIDX = rng.integers(0, len(YEARS), (B, len(YEARS)))

def doy366(idx):
    d = idx.dayofyear.values.copy(); d[(~idx.is_leap_year) & (idx.month > 2)] += 1
    return d

def hobday_clim(s, base=(1982, 2021), win=5, smooth=31, pct=90):
    """Hobday et al. (2016) climatological mean and 90th-percentile threshold, as in WP1."""
    v, d = s.values, doy366(s.index)

    inb = (s.index.year >= base[0]) & (s.index.year <= base[1])
    m, p = np.full(367, np.nan), np.full(367, np.nan)
    for k in range(1, 367):
        if k == 60: continue
        idx = np.where((d == k) & inb)[0]
        pool = np.unique(np.clip((idx[:, None] + np.arange(-win, win + 1)).ravel(), 0, len(v) -
    1))
        x = v[pool[inb[pool]]]; m[k], p[k] = np.nanmean(x), np.nanpercentile(x, pct)
    out = []
    for c in (m, p):
        c[60] = (c[59] + c[61]) / 2; c = c[1:]
        c = np.convolve(np.r_[c[-(smooth // 2):], c, c[:smooth // 2]], np.ones(smooth) / smooth,
    'valid')
        out.append(pd.Series(c[d - 1], index=s.index))
    return out

def detect(exc, min_dur=5, max_gap=2):
    e = np.asarray(exc, int); n = len(e)
    b = np.diff(np.r_[0, e, 0]); st, en = np.where(b == 1)[0], np.where(b == -1)[0] - 1
    keep = en - st + 1 >= min_dur; st, en = st[keep], en[keep]
    causal, nojoin = np.zeros(n, int), np.zeros(n, int)
    for s0, e0 in zip(st, en): nojoin[s0:e0 + 1] = 1; causal[s0 + min_dur - 1:e0 + 1] = 1
    label = nojoin.copy()
    for k in range(len(st) - 1):
        if st[k + 1] - en[k] - 1 <= max_gap: label[en[k] + 1:st[k + 1]] = 1
    return causal, label

def mcc(tp, fp, fn, tn):
    tp, fp, fn, tn = (np.asarray(v, float) for v in (tp, fp, fn, tn))
    den = np.sqrt((tp + fp) * (tp + fn) * (tn + fp) * (tn + fn))
    return np.where(den > 0, (tp * tn - fp * fn) / np.where(den > 0, den, 1), 0.0)

def counts(p, y, yr):
    g = pd.DataFrame({'tp': (p == 1) & (y == 1), 'fp': (p == 1) & (y == 0), 'fn': (p == 0) & (y ==
    1),
                      'tn': (p == 0) & (y == 0), 'yr': yr}).groupby('yr').sum()
    return g.reindex(YEARS, fill_value=0).values                       # years x 4

point = lambda c: float(mcc(*c.sum(0)))
boot = lambda c: mcc(*c[BIDX].sum(1).T)
ci    = lambda d: f'{np.percentile(d, 2.5):+.3f} to {np.percentile(d, 97.5):+.3f}'

D = {}
for name in ('OISST', 'CCI'):
    s = S[name]; tc = np.arange(len(s)) - (len(s) - 1) / 2
    slope = (s.values * tc).sum() / (tc ** 2).sum()
    sd = s - slope * tc
    clim, thr = hobday_clim(sd)
    exc = (sd > thr).astype(int).values
    causal, label = detect(exc)
    D[name] = dict(sst=s, sdet=sd, clim=clim, thr=thr, exc=exc, causal=causal, label=label,
    trend=slope * 3652.5)

def events(lab):
    b = np.diff(np.r_[0, lab, 0]); st, en = np.where(b == 1)[0], np.where(b == -1)[0] - 1
    return len(st), int(np.median(en - st + 1))
a, b = D['OISST'], D['CCI']
anomA, anomB = a['sdet'] - a['clim'], b['sdet'] - b['clim']
agree = counts(b['label'], a['label'], S.index.year.values)
cmp = {'mean SST (degC)': (a['sst'].mean(), b['sst'].mean()), 'trend (degC/decade)': (a['trend'],
```

```
    b['trend']),
       'MHW events (detrended)': (events(a['label'])[0], events(b['label'])[0]),
       'MHW days (detrended)': (int(a['label'].sum()), int(b['label'].sum())),
       'median event duration (days)': (events(a['label'])[1], events(b['label'])[1]),
       'MHW days known only in hindsight (%)': (100 * ((a['label'] == 1) & (a['causal'] ==
    0)).sum() / a['label'].sum(),
                                                100 * ((b['label'] == 1) & (b['causal'] ==
    0)).sum() / b['label'].sum())}
CMP = pd.DataFrame(cmp, index=['OISST', 'CCI']).T.round(3)
CMP.to_csv(f'{PROC}/v2_dataset_comparison.csv')
print(CMP.to_string())
print(f'\ncorrelation of daily SST {np.corrcoef(a["sst"], b["sst"])[0, 1]:.3f} | of detrended
    daily anomalies {np.corrcoef(anomA, anomB)[0, 1]:.3f}')
print(f'mean difference CCI-OISST {(b["sst"] - a["sst"]).mean():+.3f} degC | RMS difference of
    anomalies {np.sqrt(((anomB - anomA) ** 2).mean()):.3f} degC')
print(f'agreement of daily MHW labels: MCC {point(agree):.3f} | '
      f'{100 * ((a["label"] == 1) & (b["label"] == 1)).sum() / a["label"].sum():.1f}% of OISST MHW
    days are also MHW days in CCI')

# V2-2 benchmark leakage with both datasets
LEADS1 = [1, 2, 3, 5, 7, 10, 14]; rows = []

for name in ('OISST', 'CCI'):
    L = D[name]
    for h in LEADS1:
        y, yr = L['label'][h:], S.index.year.values[h:]
        cl, cc = counts(L['label'][:-h], y, yr), counts(L['causal'][:-h], y, yr)
        d = boot(cl) - boot(cc)
        rows.append({'dataset': name, 'lead': h, 'label-based': round(point(cl), 3), 'honest':
    round(point(cc), 3),
                     'inflation': round(point(cl) - point(cc), 3), 'inflation 95% CI': ci(d),
    'lo': np.percentile(d, 2.5),
                     'hi': np.percentile(d, 97.5)})
LK = pd.DataFrame(rows); LK.to_csv(f'{PROC}/v2_leakage_both.csv', index=False)
print(LK.drop(columns=['lo', 'hi']).pivot(index='lead', columns='dataset').to_string())

# V2-3 honest vs exceedance persistence vs logistic SST model, both datasets
import time
def run_length(b):
    g = (~b).cumsum(); return b.astype(int).groupby(g).cumsum()

def sst_features(L):
    a_, dist = L['sdet'] - L['clim'], L['sdet'] - L['thr']
    X = pd.DataFrame(index=S.index)
    X['sst_anom'], X['sst_dist_thr'] = a_, dist
    for k in (1, 3, 7):      X[f'sst_anom_lag{k}'] = a_.shift(k)
    for w in (3, 7, 14, 30): X[f'sst_anom_rm{w}'] = a_.rolling(w).mean()
    X['sst_anom_sd7'], X['sst_anom_max7'] = a_.rolling(7).std(), a_.rolling(7).max()
    for k in (1, 3, 7):      X[f'sst_anom_chg{k}'] = a_ - a_.shift(k)
    e = dist > 0
    X['exceed_now'] = e.astype(int); X['exceed_run'] = run_length(e); X['mhw_causal'] =
    (X.exceed_run >= 5).astype(int)
    doy = S.index.dayofyear
    X['doy_sin'], X['doy_cos'] = np.sin(2 * np.pi * doy / 365.25), np.cos(2 * np.pi * doy /
    365.25)
    return X

def train_mask(tab, test_years):
    m = ~tab.year.isin(test_years) & ~tab.target_year.isin(test_years)
    after = tab.year.isin([y + 1 for y in test_years]) & (tab.index.dayofyear <= 30)
    return m & ~after

def best_threshold(y, p):
    th = np.unique(np.r_[np.arange(0.02, 0.99, 0.01), p]); th = th[(th > 0) & (th < 1)]
    P = p[None, :] >= th[:, None]
    m = mcc((P & (y == 1)).sum(1), (P & (y == 0)).sum(1), ((~P) & (y == 1)).sum(1), ((~P) & (y ==
    0)).sum(1))
    return th[np.argmax(m)], m.max()

GRID = [(C, lambda C=C: make_pipeline(StandardScaler(), LogisticRegression(C=C,
    class_weight='balanced', max_iter=3000)))
        for C in (0.01, 0.1, 1.0)]

def loyo_logit(tab, cols, inner_folds=4):
    """Same design as the main study: inner year-grouped CV for C, isotonic calibration, MCC-
    optimal threshold."""
    preds = []
    for y in YEARS:
        tr, te = tab[train_mask(tab, [y])], tab[tab.year == y]
        chunks = np.array_split(np.array(sorted(tr.year.unique())), inner_folds)
        best = None
        for C, make in GRID:
            oof = pd.Series(np.nan, index=tr.index)
            for ch in chunks:
                itr, iva = tr[train_mask(tr, list(ch))], tr[tr.year.isin(ch)]
                oof.loc[iva.index] = make().fit(itr[cols], itr.y).predict_proba(iva[cols])[:, 1]
            ok = oof.notna(); sc = best_threshold(tr.y[ok].values, oof[ok].values)[1]
```

```
            if best is None or sc > best[0]: best = (sc, make, oof[ok], tr.y[ok])
        _, make, oof, yo = best
        iso = IsotonicRegression(out_of_bounds='clip', y_min=0, y_max=1).fit(oof.values,
    yo.values)
        th, _ = best_threshold(yo.values, iso.predict(oof.values))
        p = iso.predict(make().fit(tr[cols], tr.y).predict_proba(te[cols])[:, 1])
        preds.append(pd.Series((p >= th).astype(int), index=te.index))
    return pd.concat(preds)

LEADS2 = [1, 3, 5, 7, 10, 14]; rows = []; t0 = time.time()
for name in ('OISST', 'CCI'):
    L = D[name]; X = sst_features(L); cols = list(X.columns)
    lab = pd.Series(L['label'], index=S.index)
    for h in LEADS2:
        tb = X.assign(y=lab.shift(-h), year=S.index.year, target_year=(S.index +
    pd.Timedelta(days=h)).year).dropna()

        tb['y'] = tb.y.astype(int)
        logit = loyo_logit(tb, cols)
        c = {n: counts(p.values, tb.y.values, tb.year.values) for n, p in
             (('honest', tb.mhw_causal), ('exceedance', tb.exceed_now), ('logistic',
    logit.reindex(tb.index)))}
        bt = {n: boot(v) for n, v in c.items()}
        rows.append({'dataset': name, 'lead': h, 'honest': round(point(c['honest']), 3),
    'exceedance': round(point(c['exceedance']), 3),
                     'logistic': round(point(c['logistic']), 3),
                     'logistic vs honest': ci(bt['logistic'] - bt['honest']), 'logistic vs
    exceedance': ci(bt['logistic'] - bt['exceedance']),
                     'lo_h': np.percentile(bt['logistic'] - bt['honest'], 2.5), 'lo_e':
    np.percentile(bt['logistic'] - bt['exceedance'], 2.5),
                     'hi_e': np.percentile(bt['logistic'] - bt['exceedance'], 97.5)})
        print(f'{name} lead {h} d done ({time.time() - t0:.0f} s)')
SK = pd.DataFrame(rows); SK.to_csv(f'{PROC}/v2_skill_both.csv', index=False)
pd.set_option('display.width', 250); pd.set_option('display.max_columns', 20)
print(SK.drop(columns=['lo_h', 'lo_e', 'hi_e']).to_string(index=False))

c = SK[SK.dataset == 'CCI']; lk = LK[(LK.dataset == 'CCI') & (LK.lead <= 7)]
r1 = bool((lk.lo > 0).all()); r2 = bool((c.lo_h > 0).all())
r3 = bool((c[c.lead.isin([1, 3])].lo_e <= 0).all())
print(f'\nRULE with CCI: (1) inflation CI > 0 at 1-7 d: {r1} | (2) logistic beats honest at every
    lead: {r2} | '
      f'(3) logistic not significantly better than exceedance at 1 and 3 d: {r3} -> {"REPRODUCED"
    if r1 and r2 and r3 else "NOT REPRODUCED"}')

# V2-5 why do OISST and CCI disagree?
import numpy as np, pandas as pd
A, C = anomA, anomB                                  # detrended daily anomalies from V2-1
# 1. time alignment: the correlation should peak at lag 0
print('anomaly correlation, CCI shifted by k days:', {k: round(float(A.corr(C.shift(k))), 3) for k
    in range(-3, 4)})
# 2. agreement by time scale
print('correlation of 1-, 7- and 30-day mean anomalies:',
      [round(float(A.rolling(w).mean().corr(C.rolling(w).mean())), 3) for w in (1, 7, 30)])
# 3. persistence and smoothness of each product
for n, x in (('OISST', A), ('CCI', C)):
    print(f'{n:5s} anomaly SD {x.std():.3f} | SD of day-to-day change {x.diff().std():.3f} | '
          f'autocorrelation lag 1/7/14/30: {[round(float(x.autocorr(k)), 3) for k in (1, 7, 14,
    30)]}')
# 4. by calendar month
g = pd.DataFrame({'O': A, 'C': C, 'lo': a['label'], 'lc': b['label']}, index=S.index)
mon = g.groupby(g.index.month).apply(lambda d: pd.Series({
    'r': d.O.corr(d.C), 'RMS diff': np.sqrt(((d.C - d.O) ** 2).mean()), 'SD OISST': d.O.std(), 'SD
    CCI': d.C.std(),
    'MHW days OISST': d.lo.sum(), 'MHW days CCI': d.lc.sum(), 'both': ((d.lo == 1) & (d.lc ==
    1)).sum()})).round(3)
print('\nby calendar month\n' + mon.to_string())
# 5. by period (raw SST difference shows where the trend difference comes from)
per = pd.cut(S.index.year, [1981, 1991, 2002, 2015, 2024], labels=['1982-1991', '1992-2002',
    '2003-2015', '2016-2024'])
pp = g.assign(raw_diff=(b['sst'] - a['sst']).values).groupby(per, observed=True).apply(lambda d:
    pd.Series({
    'mean raw diff CCI-OISST': d.raw_diff.mean(), 'r': d.O.corr(d.C), 'RMS diff': np.sqrt(((d.C -
    d.O) ** 2).mean()),
    'MHW days OISST': d.lo.sum(), 'MHW days CCI': d.lc.sum(), 'both': ((d.lo == 1) & (d.lc ==
    1)).sum()})).round(3)
print('\nby period\n' + pp.to_string())
mon.to_csv(f'{PROC}/v2_diag_month.csv'); pp.to_csv(f'{PROC}/v2_diag_period.csv')
```

C.11 Individual predictor selection with the L1 penalty

```
# Reuses train_mask, best_threshold, mcc_from, year_counts, bootstrap (C.6)
# and table_for, FSETS (C.7)
import numpy as np, pandas as pd
```

```
from sklearn.pipeline import make_pipeline
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.isotonic import IsotonicRegression

COLS = FSETS['all']                                   # all 83 predictors
l1 = lambda C: (lambda: make_pipeline(StandardScaler(),
        LogisticRegression(penalty='l1', solver='liblinear', C=C,   # scikit-learn >= 1.8:
l1_ratio=1.0
                           class_weight='balanced', max_iter=5000)))
GRID_C = {h: (0.0005, 0.001, 0.003, 0.01, 0.03) for h in (3, 5, 7, 10, 14)}
GRID_C[1] = GRID_C[3] + (0.1, 0.3)                    # every 1-day fold chose 0.03 on the first
grid

def loyo_l1(tab, grid_C, inner_folds=4):
    """Same LOYO loop as loyo() in C.6; also keeps the coefficients of each fold's final model."""
    rows, coefs = [], {}
    for y in sorted(tab.year.unique()):
        tr, te = tab[train_mask(tab, [y])], tab[tab.year == y]
        chunks = np.array_split(np.array(sorted(tr.year.unique())), inner_folds)
        best = None
        for C in grid_C:
            oof = pd.Series(np.nan, index=tr.index)
            for ch in chunks:
                itr, iva = tr[train_mask(tr, list(ch))], tr[tr.year.isin(ch)]
                oof.loc[iva.index] = l1(C)().fit(itr[COLS], itr.y).predict_proba(iva[COLS])[:, 1]
            ok = oof.notna()
            score = best_threshold(tr.y[ok].values, oof[ok].values)[1]
            if best is None or score > best[0]: best = (score, C, oof[ok], tr.y[ok])
        _, C, oof, yo = best
        iso = IsotonicRegression(out_of_bounds='clip', y_min=0, y_max=1).fit(oof.values,
yo.values)
        th, _ = best_threshold(yo.values, iso.predict(oof.values))
        fit = l1(C)().fit(tr[COLS], tr.y)
        coefs[y] = pd.Series(fit[-1].coef_[0], index=COLS)
        p = iso.predict(fit.predict_proba(te[COLS])[:, 1])
        rows.append(pd.DataFrame({'year': y, 'y': te.y.values, 'prob': p,
                                  'pred': (p >= th).astype(int), 'C': C}, index=te.index))
    return pd.concat(rows), pd.DataFrame(coefs).T

def half_sample_coefs(tab, C, n_pairs=50, seed=2026):
    """Refit on 2 x n_pairs complementary half-samples of the years; C doubled for half the
rows."""
    yrs, rng, out = np.array(sorted(tab.year.unique())), np.random.default_rng(seed), []
    for _ in range(n_pairs):
        perm = rng.permutation(yrs)
        for half in (perm[:len(yrs) // 2], perm[len(yrs) // 2:]):
            sub = tab[tab.year.isin(half)]
            out.append(l1(2 * C)().fit(sub[COLS], sub.y)[-1].coef_[0])
    return pd.DataFrame(out, columns=COLS)

for h in (1, 3, 5, 7, 10, 14):
    tab = table_for(h)
    o, cf = loyo_l1(tab, GRID_C[h])
    ref = pd.read_csv(f'{PROC}/oof_lead{h}/logit_sst_det.csv' if h != 7 else
                      f'{PROC}/oof_p5/logit_sst_det.csv', index_col=0, parse_dates=True)
    new_o, new_r = tab.mhw_causal.reindex(o.index) == 0, tab.mhw_causal.reindex(ref.index) == 0
    gain_all = bootstrap(o, {'logit_sst': ref})                       # all days
    gain_new = bootstrap(o[new_o], {'logit_sst': ref[new_r]})         # new-event days
    hs = half_sample_coefs(tab, o.groupby('year').C.first().mode()[0], seed=2026 + h)
    sel = hs.abs() > 1e-8
    freq = sel.mean()                                                  # selection frequency
    same_sign = np.maximum((hs > 1e-8).sum(), (hs < -1e-8).sum()) / sel.sum().replace(0, np.nan)
    q = sel.sum(axis=1).mean()
    bound = q ** 2 / ((2 * 0.8 - 1) * len(COLS))                       # Meinshausen-Buehlmann
bound
    stable = freq[(freq >= 0.8) & (same_sign >= 0.95)].index.tolist()
```

C.12 Upper-ocean heat memory test (GODAS)

```
# setup
# (if Colab already imported pygrib in this session: Runtime > Restart session, then run from
here)
# pip install eccodes
from google.colab import drive; drive.mount('/content/drive')
import os, re, glob, time
import numpy as np, pandas as pd, requests, eccodes
import matplotlib.pyplot as plt
from concurrent.futures import ThreadPoolExecutor
from sklearn.pipeline import make_pipeline
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.isotonic import IsotonicRegression
```

```
PROC = '/content/drive/MyDrive/Colab Notebooks/MHW_Project_Data/processed'
GOD = f'{PROC}/godas'
for d in (f'{GOD}/raw', f'{GOD}/oof', f'{PROC}/figures'): os.makedirs(d, exist_ok=True)
FIG = f'{PROC}/figures'
TMP = '/content/godas_tmp'; os.makedirs(TMP, exist_ok=True)

CPC = 'https://ftp.cpc.ncep.noaa.gov/godas/pentad'
LAT, LON = (15.0, 23.0), (85.0, 93.0)
DEPTHS = [5, 15, 25, 35, 45]            # GODAS level centres (m) -> 0-50 m layer
Y0, Y1 = 1981, 2024                     # 1981 is needed for the lagged predictors in early 1982
END = '2024-12-31'
LEADS = [7, 10, 14]
LAGS = [14, 28]                         # main run, leakage check
B = 2000

# GRIB helpers + test on one pentad file
SES = requests.Session()

def get(url, headers=None, tries=5):
    for k in range(tries):
        try:
            r = SES.get(url, headers=headers, timeout=180)
            if r.status_code in (400, 401, 404, 410):
                raise RuntimeError(f'{r.status_code} for {url}')
            r.raise_for_status(); return r
        except RuntimeError:
            raise
        except Exception as e:
            if k == tries - 1: raise
            time.sleep(10 * (k + 1))

def list_year(yr):
    return sorted(set(re.findall(r'godas\.P\.(\d{8})\.grb(?!\.)', get(f'{CPC}/{yr}/').text)))

def wanted(line):
    """True for potential temperature at 5-45 m or the isothermal-layer depth (wgrib inventory
line)."""
    k5, k6, k7 = (re.search(rf'kpds{i}=(\d+)', line) for i in (5, 6, 7))
    if k5 and k6 and k7:
        p5, p6, p7 = int(k5.group(1)), int(k6.group(1)), int(k7.group(1))
        return (p5 == 13 and p6 == 160 and p7 in DEPTHS) or (p5 == 195 and p6 == 238)
    f = line.split(':'); var = f[3].upper() if len(f) > 3 else ''
    m = re.search(r':(\d+) m below sea level', line)
    return (var == 'POT' and m is not None and int(m.group(1)) in DEPTHS) or \
           (var == 'DBSS' and 'isothermal' in line.lower())

def get_range(url, start, stop, tries=5):
    """Bytes [start, stop) of a file (stop=None: to the end). Retries with a pause only when the
server
    refuses or the connection fails; if it ignores the range and sends the whole file (200), bytes
are cut from it."""
    rng = f'bytes={start}-{"" if stop is None else stop - 1}'
    for k in range(tries):
        try:
            r = SES.get(url, headers={'Range': rng}, timeout=180, allow_redirects=False)
            why = f'HTTP {r.status_code}'
            if r.status_code == 206: return r.content
            if r.status_code == 200: return r.content[start:stop]
        except requests.RequestException as e:
            why = type(e).__name__
        wait = 15 * (k + 1)
        print(f'  {url.rsplit("/", 1)[-1]}: {why}, waiting {wait} s', flush=True)
        time.sleep(wait)
    raise RuntimeError(f'{url}: no response after {tries} tries ({why})')

def tiles(chunk, n):
    """True if chunk is exactly n complete GRIB1 messages ('GRIB' ... '7777')."""
    pos, k = 0, 0
    while pos < len(chunk):
        if chunk[pos:pos + 4] != b'GRIB': return False
        L = int.from_bytes(chunk[pos + 4:pos + 7], 'big')
        if L < 8 or chunk[pos + L - 4:pos + L] != b'7777': return False
        pos += L; k += 1
    return pos == len(chunk) and k == n

def fetch_records(stamp, full=False):
    """Download only the needed GRIB records of one pentad file (byte ranges from its .inv index).
    The whole 38 MB file is used instead when there is no index or the index does not match the
file
    (some files on the server were reprocessed after their index was written)."""
    base = f'{CPC}/{stamp[:4]}/godas.P.{stamp}.grb'
    f = f'{TMP}/{stamp}.grb'
    inv = None
    if not full:
        for k in range(3):                              # a 404 can be throttling or a genuinely
missing index
```

```
            try:
                inv = [l for l in get(base + '.inv').text.splitlines() if l.strip()]; break
            except Exception:                           # missing (404) or forbidden (403) index
                time.sleep(15 * (k + 1))
    if inv is not None:
        off = [int(l.split(':')[1]) for l in inv]
        idx = [i for i, l in enumerate(inv) if wanted(l)]
        groups, cur = [], [idx[0]]
        for i in idx[1:]:
            if i == cur[-1] + 1: cur.append(i)
            else: groups.append(cur); cur = [i]
        groups.append(cur)
        chunks = []
        for g in groups:
            stop = off[g[-1] + 1] if g[-1] + 1 < len(off) else None
            c = get_range(base, off[g[0]], stop)
            if not tiles(c, len(g)): break              # index does not match this file
            chunks.append(c)
        else:
            with open(f, 'wb') as h: h.write(b''.join(chunks))
            return f, [inv[i] for i in idx]
    with open(f, 'wb') as h: h.write(get_range(base, 0, None))
    return f, [f'{stamp}: full file used']

GRID = {}
def read_messages(f):
    """Decode each GRIB message with ecCodes, reading only grid, level and value keys
    (pygrib fails on the 5-day-average time header of the GODAS files)."""
    out = []
    with open(f, 'rb') as fh:
        while True:
            gid = eccodes.codes_grib_new_from_file(fh)
            if gid is None: break
            try:
                p = eccodes.codes_get(gid, 'indicatorOfParameter')
                lev = eccodes.codes_get(gid, 'level')
                lt = eccodes.codes_get_long(gid, 'indicatorOfTypeOfLevel')
                if 'lat' not in GRID:
                    GRID['lat'] = eccodes.codes_get_array(gid, 'latitudes')
                    GRID['lon'] = eccodes.codes_get_array(gid, 'longitudes')
                v = eccodes.codes_get_values(gid).astype(float)
                if eccodes.codes_get(gid, 'bitmapPresent'):
                    v[v == eccodes.codes_get(gid, 'missingValue')] = np.nan
                v[np.abs(v) > 1e19] = np.nan
                out.append((p, lt, lev, v))
            finally:
                eccodes.codes_release(gid)
    return out

def box_values(stamp):
    """Box means for one pentad: 0-50 m mean potential temperature (degC) and isothermal-layer
depth (m).
    Records are identified from the GRIB keys themselves (parameter 13 at depth levels; parameter
195
    at the isothermal-layer level, 238), so a wrong record can never be used."""
    for full in (False, True):
        f, lines = fetch_records(stamp, full)
        try:
            msgs = read_messages(f)
        except Exception:
            msgs = []
        finally:
            os.remove(f)
        if 'sel' not in GRID and 'lat' in GRID:
            la, lo = GRID['lat'], GRID['lon']
            GRID['sel'] = (la >= LAT[0]) & (la <= LAT[1]) & (lo >= LON[0]) & (lo <= LON[1])
            GRID['w'] = np.cos(np.deg2rad(la[GRID['sel']]))
            print('box latitudes', la[GRID['sel']].min().round(2), '-',
la[GRID['sel']].max().round(2),
                  '| longitudes', lo[GRID['sel']].min(), '-', lo[GRID['sel']].max())
        T = {int(lev): v for p, lt, lev, v in msgs if p == 13 and lt == 160 and int(lev) in
DEPTHS}
        I = [v for p, lt, lev, v in msgs if p == 195 and lt == 238]
        if sorted(T) == DEPTHS and len(I) == 1: break
    else:
        raise AssertionError(f'{stamp}: records not found, got {[(p, lt, lev) for p, lt, lev, _ in
msgs][:12]}\n'
                             + '\n'.join(lines))
    sel, w = GRID['sel'], GRID['w']
    T = np.vstack([T[d][sel] for d in DEPTHS]); ild = I[0][sel]
    if np.nanmean(T) > 200: T = T - 273.15                                  # K -> degC
    okT, okI = np.isfinite(T).all(0), np.isfinite(ild)
    return dict(time=pd.to_datetime(stamp, format='%Y%m%d'),
                t50=float((T.mean(0)[okT] * w[okT]).sum() / w[okT].sum()),
cells_t50=int(okT.sum()),
                ild=float((ild[okI] * w[okI]).sum() / w[okI].sum()), cells_ild=int(okI.sum()))
```

```
# download all pentads, one CSV per year (resumable)
def safe_box(stamp):
    """A pentad the server refuses to deliver is kept as a gap (NaN) and filled below."""
    try:
        if stamp == '20240927': raise RuntimeError('refused by the server (HTTP 403)')
        return box_values(stamp)
    except Exception as e:
        print('  SKIPPED', stamp, str(e)[:150], flush=True)
        return dict(time=pd.to_datetime(stamp, format='%Y%m%d'), t50=np.nan, cells_t50=np.nan,
                    ild=np.nan, cells_ild=np.nan)

for yr in range(Y0, Y1 + 1):
    f = f'{GOD}/raw/godas_box_{yr}.csv'
    if os.path.exists(f): continue
    t0 = time.time(); stamps = list_year(yr)
    with ThreadPoolExecutor(max_workers=8) as ex:
        rows = pd.DataFrame(list(ex.map(safe_box, stamps))).set_index('time').sort_index()
    n_bad = int(rows.t50.isna().sum())
    if n_bad > 2:                                          # server is refusing generally: do not
save, retry later
        print(yr, f'{n_bad} pentads refused - not saved; wait and rerun this cell'); break
    rows.to_csv(f)
    print(yr, len(stamps), 'pentads', f'({time.time() - t0:.0f} s)', f'| {n_bad} gap(s)' if n_bad
else '')

G = pd.concat([pd.read_csv(f, index_col=0, parse_dates=True) for f in
sorted(glob.glob(f'{GOD}/raw/godas_box_*.csv'))])
G = G[~G.index.duplicated()].sort_index()
gap = G[['t50', 'ild']].isna().any(axis=1)
print('pentads not delivered by the server (filled by linear interpolation):', [str(d.date()) for
d in G.index[gap]])
assert gap.sum() <= 5 and not (gap & gap.shift(1, fill_value=False)).any(), 'too many or
consecutive missing pentads'
G[['t50', 'ild']] = G[['t50', 'ild']].interpolate(limit=1)
G.to_csv(f'{GOD}/godas_box_pentad.csv')
print(G.describe().round(2))
print('pentads per year:', G.groupby(G.index.year).size().value_counts().to_dict())
n_py = G.groupby(G.index.year).size()
assert (n_py.iloc[:-1] == 73).all(), 'a year before the last does not have 73 pentads'
print('last pentad:', G.index[-1].date())
assert G.cells_t50.dropna().nunique() == 1 and G.cells_ild.dropna().nunique() == 1, 'ocean mask
changes between pentads'
assert G[['t50', 'ild']].notna().all().all()

# anomalies, detrending, daily series with availability lag
p = G.groupby(G.index.year).cumcount().values                              # pentad of year, 0-72
base = (G.index.year >= 1982) & (G.index.year <= 2021)
for v in ('t50', 'ild'):
    c = G[v][base].groupby(p[base]).mean().reindex(range(73))
    c = pd.Series(np.convolve(np.r_[c.values[-1:], c.values, c.values[:1]], np.ones(3) / 3,
'valid'))  # circular 3-pentad
    a = G[v] - c.reindex(p).values
    tt = (G.index - G.index[0]).days.values.astype(float)
    G[f'{v}_a'] = a - np.polyval(np.polyfit(tt, a.values, 1), tt)           # linear detrend, like
iobw_dt

def godas_daily(lag, days):
    """Value on day t = latest pentad dated s with s + lag <= t; NaN if that pentad is > 10 days
stale."""
    A = G[['t50_a', 'ild_a']].copy()
    A.index = A.index + pd.Timedelta(days=lag)
    d = A.reindex(days, method='ffill')
    used = pd.Series(A.index, index=A.index).reindex(days, method='ffill')
    d.loc[(days - pd.DatetimeIndex(used.values)) > pd.Timedelta(days=10)] = np.nan
    X = pd.DataFrame(index=days)
    X['god_t50_anom'], X['god_ild_anom'] = d.t50_a, d.ild_a
    X['god_t50_chg30'] = d.t50_a - d.t50_a.shift(30)
    X['god_ild_chg30'] = d.ild_a - d.ild_a.shift(30)
    return X

# run: SST-only vs SST + GODAS, same rows, both products, both lags
GX = {lag: godas_daily(lag, S.index) for lag in LAGS}
GCOLS = list(GX[LAGS[0]].columns)
common = pd.Series(np.logical_and.reduce([GX[l].notna().all(1).values for l in LAGS]),
index=S.index)
print(f'GODAS predictors available on {common.sum()} of {len(common)} days '
      f'({common.idxmax().date()} -> {common[common].index[-1].date()})')

REF = {('OISST', 7): 0.366, ('OISST', 10): 0.276, ('OISST', 14): 0.267,      # Table 4.13
       ('CCI', 7): 0.465, ('CCI', 10): 0.351, ('CCI', 14): 0.273}
rows, t0 = [], time.time()
for name in ('OISST', 'CCI'):
    L = D[name]; X = sst_features(L); scols = list(X.columns)
    lab = pd.Series(L['label'], index=S.index)
    for h in LEADS:
        tb = X.assign(y=lab.shift(-h), year=S.index.year, target_year=(S.index +
pd.Timedelta(days=h)).year)
```

```
        for lag in LAGS: tb = tb.join(GX[lag].add_suffix(f'_L{lag}'))
        tb = tb[common.reindex(tb.index).values].dropna()
        tb['y'] = tb.y.astype(int)
        P = {'sst': cached(f'{name}_h{h}_sst', lambda: loyo_logit(tb, scols))}
        for lag in LAGS:
            gc = [f'{c}_L{lag}' for c in GCOLS]
            P[f'L{lag}'] = cached(f'{name}_h{h}_godas_L{lag}', lambda: loyo_logit(tb, scols + gc))
        y, yr, q = tb.y.values, tb.year.values, (tb.mhw_causal == 0).values
        pr = {k: v.reindex(tb.index).values for k, v in P.items()}
        C = {k: counts(v, y, yr) for k, v in pr.items()}
        Cq = {k: counts(v[q], y[q], yr[q]) for k, v in pr.items()}
        Bt = {k: boot(c) for k, c in C.items()}; Bq = {k: boot(c) for k, c in Cq.items()}
        for lag in LAGS:
            k = f'L{lag}'; d, dq = Bt[k] - Bt['sst'], Bq[k] - Bq['sst']
            rows.append({'product': name, 'lead': h, 'lag': lag, 'rows': len(tb),
                         'MCC SST (Table 4.13)': REF[(name, h)], 'MCC SST': round(point(C['sst']),
3),
                         'MCC SST+GODAS': round(point(C[k]), 3), 'gain 95% CI': ci(d),
                         'new-event MCC SST': round(point(Cq['sst']), 3),
                         'new-event MCC SST+GODAS': round(point(Cq[k]), 3), 'new-event gain 95%
CI': ci(dq),
                         'lo': np.percentile(d, 2.5), 'hi': np.percentile(d, 97.5),
                         'lo_new': np.percentile(dq, 2.5), 'hi_new': np.percentile(dq, 97.5),
                         'gain': point(C[k]) - point(C['sst']), 'gain_new': point(Cq[k]) -
point(Cq['sst'])})
        print(f'{name} lead {h} d done ({time.time() - t0:.0f} s)')

RES = pd.DataFrame(rows); RES.to_csv(f'{PROC}/wp6_godas_heat_memory.csv', index=False)
pd.set_option('display.width', 250); pd.set_option('display.max_columns', 20)
print(RES.drop(columns=['lo', 'hi', 'lo_new', 'hi_new', 'gain',
'gain_new']).to_string(index=False))
print('\n"MCC SST" should equal Table 4.13.')

M = RES[RES.lag == LAGS[0]].set_index(['product', 'lead'])
ok = all(M.loc[(p_, h), 'lo_new'] > 0 for p_ in ('OISST', 'CCI') for h in (10, 14))
print(f'\nRULE (LAG = {LAGS[0]} d): new-event gain CI > 0 at 10 and 14 d with both products ->',
      'SUPPORTED' if ok else 'NOT SUPPORTED')
chk = RES.pivot_table(index=['product', 'lead'], columns='lag', values='lo_new')
print('lower CI bound of new-event gain by lag (a gain at 14 d that vanishes at 28 d = possible
leakage):')
print(chk.round(3).to_string())
```

C.13 Previous-year ENSO and IOD

```
# data
from google.colab import drive; drive.mount('/content/drive')
import os, numpy as np, pandas as pd, matplotlib.pyplot as plt
from scipy import stats

PROC = '/content/drive/MyDrive/Colab Notebooks/MHW_Project_Data/processed'
FIG = f'{PROC}/figures'; os.makedirs(FIG, exist_ok=True)
YRS = np.arange(1983, 2026)

df = pd.read_csv(f'{PROC}/predictor_table_1982_2025.csv', index_col=0, parse_dates=True)
assert {'dmi', 'mhw_detrended', 'mhw_raw'} <= set(df.columns), list(df.columns)

# monthly Nino 3.4 anomaly (not the one-month-shifted daily copy in the predictor table)
n = pd.read_csv('https://www.cpc.ncep.noaa.gov/data/indices/ersst5.nino.mth.91-20.ascii', sep=r'\
s+')
n34 = pd.Series(n.iloc[:, -1].values, index=pd.to_datetime(dict(year=n.YR, month=n.MON, day=1)))

def ndj(y):
    v = n34.reindex([pd.Timestamp(y - 1, 11, 1), pd.Timestamp(y - 1, 12, 1), pd.Timestamp(y, 1,
1)]).values
    assert np.isfinite(v).all(), f'Nino 3.4 missing for NDJ before {y}'
    return v.mean()

X = pd.DataFrame(index=YRS)
X['nino34_NDJ'] = [ndj(y) for y in YRS]

# ONI: find which year label the NDJ row uses by matching it to the Nino 3.4 NDJ just computed
o = pd.read_csv('https://www.cpc.ncep.noaa.gov/data/indices/oni.ascii.txt', sep=r'\s+')
o_ndj = o[o.SEAS == 'NDJ'].set_index('YR').ANOM
cand = {'Nov year': pd.Series(o_ndj.reindex(YRS - 1).values, index=YRS),
        'Jan year': pd.Series(o_ndj.reindex(YRS).values, index=YRS)}
rc = {k: v.corr(X.nino34_NDJ) for k, v in cand.items()}
key = max(rc, key=rc.get)
print('ONI NDJ year label =', key, '| r with Nino 3.4 NDJ:', {k: round(v, 3) for k, v in
rc.items()})
assert rc[key] > 0.95
X['oni_NDJ'] = cand[key]
assert X.oni_NDJ.notna().all(), X[X.oni_NDJ.isna()]
```

```
son = df.dmi[df.index.month.isin([9, 10, 11])]
X['dmi_SON'] = son.groupby(son.index.year).mean().reindex(YRS - 1).values
assert X.notna().all().all()
print(f'r(ONI NDJ, Nino 3.4 NDJ) = {X.oni_NDJ.corr(X.nino34_NDJ):.3f} | '
      f'Spearman r(Nino 3.4 NDJ, DMI SON) = {stats.spearmanr(X.nino34_NDJ, X.dmi_SON)[0]:.2f}')

def responses(lab):
    lab = lab.astype(int)
    onset = (lab == 1) & (lab.shift(1, fill_value=0) == 0)
    R = pd.DataFrame(index=YRS)
    for name, months in (('FebDec', range(2, 13)), ('MAMJ', range(3, 7))):
        m = lab.index.month.isin(list(months))
        R[f'days_{name}'] = lab[m].groupby(lab.index[m].year).sum().reindex(YRS,
fill_value=0).values
        R[f'events_{name}'] = onset[m].groupby(lab.index[m].year).sum().reindex(YRS,
fill_value=0).values
    return R

RESP = {'detrended': responses(df.mhw_detrended), 'raw': responses(df.mhw_raw)}

# correlations
def partial(x, y, z, rank=True):
    if rank: x, y, z = (stats.rankdata(v) for v in (x, y, z))
    ex = x - np.polyval(np.polyfit(z, x, 1), z)
    ey = y - np.polyval(np.polyfit(z, y, 1), z)
    r = np.corrcoef(ex, ey)[0, 1]; k = len(x) - 3
    return r, 2 * stats.t.sf(abs(r * np.sqrt(k / (1 - r ** 2))), k)

def spear(x, y): return tuple(stats.spearmanr(x, y))
def pears(x, y): return tuple(stats.pearsonr(x, y))
def loyo(f, *a):
    rs = [f(*(np.delete(np.asarray(v, float), i) for v in a))[0] for i in range(len(a[0]))]
    return min(rs), max(rs)

z = X.nino34_NDJ.values
rows = []
for lt, R in RESP.items():
    for resp in R.columns:
        y = R[resp].values.astype(float)
        for pred, f, args, pf in (
                ('ONI NDJ', spear, (X.oni_NDJ.values,), lambda: pears(X.oni_NDJ.values, y)),
                ('Nino 3.4 NDJ', spear, (X.nino34_NDJ.values,), lambda: pears(X.nino34_NDJ.values,
y)),
                ('DMI SON', spear, (X.dmi_SON.values,), lambda: pears(X.dmi_SON.values, y)),
                ('DMI SON | Nino 3.4', lambda a, b, c: partial(a, b, c), (X.dmi_SON.values,),
                 lambda: partial(X.dmi_SON.values, y, z, rank=False))):
            extra = (z,) if pred.startswith('DMI SON |') else ()
            r, p = f(*args, y, *extra)
            lo, hi = loyo(f, *args, y, *extra)
            rp, pp = pf()
            rows.append({'labels': lt, 'response': resp, 'predictor (year before)': pred,
                         'Spearman r': round(r, 2), 'p': round(p, 3),
                         'LOYO r range': f'{lo:+.2f} to {hi:+.2f}', 'Pearson r': round(rp, 2),
'Pearson p': round(pp, 3)})
T = pd.DataFrame(rows)
T.to_csv(f'{PROC}/wp7_enso_iod_lag.csv', index=False)
pd.set_option('display.width', 250)
print(T.to_string(index=False))

D = RESP['detrended']
r, p = stats.spearmanr(X.oni_NDJ, D.days_FebDec); lo, _ = loyo(spear, X.oni_NDJ.values,
D.days_FebDec.values)
print(f'\nENSO RULE: Spearman r = {r:.2f}, p = {p:.3f}, lowest LOYO r = {lo:+.2f} ->',
      'SUPPORTED' if (r > 0 and p < 0.05 and lo > 0) else 'NOT SUPPORTED')
r, p = partial(X.dmi_SON.values, D.days_FebDec.values.astype(float), z)
print(f'IOD RULE: partial Spearman r = {r:.2f}, p = {p:.3f} ->', 'OWN EFFECT' if p < 0.05 else 'NO
OWN EFFECT')

# El Nino / neutral / La Nina composites (detrended labels)
ph = pd.cut(X.oni_NDJ, [-np.inf, -0.5, 0.5, np.inf], right=False, labels=['La Nina', 'Neutral',
'El Nino'])
C = D.assign(phase=ph.values).groupby('phase', observed=True)[['days_FebDec',
'events_FebDec']].agg(['mean', 'median', 'count'])
print(C.round(1).to_string())
el = ph == 'El Nino'
u = stats.mannwhitneyu(D.days_FebDec[el], D.days_FebDec[~el], alternative='greater')
print(f'MHW days Feb-Dec, years after El Nino vs all others: Mann-Whitney one-sided p =
{u.pvalue:.3f}')
print('\nYears after El Nino (ONI NDJ >= 0.5):')
print(X.loc[el, ['oni_NDJ']].join(D.loc[el, ['days_FebDec', 'events_FebDec',
'days_MAMJ']]).round(2).to_string())

# by calendar month of year Y (as Gao et al., 2022, Fig. 6)
lab = df.mhw_detrended.astype(int)
M = []
for m in range(2, 13):
    s = lab[lab.index.month == m]
```

```
    y = s.groupby(s.index.year).sum().reindex(YRS, fill_value=0).values.astype(float)
    r1, p1 = stats.spearmanr(X.oni_NDJ, y)
    r2, p2 = partial(X.dmi_SON.values, y, z)
    M.append({'month': m, 'r ONI': r1, 'p ONI': p1, 'partial r DMI': r2, 'p DMI': p2, 'mean MHW
days': y.mean()})
M = pd.DataFrame(M); M.to_csv(f'{PROC}/wp7_enso_iod_monthly.csv', index=False)
print(M.round(3).to_string(index=False))

r_crit = (lambda t: t / np.sqrt(43 - 2 + t ** 2))(stats.t.ppf(0.975, 41))
fig, ax = plt.subplots(1, 2, figsize=(11, 4))
ax[0].scatter(X.oni_NDJ, D.days_FebDec, color='#1f6fd1')
for yr in D.days_FebDec.nlargest(4).index.union(X.oni_NDJ.nlargest(3).index):
    ax[0].annotate(str(yr), (X.oni_NDJ[yr], D.days_FebDec[yr]), fontsize=8, xytext=(3, 3),
textcoords='offset points')
ax[0].axvline(0.5, color='grey', ls=':', lw=0.8); ax[0].axvline(-0.5, color='grey', ls=':',
lw=0.8)
ax[0].set_xlabel('ONI, Nov–Jan before the year (°C)'); ax[0].set_ylabel('Detrended MHW days, Feb–
Dec')
ax[0].set_title('(a) Previous-winter ONI and MHW days')
xm = np.arange(2, 13)
ax[1].bar(xm - 0.2, M['r ONI'], 0.4, color='#1f6fd1', label='ONI NDJ (Spearman)')
ax[1].bar(xm + 0.2, M['partial r DMI'], 0.4, color='#e0601a', label='DMI SON, partial on Niño
3.4')
for s_ in (1, -1): ax[1].axhline(s_ * r_crit, color='k', ls='--', lw=0.8)
ax[1].axhline(0, color='k', lw=0.8)
ax[1].set_xticks(xm); ax[1].set_xticklabels(list('FMAMJJASOND'))
ax[1].set_xlabel('Month of the following year'); ax[1].set_ylabel('Correlation with monthly MHW
days')
ax[1].set_title('(b) By calendar month (dashed: p = 0.05)'); ax[1].legend(fontsize=8,
frameon=False, loc='lower left')
plt.tight_layout(); plt.savefig(f'{FIG}/fig_wp7_enso_iod_lag.png', dpi=300); plt.show()
```